

Temporal-State-Aware Compact Routing with Coherence-Budget Management in Quantum-Native Entanglement Overlay Networks

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Contents

1	Introduction	7
1.1	Research Background	7
1.2	Research Motivation and Significance	9
1.3	Problem Statement	10
1.4	Research Objectives	11
1.5	Research Questions	12
1.6	Research Contributions	13
1.7	Thesis Organization	14
2	Literature Review	15
2.1	Introduction	15
2.2	Fundamentals of Quantum Networks	16
2.3	Entanglement Distribution in Quantum Repeater Networks . .	17
2.4	Quantum Network Routing	20
2.5	Noise, Decoherence, and Fidelity Degradation	22
2.6	Temporal State of Entanglement	25
2.7	Age-Aware Scheduling and Control	27
2.8	Quantum-Native Addressing and Entanglement Overlay Net- works	30
2.9	Compact Quantum Routing	32
2.9.1	Partial-Anchor Scheme	34
2.9.2	Full-Anchor Scheme	34
2.10	Comparative Analysis of Closest Prior Work	36
2.11	Limitations of Existing Literature	37
2.11.1	Absence of Explicit Temporal State in Compact Rout- ing Tables	37
2.11.2	Static Entangling-Stretch Guarantees	38
2.11.3	Limited Architectural Scope of Age-Aware Studies . . .	38
2.11.4	Regeneration Without Compact-Overlay-Aware Prior- itization	38
2.11.5	No Joint Analysis of Temporal and Scalability Trade-offs	39

2.11.6	Lack of Temporal Route-Feasibility Analysis	39
2.11.7	Insufficient Evaluation Under Heterogeneous Temporal Conditions	39
2.12	Identified Research Gap	40
2.13	Proposed Research Direction	41
3	System Model and Problem Formulation	45
3.1	Introduction	45
3.2	Network Architecture	46
3.3	Physical Quantum Network Model	49
3.4	Artificial Entanglement Overlay Model	52
3.4.1	Partial-Anchor Overlay	54
3.4.2	Full-Anchor Overlay	55
3.5	Temporal State of Artificial E-Links	55
3.6	Fidelity and Decoherence Model	58
3.7	Remaining Coherence-Budget Model	61
3.8	Temporal Route-Feasibility Condition	64
3.9	Time-Dependent Entangling-Cost Metric	68
3.10	Temporal Entangling Stretch	72
3.11	Problem Formulation	76
3.12	Chapter Summary	79
4	Proposed Temporal-State-Aware Compact Routing Method	80
4.1	Introduction	80
4.2	Design Objectives	81
4.2.1	Temporal Reliability	82
4.2.2	Fidelity Preservation	82
4.2.3	Compact Routing-State Preservation	83
4.2.4	Low Route-Selection Cost	83
4.2.5	Controlled Regeneration Overhead	83
4.2.6	Reduced Request Rejection	83
4.2.7	Robustness to Dynamic E-Link Conditions	84
4.2.8	Scalable Decision Making	84
4.3	Overview of the Proposed Framework	85
4.3.1	Temporal-State Monitoring Module	87
4.3.2	Compact Candidate-Construction Module	87
4.3.3	Compact Candidate-Construction Module	87
4.3.4	Temporal-Feasibility Module	87
4.3.5	Time-Dependent Route-Selection Module	87
4.3.6	Route-Repair Module	88
4.3.7	Overlay-Maintenance Module	88

4.4	Temporal E-Link State Monitoring	89
4.4.1	Temporal-State Classification and Link Availability . .	90
4.5	Candidate Route Construction	93
4.5.1	Candidate Construction in the Partial-Anchor Scheme	94
4.5.2	Candidate Construction in the Full-Anchor Scheme . .	95
4.5.3	Bounded Candidate-Set Size	95
4.5.4	Availability Test	96
4.5.5	Predicted Consumption-Time Calculation	97
4.5.6	Link-Level Fidelity Test	98
4.5.7	End-to-End Fidelity Test	98
4.5.8	Route-Delay Test	99
4.5.9	Memory and E-Bit Availability Test	99
4.5.10	Overall Temporal-Feasibility Decision	100
4.6	Time-Dependent Route Selection	100
4.6.1	Route-Level Cost Components	102
4.7	Overlay Maintenance Strategy	108
4.7.1	Retention Decision	108
4.7.2	Regeneration Decision	108
4.7.3	Replacement Decision	109
4.7.4	Discard Decision	110
4.7.5	Compact-Overlay Constraints	110
4.7.6	Memory-Aware Maintenance	111
4.7.7	Maintenance Triggering	112
4.7.8	Overlay-State Update	112
4.8	Proposed Routing Algorithm	113
4.8.1	Algorithmic Procedure	113
4.8.2	Algorithm Characteristics	117
4.9	Computational Complexity Analysis	118
4.9.1	Temporal-State Monitoring Complexity	119
4.9.2	Candidate-Route Construction Complexity	119
4.9.3	Temporal-Feasibility Filtering Complexity	120
4.9.4	Time-Dependent Route-Selection Complexity	121
4.9.5	Route-Repair Complexity	121
4.9.6	Regeneration-Priority Complexity	122
4.9.7	Overlay-Maintenance Complexity	123
4.9.8	Total Per-Slot Complexity	123
4.9.9	Space Complexity	124
4.10	Chapter Summary	125

5	Simulation Methodology and Experimental Setup	127
5.1	Introduction	127
5.2	Simulation Environment	129
5.3	Network Topology Generation	132
5.3.1	Random Connected Topology	133
5.3.2	Grid Topology	133
5.3.3	Small-World Topology	133
5.3.4	Network Sizes	134
5.4	Physical and Quantum-Link Parameters	135
5.5	Artificial Overlay Configuration	138
5.5.1	Partial-Anchor Overlay Configuration	138
5.5.2	Full-Anchor Overlay Configuration	139
5.5.3	Local E-Neighbor Construction	139
5.5.4	Artificial E-Link Generation	140
5.5.5	Initial Overlay Density	140
5.5.6	Memory Allocation	140
5.5.7	Initial E-Link Age	141
5.5.8	Overlay Initialization Procedure	141
5.6	Traffic and Entanglement-Request Model	142
5.6.1	Request-Arrival Process	143
5.6.2	Source-Destination Selection	143
5.6.3	Entanglement Demand Size	143
5.6.4	Fidelity Requirement	144
5.6.5	Request Deadline	144
5.6.6	Service Classes	145
5.6.7	Request Queue	145
5.6.8	Request Outcome	146
5.7	Benchmark Routing Methods	146
5.7.1	Shortest-Hop Routing	148
5.7.2	Static Entangling-Cost Routing	148
5.7.3	Fidelity-Only Routing	148
5.7.4	Age-Only Routing	149
5.7.5	Current-Fidelity-Aware Routing	149
5.7.6	Reactive-Regeneration Routing	149
5.7.7	Proposed Method without Regeneration	150
5.7.8	Proposed Method without Coherence-Budget Filtering	150
5.7.9	Proposed Method without Consumption-Time Prediction	150
5.8	Evaluation Metrics	152
5.8.1	Request Success Ratio	152
5.8.2	Request Rejection Ratio	152

5.8.3	Temporal Disconnection Ratio	153
5.8.4	Average End-to-End Fidelity	153
5.8.5	Fidelity-Violation Ratio	153
5.8.6	Average Completion Delay	153
5.8.7	Deadline-Violation Ratio	154
5.8.8	Average Route Length	154
5.8.9	Temporal Entangling Stretch	154
5.8.10	Stretch-Bound Violation Ratio	154
5.8.11	Regeneration Count	155
5.8.12	Regeneration Success Ratio	155
5.8.13	Regeneration Overhead	155
5.8.14	Quantum-Memory Utilization	155
5.8.15	Artificial E-Link Utilization	156
5.8.16	Stale-Link Selection Ratio	156
5.8.17	Routing Computation Time	156
5.8.18	Routing-Table Size	156
5.9	Experimental Scenarios	157
5.9.1	Scenario 1: Baseline Routing Comparison	158
5.9.2	Scenario 2: Network-Size Scalability	158
5.9.3	Scenario 3: Coherence-Time Variation	159
5.9.4	Scenario 4: Traffic-Load Variation	159
5.9.5	Scenario 5: Fidelity-Threshold Variation	160
5.9.6	Scenario 6: Quantum-Memory Capacity	160
5.9.7	Scenario 7: Regeneration-Budget Variation	160
5.9.8	Scenario 8: Partial-Anchor and Full-Anchor Comparison	161
5.9.9	Scenario 9: Hotspot and Non-Uniform Traffic	161
5.9.10	Scenario 10: Ablation Analysis	162
5.9.11	Scenario 11: Weight Sensitivity	162
5.9.12	Scenario 12: Initial-Age Distribution	163
5.10	Parameter Settings	163
5.11	Implementation Procedure	166
5.11.1	Project Organization	167
5.11.2	Topology Initialization	168
5.11.3	Artificial-Overlay Initialization	168
5.11.4	Temporal-State Update	169
5.11.5	Request Processing	169
5.11.6	Regeneration and Overlay Maintenance	170
5.11.7	Metric Collection	170
5.11.8	Matched Experimental Runs	171
5.11.9	Validation and Debugging	171
5.11.10	Large-Scale Scalability Experiment Configuration	172

5.12	Chapter Summary	172
6	Results and Discussion	174
6.1	Introduction	174
6.2	Preliminary Prototype Results	175
6.2.1	Prototype Configuration	175
6.2.2	Preliminary Request-Level Results	175
6.2.3	Controlled Route-Availability Comparison	176
6.2.4	Statistical Significance Analysis	178
6.2.5	Interpretation of the Preliminary Results	179
6.2.6	Current Limitations	179
6.2.7	Next Validation Plan	180
6.2.8	Implementation Validation	180
6.2.9	Matched Input Validation	181
6.2.10	Output Consistency Checks	182
6.2.11	Statistical Quality Control	183
6.3	Baseline Routing Comparison	184
6.4	Impact of Network Size	186
6.5	Impact of Memory Coherence Time	189
6.6	Impact of Traffic Load	191
6.7	Impact of Fidelity Threshold	193
6.8	Impact of Quantum-Memory Capacity	195
6.9	Impact of Regeneration Budget	196
6.10	Partial-Anchor and Full-Anchor Comparison	198
6.11	Ablation Study	199
6.12	Parameter Sensitivity Analysis	200
6.13	Parameter Sensitivity Analysis	201
6.14	Computational Performance	202
6.15	Discussion of Main Findings	202
6.16	Limitations of the Experimental Results	203
6.17	Chapter Summary	203
7	Conclusion and Future Work	204
7.1	Summary of Research	204
7.2	Main Contributions	204
7.3	Limitations	205
7.4	Future Research Directions	206
7.5	Conclusion	206

Chapter 1

Introduction

1.1 Research Background

The Quantum Internet is envisioned as a future communication infrastructure that connects quantum processors, sensors, memories, and users through distributed quantum entanglement. Unlike the classical Internet, where information is transmitted as bits through relatively persistent communication links, quantum networks rely on the generation, storage, distribution, and manipulation of fragile quantum states.

Quantum entanglement is one of the central resources of quantum networking. It supports important applications such as quantum teleportation, distributed quantum computing, quantum sensing, and quantum key distribution. However, direct entanglement distribution over long distances is limited by optical loss, channel noise, imperfect operations, and the no-cloning theorem.

Quantum repeaters are introduced to overcome these limitations. A long communication path is divided into shorter elementary links, where entanglement is generated between neighboring nodes. Intermediate repeater nodes then perform entanglement swapping to establish end-to-end entanglement between remote users.

Although quantum repeaters improve communication distance, they introduce new routing and resource-management challenges. Entanglement generation is probabilistic, swapping may fail, purification consumes multiple entangled pairs, and quantum memories have limited coherence times. Therefore, a route that is topologically available may not necessarily be physically or temporally usable.

The quality of an entangled state is commonly measured using fidelity. A generated pair is useful only when its fidelity remains above an application-

specific threshold. However, fidelity decreases while entanglement is stored in quantum memory because of decoherence. This means that the usefulness of an entangled link depends not only on its existence, but also on its age, initial fidelity, coherence time, and expected delay before consumption.

Recent research has introduced age-aware concepts such as Fidelity-Age and Age of Entanglement. These studies demonstrate that temporal freshness affects delivery regularity, fairness, route reliability, and end-to-end quality. Age-aware scheduling and control policies can reduce starvation and improve the timeliness of usable entanglement deliveries.

In parallel, recent quantum-native network architectures have proposed artificial entanglement overlays, quantum addressing, Entanglement-Defined Controllers, compact routing tables, and bounded entangling stretch. In these architectures, selected nodes proactively maintain entanglement with remote nodes, creating an artificial topology above the physical network.

The Partial-Anchor and Full-Anchor compact-routing schemes provide scalable routing-table structures with sublinear memory requirements. For additive entangling-cost metrics, the schemes provide worst-case entangling-stretch bounds of five and three, respectively.

However, the temporal state of each stored artificial e-link is not explicitly represented in the compact routing table. Information such as generation time, current age, effective fidelity, remaining coherence budget, expiration risk, and regeneration urgency is not directly incorporated into route selection.

This creates a significant research problem. An artificial link may be structurally present but may become unusable before the required swapping sequence is completed. Similarly, a stale anchor link may reduce overlay reachability, increase regeneration delay, or invalidate the original entangling-stretch guarantee.

Therefore, scalable quantum-native routing requires more than static connectivity and cost information. It requires temporal-state awareness that can determine whether an artificial e-link will remain usable until route completion.

This thesis investigates temporal-state-aware compact routing with coherence-budget management in quantum-native entanglement overlay networks. The study extends the compact-routing framework by introducing explicit e-link age, effective fidelity, remaining coherence budget, and regeneration cost into routing and overlay-maintenance decisions.

1.2 Research Motivation and Significance

The motivation for this research arises from the mismatch between the dynamic physical behavior of entanglement and the largely static assumptions used in existing compact-routing formulations. Artificial e-links are created proactively, stored in finite-coherence quantum memories, consumed during entanglement swapping, and continuously regenerated. Therefore, their routing value changes over time.

A compact-routing protocol may identify a structurally valid path through the artificial overlay, but the selected e-links may not remain usable until the route is completed. This can occur when stored entanglement has already accumulated significant age, when the remaining coherence time is insufficient, or when regeneration delay is ignored during route selection.

This issue is particularly important in anchor-based routing architectures. In the Partial-Anchor Scheme, a small subset of anchor nodes maintains long-range artificial links. The failure or degradation of one such link may affect many source–destination pairs. In the Full-Anchor Scheme, more nodes maintain long-range links, which can improve route quality but also increases temporal-maintenance complexity.

Existing age-aware studies demonstrate that temporal freshness affects service reliability, fairness, and delivery quality. However, these studies mainly focus on scheduling, repeater chains, or local control. They do not analyze how temporal degradation affects compact-routing tables, anchor reliability, overlay reachability, or entangling stretch.

Conversely, existing compact-routing architectures provide strong scalability and stretch guarantees, but they do not explicitly model the time-dependent state of each stored artificial e-link. This separation motivates the integration of temporal entanglement management with scalable compact routing.

The significance of this research is both theoretical and practical. From a theoretical perspective, it extends static entangling-cost and stretch formulations into time-dependent routing models. It also investigates whether compact-routing guarantees remain meaningful when links age, decohere, and require regeneration.

From a practical perspective, temporal-state-aware routing may reduce the selection of stale or near-expiration links, improve end-to-end fidelity, reduce route failure, and prioritize regeneration resources more effectively. It may also help an Entanglement-Defined Controller decide whether to retain, regenerate, replace, or avoid a particular artificial link.

The proposed coherence-budget concept provides a direct estimate of whether an e-link is expected to remain usable until route completion. This

allows routing decisions to account for both current quality and predicted future viability.

The research is also significant for large-scale quantum-network design because it aims to preserve the sublinear routing-table complexity of compact routing. A useful solution must improve temporal reliability without requiring each node to maintain full global state.

Therefore, the central significance of this work lies in developing a routing framework that jointly addresses scalability, fidelity, temporal freshness, coherence limitations, and overlay maintenance.

1.3 Problem Statement

Existing quantum-native compact-routing architectures provide scalable routing tables, artificial entanglement overlays, and bounded entangling-stretch guarantees. However, their routing decisions are primarily based on structural connectivity and generalized entangling cost, without explicitly representing the temporal condition of each stored artificial e-link.

This creates a fundamental inconsistency between routing abstraction and physical entanglement behavior. Artificial e-links are non-persistent resources. Their fidelity decreases during storage, they may expire before use, and they must be regenerated after consumption or degradation.

As a result, a route may be structurally valid but temporally infeasible. A selected path may contain one or more e-links that are too old, too close to decoherence, or unable to maintain the required fidelity until the expected completion of the swapping sequence.

The problem becomes more critical in anchor-based compact routing. In the Partial-Anchor Scheme, the degradation of one long-range anchor link may affect several source-destination paths. In the Full-Anchor Scheme, larger numbers of long-range links improve connectivity but increase maintenance and regeneration requirements.

Existing age-aware studies already model pair age, flow-level Fidelity-Age, end-to-end Age of Entanglement, discard, regeneration, and adaptive control. However, these approaches do not study quantum-native compact overlays, routing-table scalability, or entangling-stretch guarantees.

Therefore, the unresolved problem is not the absence of age-awareness or regeneration individually. The main problem is the absence of a compact-routing framework that explicitly incorporates per-e-link temporal state into route selection and overlay-maintenance decisions.

More specifically, existing work does not sufficiently answer the following questions:

- How should e-link age and effective fidelity be represented inside a compact routing table?
- How can the remaining coherence budget of an artificial e-link be estimated?
- How should a route be rejected when one or more e-links are expected to expire before completion?
- How should regeneration priorities be assigned across local, anchor, and tracked artificial links?
- How does temporal degradation affect the entangling-stretch guarantees of Partial-Anchor and Full-Anchor schemes?
- Can temporal-state awareness improve reliability without destroying sublinear routing-table scaling?

Accordingly, this thesis addresses the problem of designing a temporal-state-aware compact-routing framework that incorporates age, effective fidelity, coherence budget, and regeneration cost while preserving the scalability properties of quantum-native overlay routing.

1.4 Research Objectives

The primary objective of this research is to develop a temporal-state-aware compact-routing framework for quantum-native entanglement overlay networks.

The proposed framework aims to incorporate explicit temporal information into route selection and overlay-maintenance decisions while preserving the scalability advantages of compact routing.

The specific objectives are:

1. To define a temporal-state representation for each artificial e-link, including age, initial fidelity, effective fidelity, coherence time, remaining coherence budget, and regeneration cost.
2. To extend the existing static entangling-cost model into a time-dependent routing metric.
3. To formulate a temporal route-feasibility condition that determines whether all e-links on a selected route are expected to remain usable until route completion.

4. To design a coherence-budget-aware routing algorithm for Partial-Anchor and Full-Anchor entanglement overlays.
5. To develop an overlay-maintenance policy that determines when to retain, regenerate, replace, or avoid stale artificial e-links.
6. To define a temporal entangling-stretch metric for evaluating compact-routing performance under age-dependent fidelity degradation.
7. To analyze the impact of heterogeneous e-link ages, coherence times, fidelities, and regeneration delays on routing reliability.
8. To compare the proposed method against static compact routing, fidelity-only routing, age-only routing, fixed-threshold routing, and basic regeneration heuristics.
9. To evaluate whether temporal-state awareness improves end-to-end fidelity, delivery success probability, route reliability, and overlay robustness without significantly increasing routing-table overhead.

The overall goal is to determine whether temporal-state information can be integrated into quantum-native compact routing in a way that improves physical reliability while preserving sublinear memory scaling.

1.5 Research Questions

This research is guided by the following questions:

1. How can explicit temporal state be incorporated into quantum-native compact routing-table entries without violating their sublinear scalability?
2. How should the age, effective fidelity, coherence time, and expected completion delay of an artificial e-link be combined into a remaining coherence-budget metric?
3. How can a routing protocol determine whether a structurally valid path is also temporally feasible?
4. How does temporal degradation affect the route quality and entangling-stretch guarantees of Partial-Anchor and Full-Anchor schemes?
5. How should regeneration resources be prioritized among local e-neighbor links, anchor links, and tracked long-range links?

6. Can coherence-budget-aware routing reduce stale-link selection, route failure, and fidelity violation compared with static, fidelity-only, age-only, and fixed-threshold routing?
7. What trade-offs arise among end-to-end fidelity, delivery success probability, regeneration cost, routing-table overhead, and temporal entangling stretch?

1.6 Research Contributions

The expected contributions of this research are as follows:

1. A temporal-state extension of the quantum-native compact routing table that explicitly represents e-link age, effective fidelity, coherence time, remaining coherence budget, and regeneration cost.
2. A coherence-budget model that estimates whether an artificial e-link is expected to remain usable until the completion of an end-to-end entanglement route.
3. A time-dependent entangling-cost formulation that combines fidelity degradation, age, expiration risk, and regeneration delay.
4. A temporal route-feasibility criterion for rejecting structurally valid but temporally unreliable paths.
5. A coherence-budget-aware routing algorithm for Partial-Anchor and Full-Anchor entanglement overlays.
6. An overlay-maintenance policy for retaining, regenerating, replacing, or avoiding stale artificial e-links.
7. A temporal entangling-stretch formulation that evaluates compact-routing quality under heterogeneous age and decoherence conditions.
8. A comparative analysis of Partial-Anchor and Full-Anchor schemes under varying fidelity, coherence time, generation probability, and regeneration delay.
9. A simulation-based evaluation against static compact routing, fidelity-only routing, age-only routing, fixed-threshold routing, and basic regeneration heuristics.

These contributions are intended to connect temporal entanglement management with scalable quantum-native compact routing without claiming that age-awareness, fidelity-awareness, or regeneration are individually novel.

1.7 Thesis Organization

This thesis is organized into six chapters.

Chapter 1 introduces the research background, motivation, problem statement, objectives, research questions, and expected contributions. It explains why temporal-state awareness is necessary for scalable quantum-native compact routing.

Chapter 2 reviews the relevant literature on quantum networking, entanglement distribution, routing, noise, decoherence, fidelity degradation, temporal entanglement metrics, age-aware control, quantum-native addressing, artificial entanglement overlays, and compact routing. It also identifies the gap between temporal entanglement management and scalable compact-routing architectures.

Chapter 3 presents the system model and problem formulation. It defines the physical and artificial network topologies, e-link temporal state, fidelity-decay model, coherence budget, route-feasibility conditions, time-dependent entangling cost, and temporal entangling stretch.

Chapter 4 introduces the proposed temporal-state-aware compact-routing and overlay-maintenance algorithms. It describes the routing-table extension, coherence-budget-aware path selection, stale-link handling, regeneration prioritization, and the adaptations required for Partial-Anchor and Full-Anchor schemes.

Chapter 5 describes the simulation environment, experimental design, baseline methods, parameter settings, and evaluation metrics. It presents and analyzes the results in terms of fidelity, delivery success probability, route failure, regeneration cost, memory overhead, and temporal entangling stretch.

Chapter 6 concludes the thesis by summarizing the main findings, discussing the limitations of the proposed framework, and outlining directions for future research.

Chapter 2

Literature Review

2.1 Introduction

This chapter reviews the theoretical and technical foundations required for temporal-state-aware compact routing in quantum networks. The discussion begins with the basic principles of quantum communication, including qubits, entanglement, quantum memories, entanglement swapping, purification, and probabilistic link generation. These concepts are essential because routing in a quantum network differs fundamentally from routing in a classical packet-switched network.

The review then examines existing approaches to entanglement distribution and quantum network routing. Particular attention is given to fidelity-aware routing, decoherence-aware control, adaptive scheduling, and resource-constrained repeater networks. These research areas establish that route quality depends not only on connectivity, but also on the physical quality and temporal condition of the entangled resources used along a route.

Recent studies have introduced explicit temporal metrics for quantum networks, including pair age, Fidelity-Age, and Age of Entanglement [1, 2]. These metrics capture the fact that stored entanglement degrades over time and that delayed delivery may reduce application-level usefulness. However, most age-aware studies focus on scheduling, repeater chains, or local control policies rather than scalable quantum-native compact routing.

In parallel, quantum-native routing architectures have introduced artificial entanglement overlays, quantum addressing, compact routing tables, and bounded entangling stretch. The proposed architecture in [3] demonstrates how Partial-Anchor and Full-Anchor schemes can achieve sublinear routing-table growth while preserving constant stretch. Nevertheless, the temporal state of individual artificial e-links is not explicitly represented in the routing

table or in the stretch analysis.

Therefore, this literature review focuses on the intersection of two previously separate research directions: temporal entanglement management and scalable compact routing. The objective is to identify whether age, effective fidelity, remaining coherence budget, and regeneration delay can be incorporated into a quantum-native compact routing framework without destroying its scalability properties.

2.2 Fundamentals of Quantum Networks

Quantum networks extend classical communication infrastructures by enabling the generation, storage, distribution, and manipulation of quantum states across spatially separated nodes. Their operation relies on physical principles that have no direct classical counterpart, including quantum superposition, entanglement, measurement-induced state collapse, and the no-cloning theorem. These principles fundamentally constrain how information can be transmitted and how network resources must be managed. [4, 5] A qubit is the basic unit of quantum information and can be represented as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad (2.1)$$

where α and β are complex probability amplitudes satisfying

$$|\alpha|^2 + |\beta|^2 = 1. \quad (2.2)$$

Unlike a classical bit, which takes either the value 0 or 1, a qubit may exist in a coherent superposition of both basis states until measurement.

Quantum entanglement is a non-classical correlation shared between two or more quantum systems. A widely used example is the Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle). \quad (2.3)$$

The two qubits in this state cannot be described independently, even when they are stored at distant nodes. This property makes entanglement a fundamental communication resource for quantum teleportation, distributed quantum computing, quantum sensing, and quantum key distribution [4, 5]. Direct entanglement distribution over long distances is strongly limited by photon loss, channel noise, and imperfect hardware [6, 7] Quantum repeaters are therefore introduced to divide a long communication path into shorter elementary links. Entanglement is first generated between neighboring nodes and then extended over multiple links through entanglement swapping. If two

adjacent entangled pairs are available, an intermediate repeater can perform a Bell-state measurement to establish entanglement between the remote end nodes.

The quality of an entangled state is commonly measured using fidelity [6, 8]. For a quantum state described by density matrix ρ and a target pure state $|\psi_{\text{tar}}\rangle$, fidelity is defined as

$$F = \langle \psi_{\text{tar}} | \rho | \psi_{\text{tar}} \rangle. \quad (2.4)$$

A high fidelity value indicates that the generated state is close to the intended target state. In practical quantum networks, fidelity decreases due to channel noise, imperfect gates, measurement errors, and storage-induced decoherence.

Quantum memories temporarily store entangled qubits until other links or operations become available [9, 5]. However, stored states are not persistent resources. Their quality deteriorates over time because of decoherence, and an entangled pair may eventually become unusable if its fidelity falls below an application-specific threshold. Consequently, quantum-network control must account not only for whether an entangled pair exists, but also for its current physical and temporal condition.

Entanglement purification can improve fidelity by consuming multiple low-quality entangled pairs to probabilistically produce a smaller number of higher-quality pairs. This creates an important trade-off between fidelity improvement and resource consumption. Similarly, entanglement swapping extends communication distance but may reduce end-to-end fidelity and introduces additional probabilistic failure.

These characteristics make quantum networking fundamentally stateful. Routing and resource-allocation decisions must consider link-generation probability, memory occupancy, fidelity, decoherence, swapping success, purification cost, and the age of stored entanglement. This statefulness motivates the temporal-state-aware routing framework developed in this thesis.

2.3 Entanglement Distribution in Quantum Repeater Networks

Entanglement distribution is the process of establishing shared entangled states between spatially separated quantum nodes [6, 5]. In a practical quantum network, direct end-to-end entanglement generation becomes increasingly difficult as the communication distance grows because optical loss, im-

perfect photon transmission, detector inefficiency, and hardware noise reduce the probability of successful link establishment.

To overcome these limitations, a long communication path is divided into a sequence of shorter elementary links connected through quantum repeaters [6, 7]. Entanglement is first generated independently over each elementary link. After adjacent links have successfully established entangled pairs, intermediate repeater nodes perform entanglement swapping to extend the entanglement over a longer distance.

Let the quantum network be represented by an undirected graph [7, 10].

$$G = (V, E), \quad (2.5)$$

where V denotes the set of quantum nodes and E denotes the set of physical quantum links. For an elementary link $e \in E$, the outcome of an entanglement-generation attempt in time slot t may be modeled as

$$X_t(e) \sim \text{Bernoulli}(p_e), \quad (2.6)$$

where p_e is the probability of successfully generating an entangled pair over link e .

When multiple transmission modes are available, the probability that at least one entangled pair is successfully generated over a link can be expressed as

$$p_{\text{link}}(e) = 1 - (1 - p_e)^{S(e)}, \quad (2.7)$$

where $S(e)$ denotes the number of parallel generation attempts or transmission modes allocated to the link.

Consider an end-to-end path

$$\pi = (e_1, e_2, \dots, e_k), \quad (2.8)$$

consisting of k elementary links. If each elementary link is available and each intermediate Bell-state measurement succeeds independently with probability q , the approximate end-to-end entanglement-success probability is

$$P_{\text{e2e}}(\pi) = \left(\prod_{j=1}^k p_{\text{link}}(e_j) \right) q^{k-1}. \quad (2.9)$$

This expression illustrates why longer paths are generally less reliable. As the number of links increases, more elementary-link generations and swapping operations must succeed simultaneously.

Entanglement swapping is essential for extending the communication distance. Suppose node B shares an entangled pair with node A and another

entangled pair with node C . By performing a Bell-state measurement on its local qubits, node B can establish entanglement between A and C . The original elementary pairs are consumed during this operation, demonstrating that entanglement is a non-persistent network resource.

The quality of the resulting end-to-end entangled state depends on the fidelities of all elementary pairs and the errors introduced by swapping operations. For Werner-state approximations, a transformed parameter may be used to compose link fidelities along a path. If

$$p_{W,e} = \frac{4F_e - 1}{3}, \quad (2.10)$$

then the end-to-end Werner parameter for a path can be approximated by

$$p_{W,\text{end}} = \prod_{e \in \pi} p_{W,e}, \quad (2.11)$$

and the resulting end-to-end fidelity becomes

$$F_{\text{end}} = \frac{1 + 3p_{W,\text{end}}}{4}. \quad (2.12)$$

A delivered entangled state is considered usable only when

$$F_{\text{end}} \geq F_{\min}, \quad (2.13)$$

where $F_{\min}$ is an application-specific fidelity threshold.

Quantum memories allow successfully generated elementary pairs to be stored while the network waits for other links to become available. However, storage introduces a temporal dependency because the fidelity of stored entanglement decreases due to decoherence. Therefore, two links that are simultaneously available may have different ages and different effective fidelities.

Entanglement purification may be applied when the fidelity of available pairs is insufficient. Purification consumes two or more imperfect entangled pairs and probabilistically produces a smaller number of higher-fidelity pairs. Although purification improves quality, it also increases resource consumption, memory occupancy, and delivery delay.

Consequently, entanglement distribution involves several coupled decisions, including which links should attempt generation, which pairs should be stored or discarded, where purification should be performed, when swapping should occur, and which end-to-end path should be selected. These decisions directly influence throughput, fidelity, latency, resource consumption, and entanglement freshness.

In quantum-native overlay architectures, proactively established entanglement creates artificial links between nodes that may not be physically adjacent. These artificial links can reduce distribution delay and routing complexity, but they must be continuously monitored, replenished, and re-configured because they are consumed during use and degraded by decoherence [3]. This motivates the need for routing methods that consider both the structural role and temporal condition of each entangled link.

2.4 Quantum Network Routing

Quantum network routing determines how entanglement should be established between remote source and destination nodes through a sequence of physical or artificial links. Unlike classical routing, the objective is not merely to forward packets through a connected topology. Instead, the routing process must identify and coordinate a sequence of entanglement-generation, storage, purification, and swapping operations capable of producing an end-to-end entangled state of sufficient quality [7, 10, 8]. [7, 10, 8] A routing path between source node s and destination node d may be represented as

$$\pi_{s,d} = (v_0 = s, v_1, \dots, v_k = d), \quad (2.14)$$

where consecutive nodes are connected through physical or pre-established entanglement links.

In classical networks, routing cost is commonly based on hop count, delay, bandwidth, or packet-loss probability. In quantum networks, routing decisions must additionally consider quantum-specific resources [8, 11].

- elementary-link generation probability,
- quantum-memory availability,
- storage time and decoherence,
- link and end-to-end fidelity,
- entanglement-swapping reliability,
- purification requirements,
- resource consumption, and
- the age of stored entanglement.

A generic quantum-routing problem may be expressed as

$$\pi_{s,d}^* = \arg \min_{\pi \in \mathcal{P}_{s,d}} C(\pi), \quad (2.15)$$

where $\mathcal{P}_{s,d}$ is the set of feasible paths between s and d , and $C(\pi)$ is a quantum-specific route-cost function [7, 8].

Early quantum-routing approaches adapted classical shortest-path algorithms by assigning quantum-aware weights to physical links. A simple additive path cost can be written as

$$C(\pi) = \sum_{e \in \pi} w(e), \quad (2.16)$$

where $w(e)$ may reflect distance, generation delay, memory consumption, fidelity loss, or another link-level property. Although this formulation is computationally convenient, it may be insufficient when route quality depends on multiplicative success probabilities or non-additive fidelity composition.

For example, if link-generation and swapping events are independent, an end-to-end success-oriented route may be selected by maximizing

$$P_{\text{e2e}}(\pi) = \left(\prod_{e \in \pi} p_{\text{link}}(e) \right) \left(\prod_{r \in \pi} q_r \right), \quad (2.17)$$

where q_r denotes the swapping-success probability at repeater r . By applying a negative logarithm, this multiplicative objective can be transformed into an additive routing cost.

Fidelity-aware routing explicitly selects paths according to the quality of the expected end-to-end entangled state. A path is feasible only if

$$F_{\text{end}}(\pi) \geq F_{\text{min}}. \quad (2.18)$$

Among feasible paths, a routing protocol may maximize expected fidelity, minimize purification cost, maximize delivery rate, or balance multiple objectives. However, a route with high initial fidelity may become unsuitable if the required entangled pairs remain in memory for too long.

Reactive routing computes or updates a path after a demand arrives, whereas proactive routing creates and maintains entanglement resources before requests are received. Reactive approaches may reduce idle resource consumption but introduce setup delay. Proactive approaches can reduce service latency by maintaining pre-established entanglement, but they require continuous memory allocation, monitoring, and regeneration.

Adaptive routing responds to changes in link state, memory availability, noise, fidelity, and request load. Recent approaches formulate the routing

process as a Markov decision process, partially observable Markov decision process, or reinforcement-learning problem. In these models, the controller selects actions based on the current or estimated network state rather than relying on a fixed link metric.

Routing becomes especially challenging when multiple source–destination requests compete for the same stored entangled pairs. Since each pair can be consumed by at most one swapping or purification operation, path selection must be coordinated with scheduling and resource allocation. Therefore, routing and scheduling cannot always be treated as independent layers.

Quantum-native routing introduces a further conceptual shift by operating over an artificial topology formed by pre-established entanglement. In this setting, an artificial link may connect physically remote nodes and can be consumed through entanglement swapping. The routing table therefore stores information about currently available entangled resources rather than only physical next hops.

The quantum-native compact-routing architecture in [3] defines a general entangling-cost metric $w(i, j)$ and uses it to select e-neighbors, anchors, and artificial entanglement paths. The framework supports a wide class of possible metrics, including fidelity, memory usage, and decoherence-related cost. Nevertheless, the proposed routing-table structure does not explicitly track the temporal state of each stored e-link.

This limitation is important because the same artificial link may have different routing value at different times. A recently generated link with high effective fidelity and sufficient remaining coherence may be preferable to an older link with lower generation cost. Consequently, routing decisions should consider not only static structural connectivity but also time-dependent entanglement usability.

2.5 Noise, Decoherence, and Fidelity Degradation

Noise and decoherence are fundamental limitations in practical quantum networks. Unlike classical information, quantum states are highly sensitive to environmental interaction, imperfect control operations, photon loss, detector errors, and finite memory coherence. These effects reduce the quality and reliability of distributed entanglement and therefore directly influence routing and resource-allocation decisions [6, 5].

Quantum noise can arise from several physical mechanisms. Common models include depolarizing noise, dephasing, amplitude damping, gate er-

rors, and measurement errors [12, 9]. Although these models differ physically, their common consequence is a reduction in the fidelity of the stored or transmitted quantum state.

For an ideal target state $|\psi_{\text{tar}}\rangle$ and an actual density matrix ρ , fidelity is defined as

$$F = \langle \psi_{\text{tar}} | \rho | \psi_{\text{tar}} \rangle. \quad (2.19)$$

The value of F lies between zero and one, with larger values indicating that the actual state is closer to the target state. In entanglement-distribution systems, fidelity is commonly used as the main quality metric for determining whether an entangled pair is suitable for communication tasks [7, 8].

A practical application usually imposes a minimum acceptable fidelity threshold:

$$F \geq F_{\min}. \quad (2.20)$$

An entangled pair whose fidelity falls below $F_{\min}$ is treated as unusable, even if the pair still physically exists in quantum memory.

One important source of degradation is memory decoherence. When an entangled pair is stored, its quality decreases as a function of storage duration. Quantum memories therefore introduce a temporal dependency into entanglement management because stored resources gradually lose their usefulness over time [9, 1]. A simplified exponential decay model may be written as

$$F_{ij}(t) = F_{ij}^0 \exp\left(-\frac{A_{ij}(t)}{T_{2,ij}}\right), \quad (2.21)$$

where F_{ij}^0 is the initial fidelity of the pair shared by nodes i and j , $A_{ij}(t)$ is its age at time t , and $T_{2,ij}$ is the corresponding coherence time.

This model captures the basic principle that older entangled pairs generally have lower effective fidelity. More detailed physical models may include dephasing, amplitude damping, correlated noise, or hardware-specific memory behavior.

For Werner-state approximations, the fidelity may instead be represented through the Werner parameter

$$p_W = \frac{4F - 1}{3}. \quad (2.22)$$

Under storage-induced decoherence, this parameter may evolve according to

$$p_W(t + \Delta) = p_W(t) \exp \left(-\frac{\Delta}{T_2} \right). \quad (2.23)$$

The resulting fidelity can then be recovered using

$$F(t) = \frac{1 + 3p_W(t)}{4}. \quad (2.24)$$

Noise also accumulates during entanglement swapping. If multiple elementary pairs are combined along a path, the final end-to-end state depends on the fidelity of every stored pair and the reliability of each Bell-state measurement. Therefore, a path that is feasible at the beginning of a time slot may become infeasible after waiting, additional generation delay, or repeated swapping attempts [6, 7].

Purification can partially compensate for fidelity degradation. However, purification consumes multiple entangled pairs and succeeds only probabilistically. As a result, it introduces a trade-off between fidelity improvement, memory usage, throughput, and delay [6].

The temporal effect of decoherence is particularly important for proactive entanglement overlays. Artificial e-links may be generated in advance and stored until they are needed. During this waiting period, their effective fidelity decreases. Consequently, the routing value of an e-link is not static but changes with its temporal condition [3, 1].

A more realistic time-dependent link state can therefore be represented as

$$\mathcal{L}_{ij}(t) = [F_{ij}^0, A_{ij}(t), F_{ij}^{\text{eff}}(t), T_{2,ij}]. \quad (2.25)$$

The effective fidelity may be modeled as

$$F_{ij}^{\text{eff}}(t) = f \left(F_{ij}^0, A_{ij}(t), T_{2,ij}, \eta_{ij}, \epsilon_{ij} \right), \quad (2.26)$$

where η_{ij} and ϵ_{ij} represent channel and operational noise parameters.

Existing quantum-native compact-routing work allows the entangling-cost metric to represent fidelity degradation, memory use, or decoherence abstractly [3]. However, it does not provide an explicit temporal-state model or derive how age-dependent fidelity affects route feasibility and entangling stretch.

This gap motivates the development of a routing framework in which each artificial e-link is evaluated according to its current effective fidelity, age, and remaining coherence budget rather than by static connectivity alone [1, 2].

2.6 Temporal State of Entanglement

The temporal state of entanglement describes how the usefulness of a stored entangled resource changes over time. Unlike classical links, an entangled link is not a persistent resource. Its fidelity may decrease during storage, it may expire after a finite coherence interval, and it is consumed when used in swapping, purification, or application-level delivery.

For an entangled pair generated between nodes i and j at time t_{ij}^{gen} , its age at time t can be defined as

$$A_{ij}(t) = t - t_{ij}^{\text{gen}}. \quad (2.27)$$

A newly generated pair has a small age, whereas a pair that has remained in quantum memory for several slots has a larger age. Since decoherence accumulates during storage, age provides an indirect indicator of the physical quality of the pair [1, 2].

However, age and fidelity are not identical. Two pairs with the same age may have different fidelities because of differences in initial fidelity, coherence time, channel noise, or hardware quality. Therefore, a temporal-state model should combine age with physical parameters rather than using age alone.

A basic temporal link state may be expressed as

$$\mathcal{L}_{ij}(t) = [A_{ij}(t), F_{ij}^0, F_{ij}^{\text{eff}}(t), T_{2,ij}], \quad (2.28)$$

where F_{ij}^0 is the initial fidelity, $F_{ij}^{\text{eff}}(t)$ is the current effective fidelity, and $T_{2,ij}$ is the memory coherence time.

A stored pair remains usable only while its effective fidelity satisfies

$$F_{ij}^{\text{eff}}(t) \geq F_{\min}. \quad (2.29)$$

This condition induces a finite usability interval. Once the pair is expected to fall below the required threshold, it should be discarded, purified, regenerated, or excluded from candidate routes [7, 8].

A useful quantity for routing is the remaining coherence budget. It may be approximated as

$$B_{ij}(t) = T_{2,ij} - A_{ij}(t) - \widehat{D}_{ij}, \quad (2.30)$$

where $\widehat{D}_{ij}$ denotes the expected additional delay required before the entangled link is consumed in an end-to-end operation.

The value $B_{ij}(t)$ estimates whether the stored resource is likely to remain usable until route completion. A negative or very small coherence budget

indicates a high risk of expiration before the required swapping or delivery operations are completed. Such temporal constraints motivate routing strategies that jointly consider resource age, fidelity, and future operation delay [1].

Temporal freshness has also been studied at the flow level. Fidelity-Age measures the time elapsed since the most recent delivery of an entangled pair whose fidelity exceeds a prescribed threshold. For a source–destination request (s, d) , it can be represented as

$$A_{s,d}(t) = t - \max \left\{ t_n^{(s,d)} \leq t \right\}, \quad (2.31)$$

where $t_n^{(s,d)}$ denotes a successful usable-delivery epoch.

This flow-level metric differs from pair age. Pair age describes the current storage duration of an individual entangled resource, whereas Fidelity-Age describes how long a flow has waited since its last usable delivery [1].

Another related concept is the Age of Entanglement of an end-to-end link. In repeater-chain models, the age may reset after successful swapping according to the accumulated ages of the elementary links used to create the end-to-end state. This captures the fact that a newly established end-to-end pair may already contain storage-induced degradation inherited from its component links [2]. These temporal quantities can therefore be classified into three levels:

- **pair age**, describing the age of an individual stored Bell pair;
- **flow-level age**, describing the time since the last usable delivery for a request; and
- **end-to-end entanglement age**, describing the freshness of the most recently generated remote entangled state.

These different temporal metrics provide complementary views of entanglement freshness and have recently been considered for adaptive scheduling and resource management in quantum networks [1, 2].

For compact routing over artificial entanglement overlays, pair-level or e-link-level temporal state is particularly important. The routing protocol must determine whether each stored artificial link will remain usable throughout the expected route-completion interval.

A temporally feasible route π may therefore be required to satisfy

$$B_{ij}(t) > 0, \quad \forall (i, j) \in \pi. \quad (2.32)$$

Alternatively, feasibility may be expressed as

$$F_{ij}^{\text{eff}}(t + \widehat{D}_\pi) \geq F_{\min}, \quad \forall (i, j) \in \pi, \quad (2.33)$$

where $\widehat{D}_\pi$ is the predicted completion delay of the selected route.

This perspective changes routing from a static connectivity problem into a time-dependent resource-selection problem. A route may be topologically available but temporally infeasible because one or more e-links are too old or too close to expiration. Therefore, future route quality depends not only on current link availability but also on the expected evolution of quantum resources during execution [7, 8].

The quantum-native compact-routing architecture in [3] recognizes the stateful and ephemeral nature of entanglement and discusses continuous overlay regeneration. Nevertheless, its routing-table entries do not explicitly include age, current fidelity, generation time, or remaining coherence budget.

The proposed thesis addresses this limitation by extending compact quantum-routing entries with explicit temporal state and by studying how stale or degrading artificial links affect route feasibility, fidelity, and entangling stretch [1, 2].

2.7 Age-Aware Scheduling and Control

Age-aware scheduling has recently emerged as an important research direction in quantum networking because entangled resources are both probabilistic and time-sensitive. A scheduler must decide which flow, link, or swapping operation should receive limited network resources while accounting for fidelity degradation, memory constraints, and service regularity [1, 2].

One recent approach introduces the Fidelity-Age metric, which measures the time elapsed since the most recent successful delivery of an entangled pair whose fidelity satisfies a prescribed threshold [1]. For a source–destination request (s, d) , the Fidelity-Age process can be written as

$$A_{s,d}(t) = t - \max \{ t_n^{(s,d)} \leq t \}, \quad (2.34)$$

where $t_n^{(s,d)}$ denotes the time of the most recent usable entanglement delivery.

The age increases when no usable delivery occurs and resets when a newly delivered end-to-end pair satisfies

$$F_{\text{end}} \geq F_{\min}. \quad (2.35)$$

This formulation jointly captures delivery regularity and fidelity-conditioned usability. A flow may have high throughput on average but still experience

long periods without a usable entangled pair. Therefore, mean throughput alone is insufficient for evaluating temporal service quality in quantum networks [1].

The corresponding long-run average Fidelity-Age can be expressed as

$$\bar{A}_{s,d} = \limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T A_{s,d}(t). \quad (2.36)$$

The system objective is then to minimize a weighted sum of average ages:

$$\min_{\phi \in \Phi} \sum_{(s,d) \in \mathcal{P}} w_{s,d} \bar{A}_{s,d}, \quad (2.37)$$

subject to attempt-budget, memory, link-exclusivity, swapping, purification, and fidelity constraints.

The Fidelity-Age study proposes lightweight scheduling methods, including threshold-based and index-based policies. A threshold policy prioritizes flows whose age exceeds a predefined value, whereas an index policy ranks flows using a combination of instantaneous utility and current age [1].

These scheduling concepts are closely related to adaptive routing because routing decisions should consider not only the current availability of entangled resources but also their expected future quality and freshness. Therefore, age-aware scheduling provides an important foundation for routing frameworks that explicitly model temporal link states [8, 1]. A generic age-aware scheduling score may be represented as

$$I_{s,d}(t) = \frac{U_{s,d}(t)}{1 + \beta A_{s,d}(t)}, \quad (2.38)$$

where $U_{s,d}(t)$ represents an instantaneous utility, such as expected success probability or fidelity, and β controls the influence of age.

The simulations reported in [1] show that age-aware scheduling can substantially reduce extreme-age events, starvation, and unfairness while preserving throughput. However, the study focuses primarily on flow scheduling over a repeater grid rather than on compact routing over a quantum-native artificial overlay.

A related approach introduces the Age of Entanglement for satellite-assisted repeater chains with intermittent availability [2]. In this model, the network state includes satellite visibility, elementary-link ages, and the end-to-end entanglement age:

$$s_t = (v_{12}^t, v_{23}^t, m_{12}^t, m_{23}^t, \Delta_t^e). \quad (2.39)$$

The action space includes elementary-link generation, waiting, and entanglement swapping:

$$a_t = (a_{12}^t, a_{23}^t, a_{\text{sw}}^t). \quad (2.40)$$

A successful swapping operation resets the end-to-end Age of Entanglement according to the ages of the two elementary links:

$$\Delta_{t+1}^e = m_{12}^t + m_{23}^t. \quad (2.41)$$

If swapping does not succeed, the age increases:

$$\Delta_{t+1}^e = \Delta_t^e + 1. \quad (2.42)$$

The problem is formulated as an average-cost Markov decision process, where the controller jointly decides whether to generate, wait, discard, regenerate, or swap entanglement. Relative Value Iteration is then used to obtain an optimal stationary policy [2].

This work is important because it demonstrates that generation, storage, discard, regeneration, and swapping decisions can be optimized jointly under temporal degradation. Therefore, regeneration or age-aware control alone cannot be considered a sufficient research contribution.

Nevertheless, the satellite-based Age of Entanglement model is limited to a three-node linear repeater chain. It does not consider general graph routing, multiple competing routes, quantum addressing, compact routing tables, artificial ESP overlays, Partial-Anchor or Full-Anchor structures, or entangling-stretch guarantees.

The distinction between scheduling and routing is therefore essential. Age-aware scheduling determines which request or operation should receive service, while compact routing determines which sequence of artificial e-links should be selected to establish end-to-end entanglement.

Existing age-aware studies provide strong temporal-control mechanisms, but they do not examine how stale e-links affect the structural guarantees of a compact routing architecture. In particular, they do not analyze whether aging may invalidate an anchor link, increase effective entangling stretch, reduce overlay reachability, or require proactive replacement of specific artificial links.

This motivates a joint perspective in which temporal state influences both route selection and overlay maintenance. The proposed thesis therefore extends existing age-aware control by embedding temporal e-link information into a scalable quantum-native compact-routing framework.

2.8 Quantum-Native Addressing and Entanglement Overlay Networks

Quantum-native network architecture departs from the classical assumption that network nodes only forward information over persistent physical links. In the Quantum Internet, entanglement itself acts as the primary communication resource, and network nodes must generate, store, monitor, manipulate, and replenish entangled states [3, 5].

The architecture proposed in [3] introduces a two-tier hierarchy. The upper tier is composed of Entanglement Service Providers, referred to as ESPs, while the lower tier contains quantum edge nodes that consume entanglement resources for applications such as teleportation, distributed quantum computing, sensing, and quantum key distribution.

The ESPs form an entangled backbone by proactively generating and maintaining entanglement with selected remote ESPs. This creates an artificial topology over the underlying physical network.

Let the physical quantum network be represented as

$$G = (V, E), \quad (2.43)$$

where V is the set of network nodes and E is the set of physical quantum links.

The artificial entanglement overlay may be represented as

$$G_E(t) = (V_E, E_E(t)), \quad (2.44)$$

where V_E denotes the set of ESPs and $E_E(t)$ denotes the set of artificial entanglement links available at time t .

Unlike the physical topology, the artificial topology is dynamic. An artificial e-link may be created through elementary-link generation and entanglement swapping, consumed by an application, degraded by decoherence, or removed after failure. Therefore, the state of an artificial overlay depends not only on connectivity but also on the temporal condition of the underlying entanglement resources [3, 1].

The architecture includes an Entanglement-Defined Controller, or EDC, which performs three main functions:

- monitoring the availability and fidelity of entangled resources;
- reconfiguring the artificial topology; and
- enforcing routing, recovery, and resource-allocation policies.

The EDC is conceptually similar to a software-defined networking controller, but its role is broader because it must manage non-persistent and stateful quantum resources.

The proposed architecture also separates the control plane from the data plane. The control plane coordinates routing and entanglement-resource management, whereas the data plane performs entanglement generation, purification, swapping, forwarding, and overlay repair.

A key feature of the architecture is continuous overlay maintenance. Artificial links are consumed during end-to-end entanglement establishment and must therefore be regenerated. This means that overlay refresh, link regeneration, and resource replenishment are already recognized as necessary architectural functions [3].

Consequently, overlay maintenance or regeneration alone cannot be treated as a novel research contribution. The research opportunity lies in determining how maintenance decisions should be made under explicit temporal-state information, including link age, effective fidelity, and remaining coherence time.

The same architecture introduces a quantum addressing scheme in which each node is assigned a quantum address represented by a computational-basis state. For a network containing n nodes, the address length is

$$N = \lceil \log_2 n \rceil. \quad (2.45)$$

A quantum address can be represented as

$$|v_i\rangle \in \{|x\rangle \mid x \in \{0, 1\}^N\}. \quad (2.46)$$

Unlike a classical address, a quantum address may also appear in a superposition representing a set of nodes. For example,

$$|A\rangle = \frac{1}{\sqrt{M}} \sum_{i=1}^M |v_i\rangle \quad (2.47)$$

may represent a set of M quantum nodes.

This allows routing information about multiple ESPs to be encoded compactly inside a quantum routing table [3].

The quantum routing table does not store only physical next hops. Instead, it contains information about locally stored e-bits, the quantum identities of the remote ESPs sharing those e-bits, and superposed addresses representing e-neighborhood information.

A conceptual routing-table entry can be written as

$$\mathcal{R}_{ij} = [\text{e-bits}, |v_j\rangle, |\mathcal{N}(v_j)\rangle, \text{anchor status}]. \quad (2.48)$$

This structure enables routing decisions over the artificial entanglement topology rather than only over the physical graph [3].

However, the routing-table design in [3] does not explicitly include

- entanglement age,
- generation time,
- current effective fidelity,
- remaining coherence time,
- expiration probability, or
- regeneration urgency.

This omission is significant because two e-links with the same structural role may have very different temporal usability due to different storage durations, coherence properties, and degradation rates [1, 2].

A temporally extended routing-table entry may therefore be represented as

$$\mathcal{R}_{ij}(t) = [\text{e-bits}, |v_j\rangle, |\mathcal{N}(v_j)\rangle, A_{ij}(t), F_{ij}^{\text{eff}}(t), B_{ij}(t)], \quad (2.49)$$

where $A_{ij}(t)$ is the e-link age, $F_{ij}^{\text{eff}}(t)$ is the current effective fidelity, and $B_{ij}(t)$ is the remaining coherence budget.

The integration of this temporal information creates a direct connection between quantum-native addressing, compact routing, and age-dependent entanglement management. This connection forms the central research direction of the present thesis.

2.9 Compact Quantum Routing

Compact routing aims to reduce routing-table size while preserving bounded route quality. In a large-scale quantum network, maintaining entanglement between every pair of ESPs would require excessive quantum memory, control signaling, and continuous regeneration. Therefore, a scalable routing protocol must operate over a sparse artificial entanglement overlay [3].

The quantum-native routing architecture in [3] introduces compact routing based on two design objectives:

- sublinear routing-table size; and

- constant entangling stretch.

For a network with n_e Entanglement Service Providers, each ESP maintains approximately

$$\tilde{O}(\sqrt{n_e}) \quad (2.50)$$

routing-table entries, where the soft- O notation suppresses logarithmic factors.

The routing framework defines a general link-entanglement cost

$$w(i, j), \quad (2.51)$$

which represents the cost of establishing entanglement between ESPs v_i and v_j [3].

The cost metric is assumed to support an order relation and a composition operator $\oplus$. For a multi-hop artificial route

$$R = (v_i, v_{r_1}, \dots, v_{r_k}, v_j), \quad (2.52)$$

the accumulated entangling cost is represented as

$$w_R(i, j) = w(i, r_1) \oplus w(r_1, r_2) \oplus \dots \oplus w(r_k, j). \quad (2.53)$$

The optimal entangling cost between two ESPs is

$$w^*(i, j) = \min_{R \in \mathcal{R}_{i,j}} w_R(i, j), \quad (2.54)$$

where $\mathcal{R}_{i,j}$ is the set of feasible entanglement routes between v_i and v_j .

The entangling stretch of a routing protocol is defined as

$$ES(i, j) = \frac{w_{R_{QR}}(i, j)}{w^*(i, j)}, \quad (2.55)$$

where R_{QR} is the route selected by the quantum-routing protocol.

The worst-case entangling stretch is

$$ES = \max_{v_i, v_j \in V_2} ES(i, j). \quad (2.56)$$

A routing protocol is optimal when

$$ES = 1. \quad (2.57)$$

A value greater than one indicates that the protocol may choose a route with higher entangling cost than the globally optimal route.

The compact-routing architecture proposes two variants: the Partial-Anchor Scheme and the Full-Anchor Scheme [3].

2.9.1 Partial-Anchor Scheme

In the Partial-Anchor Scheme, each ESP proactively maintains low-cost artificial links with the nodes in its e-neighborhood. A smaller subset of ESPs, called anchor nodes, maintains additional long-range artificial links [3].

The e-neighborhood of ESP v_i is denoted by

$$\mathcal{N}(v_i), \quad (2.58)$$

and contains the ESPs with the lowest entangling cost relative to v_i .

The number of e-neighbors is selected approximately as

$$k = (1 + m)\sqrt{n_e} \log n_e, \quad (2.59)$$

where $m > 1$ is a design parameter.

The anchor set is denoted by

$$\mathcal{T} \subseteq V_2, \quad (2.60)$$

with cardinality approximately

$$|\mathcal{T}| = \sqrt{n_e}. \quad (2.61)$$

Each anchor proactively establishes and maintains entanglement with the other anchors. This creates a sparse long-range backbone over the local e-neighborhood structure [3].

Under this design, any ESP can reach another ESP through at most three artificial entanglement hops with high probability.

For additive entangling-cost metrics, the Partial-Anchor Scheme guarantees

$$ES \leq 5. \quad (2.62)$$

The scheme provides low memory overhead but may produce larger routing cost because only a subset of ESPs maintains long-range artificial links.

However, this architecture assumes that the entanglement resources forming the artificial overlay remain sufficiently available and does not explicitly model the temporal evolution of individual e-links, including aging, fidelity degradation, and remaining coherence budget.

2.9.2 Full-Anchor Scheme

In the Full-Anchor Scheme, every ESP maintains both local e-neighborhood links and selected long-range artificial links [3].

The ESP set is divided into tracked subsets, and each ESP proactively maintains entanglement with the nodes belonging to one assigned tracked set.

This design increases artificial-overlay connectivity while maintaining sub-linear routing-table size.

Under the Full-Anchor Scheme, any ESP can reach another ESP through at most two artificial entanglement hops with high probability.

For additive entangling-cost metrics, the Full-Anchor Scheme guarantees

$$ES \leq 3. \quad (2.63)$$

Therefore, the Full-Anchor Scheme provides lower worst-case entangling stretch than the Partial-Anchor Scheme, but it requires more ESPs to maintain long-range entanglement resources.

The main trade-off can be summarized as

$$\text{lower overlay-maintenance complexity} \longleftrightarrow \text{lower entangling stretch.} \quad (2.64)$$

The existing stretch guarantees are primarily structural. They assume that the artificial links required by the routing scheme are available and usable when selected [3].

However, an artificial link may become stale, lose fidelity, expire, or require regeneration before it can be used. Consequently, the static stretch bound may not accurately reflect the actual routing cost under temporal degradation. This limitation motivates the integration of temporal resource information into compact quantum routing.

To address this issue, a time-dependent entangling cost can be introduced:

$$w_t(i, j) = \alpha C_F(i, j, t) + \beta C_A(i, j, t) + \gamma C_B(i, j, t) + \delta C_R(i, j, t), \quad (2.65)$$

where

- $C_F(i, j, t)$ represents fidelity-related cost;
- $C_A(i, j, t)$ represents entanglement-age cost;
- $C_B(i, j, t)$ represents remaining-coherence-budget risk; and
- $C_R(i, j, t)$ represents regeneration cost or delay.

The corresponding temporal entangling stretch may be defined as

$$ES_t(i, j) = \frac{w_t(R_{QR}(i, j))}{w_t(R_t^*(i, j))}, \quad (2.66)$$

where $R_t^*(i, j)$ is the optimal temporally feasible route at time t .

This extension allows compact-routing performance to be evaluated when artificial links have heterogeneous ages, coherence times, fidelities, and regeneration delays.

The central research question is whether temporal-state awareness can improve fidelity and route reliability while preserving the original compact-routing property

$$\tilde{O}(\sqrt{n_e}). \quad (2.67)$$

2.10 Comparative Analysis of Closest Prior Work

The closest prior studies can be grouped into four research lines: quantum-native compact routing, Fidelity-Age scheduling, Age of Entanglement control, and adaptive belief-state routing. These studies address different aspects of temporal quantum-resource management, but none of them fully integrates explicit e-link temporal state with scalable compact routing over an artificial entanglement overlay.

Table 2.1: Comparison of Closest Prior Work

Research Line	Age	Fidelity	Decoherence	Routing	Compact Overlay	Stretch
Quantum-native compact routing	No explicit age	Abstractly supported	Abstractly supported	Yes	Yes	Yes
Fidelity-Age scheduling	Flow-level	Yes	Yes	Scheduling-oriented	No	No
Satellite Age of Entanglement	Pair and end-to-end age	Age-based quality	Yes	Chain control	No	No
Belief-state adaptive routing	Pair-level state	Yes	Yes	Yes	No	No
Proposed thesis	Explicit e-link age	Effective fidelity	Yes	Yes	Yes	Temporal stretch

The quantum-native architecture in [3] provides artificial entanglement overlays, quantum addressing, compact routing tables, Partial-Anchor and Full-Anchor schemes, and bounded entangling stretch. However, it does not explicitly represent e-link age, generation time, remaining coherence budget, or time-varying fidelity inside the routing table.

The Fidelity-Age framework in [1] explicitly combines delivery freshness with a minimum fidelity threshold. It also considers finite memory, probabilistic generation, purification, swapping, and scheduling fairness. Nev-

ertheless, its primary focus is flow scheduling rather than compact route construction over a quantum-native overlay.

The satellite-assisted Age of Entanglement model in [2] jointly controls generation, waiting, swapping, discard, and regeneration under intermittent visibility and memory degradation. However, it is limited to a three-node repeater chain and does not study general graph routing, artificial overlays, compact routing tables, or entangling stretch.

Existing adaptive routing approaches also investigate dynamic quantum-resource states and uncertainty-aware decision making, but they do not combine temporal e-link modeling with scalable compact overlay routing.

The comparison shows that age-awareness, fidelity-awareness, decoherence modeling, regeneration, and adaptive control already exist in the literature. Therefore, none of these components can be claimed as individually novel.

The remaining research opportunity lies in their integration with scalable quantum-native compact routing. More specifically, existing work does not sufficiently characterize how stale artificial e-links affect route feasibility, overlay reachability, anchor reliability, memory overhead, and entangling stretch.

This motivates a temporal-state-aware routing framework that extends compact quantum routing by incorporating dynamic e-link age, effective fidelity, and coherence-aware route selection.

2.11 Limitations of Existing Literature

Although recent research has significantly advanced quantum routing, entanglement scheduling, fidelity management, and age-aware control, several limitations remain at the intersection of temporal entanglement management and scalable compact routing.

2.11.1 Absence of Explicit Temporal State in Compact Routing Tables

Existing quantum-native compact-routing tables primarily store entangled resources, e-hop identities, e-neighborhood information, and anchor-related metadata [3]. However, they do not explicitly include the generation time, age, current effective fidelity, remaining coherence budget, or expiration risk of each stored e-link.

This omission prevents the routing protocol from distinguishing between

two structurally equivalent artificial links that have different temporal usability.

2.11.2 Static Entangling-Stretch Guarantees

The Partial-Anchor and Full-Anchor schemes provide constant worst-case entangling-stretch bounds. For additive metrics, these bounds are

$$ES \leq 5 \quad (2.68)$$

for the Partial-Anchor Scheme and

$$ES \leq 3 \quad (2.69)$$

for the Full-Anchor Scheme.

However, these guarantees are derived under the assumption that the required artificial links are available and usable when selected [3]. They do not characterize the effect of stale links, fidelity decay, regeneration delay, or temporal route infeasibility.

2.11.3 Limited Architectural Scope of Age-Aware Studies

Recent age-aware studies explicitly model pair age, Fidelity-Age, end-to-end Age of Entanglement, discard, regeneration, and adaptive control [1, 2]. Nevertheless, these works mainly focus on scheduling, small repeater chains, or flow-level control.

They do not investigate quantum addressing, compact routing tables, artificial ESP overlays, anchor structures, or entangling-stretch guarantees.

2.11.4 Regeneration Without Compact-Overlay-Aware Prioritization

Existing work already recognizes that depleted or stale entangled resources must be discarded or regenerated [1, 2]. Therefore, regeneration alone is not a new contribution.

The unresolved issue is how regeneration resources should be prioritized across a compact artificial overlay. Regenerating one anchor link may restore global reachability, while regenerating a local e-neighbor link may improve only a small part of the network. Existing studies do not sufficiently quantify this structural difference.

2.11.5 No Joint Analysis of Temporal and Scalability Trade-offs

Current research does not jointly characterize the relationship among

$$\begin{aligned} &\text{fidelity, age, coherence budget,} \\ &\text{regeneration cost, entangling stretch,} \\ &\text{routing-table overhead.} \end{aligned} \tag{2.70}$$

This trade-off is central to large-scale quantum-native routing because improved temporal freshness may require additional memory, control updates, or regeneration attempts.

2.11.6 Lack of Temporal Route-Feasibility Analysis

A route may be structurally available but temporally infeasible. For example, one or more artificial links may be expected to decohere below the required fidelity threshold before the end-to-end swapping sequence is completed.

Existing compact-routing schemes do not explicitly define a route-feasibility condition such as

$$F_{ij}^{\text{eff}}(t + \widehat{D}_\pi) \geq F_{\min}, \quad \forall (i, j) \in \pi. \tag{2.71}$$

Without this condition, a selected route may fail despite being topologically valid.

Therefore, a key open problem is how to integrate temporal-state information into compact quantum routing while preserving scalability, bounded routing-table complexity, and low entangling stretch.

2.11.7 Insufficient Evaluation Under Heterogeneous Temporal Conditions

Artificial e-links may differ in initial fidelity, coherence time, age, generation probability, and regeneration delay. Existing compact-routing analysis largely abstracts these differences through a general cost metric [3].

A detailed evaluation is still required to determine how Partial-Anchor and Full-Anchor schemes behave when e-links have heterogeneous temporal states, including different ages, fidelities, and remaining coherence times.

These limitations indicate that the main unresolved problem is not the absence of age-awareness or regeneration individually. Rather, it is the absence of an explicit temporal-state framework integrated with scalable quantum-native compact routing and its analytical guarantees.

2.12 Identified Research Gap

Existing research has independently advanced fidelity-aware routing, decoherence-aware scheduling, explicit entanglement-age metrics, adaptive generation and swapping control, and quantum-native compact routing. Recent studies already model pair age, flow-level Fidelity-Age, end-to-end Age of Entanglement, discard decisions, regeneration, and adaptive control [1, 2]. Therefore, age-awareness, fidelity-awareness, regeneration, or adaptive routing alone cannot be claimed as novel contributions.

At the same time, quantum-native compact-routing architectures provide scalable routing tables, artificial entanglement overlays, quantum addressing, Partial-Anchor and Full-Anchor schemes, and bounded entangling stretch [3]. However, their routing tables do not explicitly represent the temporal state of each stored artificial e-link.

In particular, the following e-link attributes are not directly incorporated into the compact routing-table structure:

- generation time;
- current age;
- current effective fidelity;
- remaining coherence budget;
- expiration probability;
- predicted route-completion risk; and
- regeneration urgency.

Although the general entangling-cost metric can abstractly represent fidelity, memory use, or decoherence, the existing framework does not define an explicit time-dependent cost model.

Consequently, the effect of temporal degradation on compact-routing guarantees remains insufficiently characterized. A route may satisfy the structural conditions of the Partial-Anchor or Full-Anchor scheme but fail because one or more required e-links become unusable before completion.

This leads to the central research gap:

Existing quantum-network studies do not sufficiently integrate explicit per-e-link temporal state with scalable quantum-native

compact routing. In particular, the impact of age-dependent fidelity degradation, remaining coherence budget, and regeneration delay on route feasibility, overlay reliability, and entangling stretch remains insufficiently studied.

The gap lies at the intersection of two established research directions:

temporal entanglement management + quantum-native compact routing. (2.72)

Addressing this gap requires a framework that extends compact routing-table entries with temporal state while preserving sublinear routing-table scaling and bounded routing overhead.

A time-dependent routing entry may be defined as

$$\mathcal{R}_{ij}(t) = [|v_j\rangle, A_{ij}(t), F_{ij}^{\text{eff}}(t), T_{2,ij}, B_{ij}(t), C_{ij}^{\text{regen}}]. \quad (2.73)$$

The remaining coherence budget may be represented as

$$B_{ij}(t) = T_{2,ij} - A_{ij}(t) - \widehat{D}_{ij}, \quad (2.74)$$

where $\widehat{D}_{ij}$ is the expected additional delay before the e-link is consumed.

This temporal state can then support route rejection, rerouting, selective regeneration, anchor-link replacement, and overlay repair.

The research gap is therefore not the absence of temporal concepts in quantum networking. Rather, it is the lack of a rigorous integration of temporal e-link state into compact overlay routing and stretch analysis.

2.13 Proposed Research Direction

Based on the identified gap, this thesis proposes a temporal-state-aware extension of quantum-native compact routing for artificial entanglement overlay networks. The main objective is to incorporate explicit e-link temporal information into routing and maintenance decisions while preserving the scalability advantages of compact routing.

The proposed research title is

Temporal-State-Aware Quantum Routing for Fidelity-Preserving Entanglement Distribution in Quantum Networks

The proposed framework extends each artificial e-link with a temporal state defined as

$$\mathcal{L}_{ij}(t) = [A_{ij}(t), F_{ij}^0, F_{ij}^{\text{eff}}(t), T_{2,ij}, B_{ij}(t), C_{ij}^{\text{regen}}], \quad (2.75)$$

where

- $A_{ij}(t)$ is the age of the e-link;
- F_{ij}^0 is the initial fidelity;
- $F_{ij}^{\text{eff}}(t)$ is the current effective fidelity;
- $T_{2,ij}$ is the coherence time;
- $B_{ij}(t)$ is the remaining coherence budget; and
- C_{ij}^{regen} is the regeneration cost or delay.

The remaining coherence budget is defined as

$$B_{ij}(t) = T_{2,ij} - A_{ij}(t) - \widehat{D}_{ij}, \quad (2.76)$$

where $\widehat{D}_{ij}$ denotes the predicted additional delay before the link is consumed.

A route is considered temporally feasible only if every e-link remains usable until route completion. This can be expressed as

$$F_{ij}^{\text{eff}}(t + \widehat{D}_{\pi}) \geq F_{\min}, \quad \forall (i, j) \in \pi. \quad (2.77)$$

The static entangling cost is extended into a time-dependent cost:

$$w_t(i, j) = \alpha C_F(i, j, t) + \beta C_A(i, j, t) + \gamma C_B(i, j, t) + \delta C_R(i, j, t), \quad (2.78)$$

where the terms represent fidelity cost, age cost, coherence-budget risk, and regeneration cost, respectively.

The corresponding temporal entangling stretch is defined as

$$ES_t(i, j) = \frac{w_t(R_{\text{QR}}(i, j))}{w_t(R_t^*(i, j))}. \quad (2.79)$$

The proposed framework supports the following temporal resource-management actions:

- retain a usable e-link;
- reject a temporally infeasible route;

- reroute through fresher links;
- regenerate a stale e-link;
- replace an unreliable anchor link; and
- repair the artificial overlay.

The main research questions are:

1. How can explicit e-link temporal state be incorporated into a quantum-native compact routing table while preserving sublinear memory scaling?
2. How do age, fidelity degradation, coherence budget, and regeneration delay affect route feasibility in Partial-Anchor and Full-Anchor overlays?
3. How does temporal degradation affect entangling stretch and end-to-end reliability?
4. Can coherence-budget-aware routing outperform static compact routing, fidelity-only routing, age-only routing, and fixed-threshold approaches?

The expected contributions of this thesis are:

1. a temporal-state extension of the compact quantum routing table;
2. a coherence-budget model for artificial e-links;
3. a time-dependent entangling-cost formulation;
4. a temporal route-feasibility criterion;
5. a coherence-budget-aware routing and overlay-maintenance algorithm;
6. a temporal entangling-stretch formulation; and
7. a comparative evaluation of Partial-Anchor and Full-Anchor schemes under heterogeneous temporal conditions.

The evaluation will compare the proposed method against static compact routing, fidelity-only routing, age-only routing, fixed-cutoff routing, and regeneration heuristics. Performance will be measured using end-to-end fidelity, delivery success probability, throughput, route failure rate, memory overhead, regeneration cost, and temporal entangling stretch.

This research direction preserves the key advantage of quantum-native compact routing, namely scalable overlay-based communication, while addressing its limitation in representing time-dependent entanglement-resource states.

Chapter 3

System Model and Problem Formulation

3.1 Introduction

This chapter presents the system model and formal problem formulation for temporal-state-aware compact routing in quantum-native entanglement overlay networks. The objective is to extend the existing quantum-native compact-routing architecture by incorporating explicit temporal information associated with each artificial entanglement link (e-link).

The proposed model distinguishes between the underlying physical quantum network and the artificial entanglement overlay. The physical network represents the actual quantum communication infrastructure, including quantum nodes, repeaters, quantum memories, and physical quantum channels. In contrast, the artificial overlay represents proactively established entangled connections among Entanglement Service Providers (ESPs), enabling routing over an abstract entanglement topology rather than only over physical connectivity.

Unlike classical network links, artificial e-links are dynamic and time-dependent resources. Their availability and quality are affected by probabilistic entanglement generation, finite quantum-memory coherence, decoherence-induced fidelity degradation, entanglement consumption during swapping operations, and regeneration processes. Therefore, the routing cost of an artificial e-link cannot be considered static and must be evaluated according to its current temporal state.

To capture this dynamic behavior, each artificial e-link is associated with a temporal state containing its generation time, age, initial fidelity, effective fidelity, coherence time, remaining coherence budget, and regeneration-

related cost. These temporal attributes enable the routing protocol to determine whether a structurally available route can remain physically usable throughout the expected end-to-end entanglement establishment process.

This chapter further defines the time-dependent entangling-cost model and introduces the concept of temporal entangling stretch. These extensions provide a framework for evaluating compact-routing schemes, including the Partial-Anchor and Full-Anchor architectures, under heterogeneous conditions involving different link ages, fidelity levels, coherence characteristics, and regeneration requirements.

Finally, the routing problem is formulated as a temporal resource-selection problem. The objective is to select a temporally feasible path that minimizes the time-dependent entangling cost while satisfying fidelity requirements, coherence-budget constraints, memory limitations, and compact-routing scalability conditions.

3.2 Network Architecture

The proposed system follows a two-tier quantum-network architecture composed of quantum edge nodes and Entanglement Service Providers (ESPs). The edge nodes generate application-level entanglement requests, while the ESPs maintain, manage, and route shared entanglement resources across the quantum-network core.

Figure 3.1 illustrates the overall layered architecture considered in this thesis.

The complete quantum network is represented as

$$G = (V, E), \quad (3.1)$$

where V denotes the set of quantum nodes and E represents the set of physical quantum communication links.

The node set is partitioned into two disjoint subsets:

$$V = V_1 \cup V_2, \quad V_1 \cap V_2 = \emptyset, \quad (3.2)$$

where V_1 represents quantum edge nodes and V_2 represents Entanglement Service Providers (ESPs).

The numbers of edge nodes and ESPs are defined as

$$|V_1| = n_u, \quad |V_2| = n_e, \quad (3.3)$$

such that

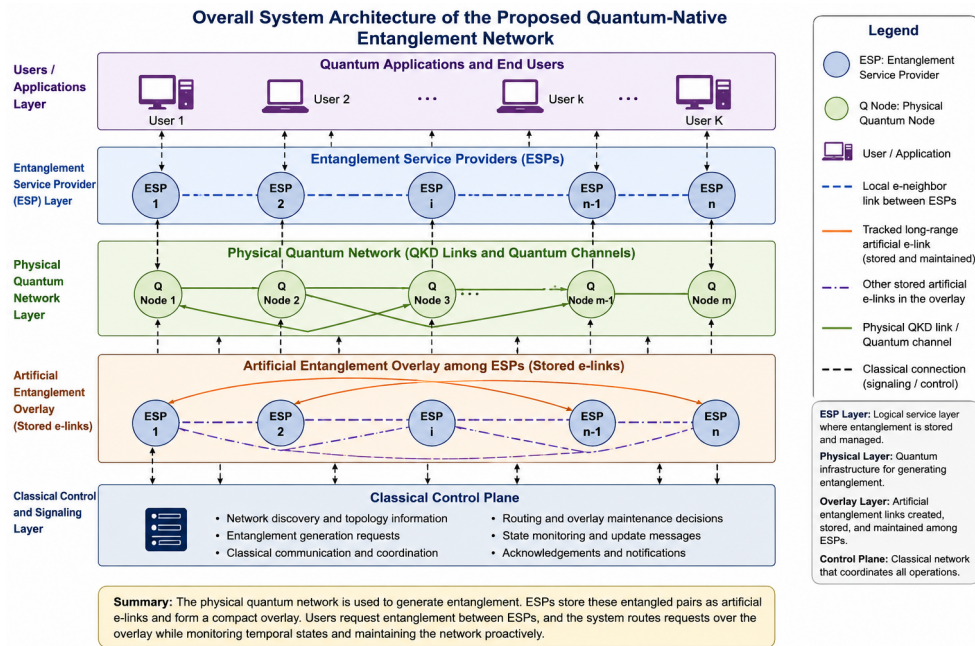


Figure 3.1: Overall architecture of the quantum-native entanglement network, including the user layer, Entanglement Service Provider layer, physical quantum network, artificial entanglement overlay, and classical control plane.

$$|V| = n = n_u + n_e. \quad (3.4)$$

Each edge node is connected to a serving ESP through a local quantum connection. When an application requests end-to-end entanglement, the corresponding ESP coordinates entanglement generation, routing, and resource allocation operations over the ESP-level network.

The ESPs constitute the quantum-network core. Each ESP is assumed to contain quantum memories, entanglement-generation interfaces, Bell-state measurement capabilities, and classical communication channels required for entanglement establishment, heralding, synchronization, and control operations.

An Entanglement-Defined Controller (EDC), denoted by $\mathcal{C}_{\text{EDC}}$, manages the control plane of the artificial entanglement overlay. The controller maintains information about artificial e-link availability, routing-table states, temporal link conditions, regeneration status, and network constraints.

The main functions of the EDC include:

- monitoring artificial e-link availability and temporal state;
- constructing and updating e-neighborhood information;
- assigning anchor and tracked-link responsibilities;
- coordinating route-selection decisions;
- prioritizing stale-link regeneration;
- repairing depleted artificial-overlay resources; and
- enforcing fidelity, memory, and coherence constraints.

The architecture separates the physical quantum network from the artificial entanglement overlay. The physical network represents the available quantum communication infrastructure, while the artificial overlay represents dynamically maintained entanglement resources established through entanglement generation and swapping operations.

The physical ESP-level network is represented as

$$G_P = (V_2, E_P), \quad (3.5)$$

where E_P denotes the set of physical quantum links connecting ESPs.

The artificial entanglement overlay at time t is represented as

$$G_A(t) = (V_2, E_A(t)), \quad (3.6)$$

where $E_A(t)$ represents the set of currently available artificial e-links. An artificial e-link is defined as

$$e_{ij}^A(t) \in E_A(t), \quad (3.7)$$

and represents an entangled resource shared between ESPs v_i and v_j . The two ESPs may be physically adjacent or may be connected through multi-hop entanglement generation and swapping operations.

Because artificial e-links are generated, consumed, degraded, and regenerated dynamically, the artificial overlay is time varying:

$$E_A(t+1) \neq E_A(t). \quad (3.8)$$

The proposed framework considers both Partial-Anchor and Full-Anchor compact-routing schemes. In the Partial-Anchor Scheme, every ESP maintains local e-neighbor links, while only selected anchor ESPs maintain long-range artificial links. In the Full-Anchor Scheme, every ESP maintains local e-neighbor links together with additional long-range tracked links.

Routing decisions are performed using local compact routing-table information combined with temporal-state updates provided by the EDC. Therefore, each ESP does not require complete global network knowledge, which enables scalable routing for large quantum networks.

This architecture provides the structural foundation for defining the physical network model, artificial entanglement overlay, temporal e-link state, and coherence-budget-aware routing problem in the following sections.

3.3 Physical Quantum Network Model

The physical quantum network represents the underlying hardware-level infrastructure that enables entanglement generation and distribution among Entanglement Service Providers. It includes quantum communication channels, quantum memories, entanglement-generation devices, Bell-state measurement modules, and classical communication links required for synchronization and coordination.

The ESP-level physical quantum network is modeled as an undirected graph

$$G_P = (V_2, E_P), \quad (3.9)$$

where V_2 denotes the set of Entanglement Service Providers and E_P represents the set of physical quantum links connecting them.

A physical quantum link between ESPs v_i and v_j is defined as

$$e_{ij}^P = (v_i, v_j) \in E_P. \quad (3.10)$$

Each physical link is characterized by a parameter vector

$$\Theta_{ij}^P = [L_{ij}, p_{ij}^{\text{gen}}, F_{ij}^0, T_{2,ij}, q_{ij}, M_{ij}, D_{ij}^{\text{gen}}], \quad (3.11)$$

where

- L_{ij} represents the physical distance between ESPs v_i and v_j ;
- p_{ij}^{gen} denotes the probability of successful elementary-entanglement generation;
- F_{ij}^0 represents the initial fidelity of a successfully generated entangled pair;
- $T_{2,ij}$ denotes the coherence time of the quantum memory;
- q_{ij} represents the success probability of local operations, including Bell-state measurements;
- M_{ij} denotes the available quantum-memory capacity; and
- D_{ij}^{gen} represents the expected elementary-link generation delay.

The network evolution is considered in discrete time slots:

$$t \in \mathbb{N}. \quad (3.12)$$

For physical link e_{ij}^P , the result of an elementary entanglement-generation attempt at slot t is modeled as a Bernoulli random variable:

$$X_{ij}(t) \sim \text{Bernoulli}(p_{ij}^{\text{gen}}). \quad (3.13)$$

Therefore,

$$X_{ij}(t) = \begin{cases} 1, & \text{if elementary entanglement generation succeeds,} \\ 0, & \text{otherwise.} \end{cases} \quad (3.14)$$

When multiple parallel generation modes are available, denoted by S_{ij} , the probability of obtaining at least one successful elementary pair is

$$p_{ij}^{\text{link}} = 1 - \left(1 - p_{ij}^{\text{gen}}\right)^{S_{ij}}. \quad (3.15)$$

The elementary-generation probability is affected by channel loss and can be approximated using an exponential attenuation model:

$$p_{ij}^{\text{gen}} = \eta_{ij} = \exp(-\alpha L_{ij}), \quad (3.16)$$

where α is the channel attenuation coefficient.

Each successfully generated elementary entangled pair occupies quantum-memory resources at both endpoint ESPs. Let

$$m_i(t) \quad (3.17)$$

denote the number of occupied memory units at ESP v_i during time slot t . The memory constraint is given by

$$m_i(t) \leq M_i, \quad \forall v_i \in V_2. \quad (3.18)$$

A physical entanglement path between ESPs v_i and v_j is represented as

$$P_{ij} = (v_i, v_{r_1}, \dots, v_{r_k}, v_j). \quad (3.19)$$

If all elementary links along the path are successfully generated and intermediate Bell-state measurements occur independently, the approximate end-to-end establishment probability is

$$P_{\text{succ}}(P_{ij}) = \left(\prod_{e_{uv}^p \in P_{ij}} p_{uv}^{\text{link}} \right) \left(\prod_{v_r \in P_{ij} \setminus \{v_i, v_j\}} q_r \right). \quad (3.20)$$

The expected physical-route establishment delay is denoted by

$$D^{\text{phys}}(P_{ij}). \quad (3.21)$$

This delay includes elementary-link generation time, classical heralding latency, memory waiting time, purification overhead, and entanglement-swapping delay.

A physical route is considered feasible only when sufficient quantum-memory resources are available and the resulting entangled state satisfies the required fidelity condition:

$$F_{\text{end}}(P_{ij}) \geq F_{\text{min}}. \quad (3.22)$$

The physical quantum network provides the substrate for constructing artificial entanglement links. An artificial e-link between physically non-adjacent ESPs is established through a feasible physical path and stored at the corresponding ESP endpoints.

Therefore, the quality of an artificial e-link depends on the physical path used for its creation, including generation delay, success probability, initial fidelity, memory consumption, and accumulated operational errors.

3.4 Artificial Entanglement Overlay Model

The artificial entanglement overlay represents the set of proactively established entangled connections maintained among Entanglement Service Providers. Unlike the physical quantum network, an artificial e-link may connect ESPs that are not physically adjacent by using multi-hop entanglement generation and swapping operations.

At time slot t , the artificial entanglement overlay is represented as

$$G_A(t) = (V_2, E_A(t)), \quad (3.23)$$

where V_2 denotes the set of ESPs and $E_A(t)$ represents the set of available artificial e-links at time t .

An artificial e-link between ESPs v_i and v_j is defined as

$$e_{ij}^A(t) = (v_i, v_j) \in E_A(t). \quad (3.24)$$

The availability state of an artificial e-link is represented by

$$z_{ij}(t) = \begin{cases} 1, & \text{if a usable artificial e-link exists,} \\ 0, & \text{otherwise.} \end{cases} \quad (3.25)$$

An artificial e-link is considered usable only when the entangled resource is available in quantum memory, has not been consumed by previous operations, and satisfies the required fidelity constraint:

$$F_{ij}^{\text{eff}}(t) \geq F_{\min}. \quad (3.26)$$

The artificial e-link is established through an underlying physical entanglement-generation path. Let

$$P_{ij}^{\text{gen}} \quad (3.27)$$

denote the physical route used to create an artificial e-link between ESPs v_i and v_j .

The generation success probability of the artificial e-link is determined by the physical path:

$$P_{ij}^A = P_{\text{succ}} \left(P_{ij}^{\text{gen}} \right). \quad (3.28)$$

The corresponding expected generation delay is

$$D_{ij}^A = D^{\text{phys}} \left(P_{ij}^{\text{gen}} \right). \quad (3.29)$$

The initial fidelity of the generated artificial e-link is denoted by

$$F_{ij}^{A,0}. \quad (3.30)$$

This initial fidelity depends on the quality of elementary entangled pairs, the number of entanglement-swapping operations, purification procedures, and accumulated hardware-induced errors during link establishment.

Since artificial e-links are temporary quantum resources, the overlay topology evolves over time due to link generation, consumption, degradation, expiration, and regeneration. The evolution of the artificial overlay is expressed as

$$E_A(t+1) = (E_A(t) \setminus E_{\text{cons}}(t) \setminus E_{\text{drop}}(t)) \cup E_{\text{gen}}(t), \quad (3.31)$$

where

- $E_{\text{cons}}(t)$ denotes the set of artificial e-links consumed during routing, swapping, or application operations;
- $E_{\text{drop}}(t)$ represents expired, stale, or unusable e-links; and
- $E_{\text{gen}}(t)$ represents newly generated or regenerated artificial e-links.

Each artificial e-link requires quantum-memory resources at both end-point ESPs. Let

$$n_{ij}^{\text{ebit}}(t) \quad (3.32)$$

denote the number of stored e-bits associated with artificial link (i, j) .

The memory constraint at ESP v_i is therefore

$$\sum_{j: e_{ij}^A(t) \in E_A(t)} n_{ij}^{\text{ebit}}(t) \leq M_i. \quad (3.33)$$

A routing path over the artificial overlay between source ESP v_s and destination ESP v_d is represented as

$$R_{s,d}(t) = (v_s, v_{r_1}, \dots, v_{r_h}, v_d). \quad (3.34)$$

A feasible overlay route requires that every consecutive ESP pair along the path has an available artificial e-link:

$$e_{uv}^A(t) \in E_A(t), \quad \forall (u, v) \in R_{s,d}(t). \quad (3.35)$$

Establishing end-to-end entanglement along the route requires entanglement swapping at intermediate ESPs. Therefore, a route containing $h + 1$ artificial e-links requires h swapping operations.

Assuming independent swapping success probabilities, the route-level establishment probability is approximated as

$$P_{\text{route}}(R_{s,d}(t)) = \left(\prod_{(i,j) \in R_{s,d}(t)} z_{ij}(t) \right) \left(\prod_{v_r \in R_{s,d}(t) \setminus \{v_s, v_d\}} q_r \right). \quad (3.36)$$

The artificial overlay follows compact-routing structures based on local e-neighborhood maintenance and long-range connectivity mechanisms. In this thesis, both Partial-Anchor and Full-Anchor schemes are considered as baseline overlay architectures for evaluating temporal-state-aware routing performance.

3.4.1 Partial-Anchor Overlay

In the Partial-Anchor Scheme, each ESP maintains artificial e-links with ESPs belonging to its local e-neighborhood

$$\mathcal{N}(v_i). \quad (3.37)$$

A smaller subset of ESPs, referred to as anchor nodes, maintains additional long-range artificial links. The anchor set is defined as

$$\mathcal{T} \subseteq V_2. \quad (3.38)$$

The artificial-link set in the Partial-Anchor overlay can be represented as

$$E_A^{\text{PA}}(t) = E_{\text{local}}(t) \cup E_{\text{anchor}}(t). \quad (3.39)$$

3.4.2 Full-Anchor Overlay

In the Full-Anchor Scheme, every ESP maintains local e-neighbor links together with selected long-range tracked links.

The artificial-link set is represented as

$$E_A^{\text{FA}}(t) = E_{\text{local}}(t) \cup E_{\text{tracked}}(t). \quad (3.40)$$

Compared with the Partial-Anchor Scheme, the Full-Anchor architecture provides stronger overlay connectivity and shorter artificial routes, but requires additional quantum-memory resources and long-range entanglement maintenance.

In both compact-routing architectures, structural availability alone is insufficient for reliable routing. An artificial e-link may remain present in the routing table while its fidelity has degraded, its coherence budget has been exhausted, or its remaining lifetime is insufficient for completing an end-to-end entanglement operation.

Therefore, the following section introduces an explicit temporal-state model for artificial e-links.

3.5 Temporal State of Artificial E-Links

The temporal state of an artificial e-link describes how its routing usability evolves after link establishment. Although an e-link may remain physically stored in quantum memory, decoherence gradually reduces its fidelity, increases expiration risk, and changes its suitability for future routing operations.

For an artificial e-link shared between ESPs v_i and v_j , let

$$t_{ij}^{\text{gen}} \quad (3.41)$$

denote the time slot when the e-link was successfully generated or regenerated.

The age of the artificial e-link at time t is defined as

$$A_{ij}(t) = t - t_{ij}^{\text{gen}}. \quad (3.42)$$

For a newly generated e-link,

$$A_{ij}(t) = 0. \quad (3.43)$$

If the e-link remains stored without consumption or regeneration, its age evolves as

$$A_{ij}(t+1) = A_{ij}(t) + 1. \quad (3.44)$$

The activity state of an artificial e-link is represented by

$$x_{ij}(t) \in \{0, 1\}, \quad (3.45)$$

where

$$x_{ij}(t) = \begin{cases} 1, & \text{if the e-link is stored and available,} \\ 0, & \text{otherwise.} \end{cases} \quad (3.46)$$

The complete temporal state vector of an artificial e-link is defined as

$$\mathcal{L}_{ij}(t) = [x_{ij}(t), A_{ij}(t), F_{ij}^{A,0}, F_{ij}^{\text{eff}}(t), T_{2,ij}, B_{ij}(t), D_{ij}^{\text{regen}}, C_{ij}^{\text{regen}}]. \quad (3.47)$$

The components represent:

- $x_{ij}(t)$: current availability state of the artificial e-link;
- $A_{ij}(t)$: elapsed storage time after generation or regeneration;
- $F_{ij}^{A,0}$: initial fidelity after establishment;
- $F_{ij}^{\text{eff}}(t)$: current effective fidelity;
- $T_{2,ij}$: coherence time of the quantum memory;
- $B_{ij}(t)$: remaining coherence budget;
- D_{ij}^{regen} : expected regeneration delay; and
- C_{ij}^{regen} : regeneration resource cost.

The temporal state is updated at the beginning of each time slot. Depending on generation, aging, consumption, or expiration events, the state evolves according to

$$\mathcal{L}_{ij}(t+1) = \begin{cases} \mathcal{L}_{ij}^{\text{new}}, & \text{if generation succeeds,} \\ \mathcal{L}_{ij}^{\text{age}}, & \text{if the e-link remains stored,} \\ \mathcal{L}_{ij}^{\text{inactive}}, & \text{if consumed, discarded, or expired.} \end{cases} \quad (3.48)$$

For a newly generated e-link,

$$\mathcal{L}_{ij}^{\text{new}} = [1, 0, F_{ij}^{\text{A},0}, F_{ij}^{\text{A},0}, T_{2,ij}, B_{ij}^0, D_{ij}^{\text{regen}}, C_{ij}^{\text{regen}}]. \quad (3.49)$$

For an aged but still usable e-link,

$$\mathcal{L}_{ij}^{\text{age}} = [1, A_{ij}(t) + 1, F_{ij}^{\text{A},0}, F_{ij}^{\text{eff}}(t + 1), T_{2,ij}, B_{ij}(t + 1), D_{ij}^{\text{regen}}, C_{ij}^{\text{regen}}]. \quad (3.50)$$

An artificial e-link is classified as stale when its temporal quality becomes insufficient. This occurs if

$$F_{ij}^{\text{eff}}(t) < F_{\min}, \quad (3.51)$$

or

$$B_{ij}(t) \leq 0, \quad (3.52)$$

or

$$A_{ij}(t) \geq A_{ij}^{\max}. \quad (3.53)$$

The stale-link indicator is defined as

$$s_{ij}(t) = \begin{cases} 1, & \text{if the e-link is stale,} \\ 0, & \text{otherwise.} \end{cases} \quad (3.54)$$

A stale e-link may still physically exist in quantum memory; however, it should not participate in temporally feasible routing decisions.

The temporally usable-link indicator is therefore defined as

$$u_{ij}(t) = x_{ij}(t) (1 - s_{ij}(t)). \quad (3.55)$$

The temporally usable artificial overlay is given by

$$G_U(t) = (V_2, E_U(t)), \quad (3.56)$$

where

$$E_U(t) = \{e_{ij}^{\text{A}}(t) \in E_{\text{A}}(t) : u_{ij}(t) = 1\}. \quad (3.57)$$

This distinction between stored and temporally usable artificial links is fundamental for the proposed routing framework. A routing table may contain an e-link entry that is structurally valid but temporally unreliable. Therefore, route selection is performed over the temporally usable overlay $G_U(t)$ rather than the complete stored overlay $G_{\text{A}}(t)$.

3.6 Fidelity and Decoherence Model

The fidelity and decoherence model describes the evolution of artificial e-link quality during storage, generation, purification, and entanglement-swapping operations. Since the proposed routing framework relies on stored entanglement resources, the temporal degradation of fidelity must be explicitly considered in routing decisions.

For an artificial e-link shared between ESPs v_i and v_j , let

$$F_{ij}^{A,0} \quad (3.58)$$

denote the initial fidelity immediately after successful e-link generation or regeneration.

The initial fidelity depends on several physical factors, including the quality of elementary entangled pairs, the number of entanglement-swapping operations, purification performance, and accumulated hardware imperfections.

The current effective fidelity of the stored artificial e-link at time t is represented as

$$F_{ij}^{\text{eff}}(t). \quad (3.59)$$

A general temporal fidelity model can be expressed as

$$F_{ij}^{\text{eff}}(t) = f_{\text{dec}} \left(F_{ij}^{A,0}, A_{ij}(t), T_{2,ij}, \epsilon_{ij}^{\text{op}} \right), \quad (3.60)$$

where $\epsilon_{ij}^{\text{op}}$ represents accumulated operational errors caused by imperfect quantum operations.

For analytical modeling and simulation, an exponential decoherence model is adopted:

$$F_{ij}^{\text{eff}}(t) = F_{\infty} + \left(F_{ij}^{A,0} - F_{\infty} \right) \exp \left(-\frac{A_{ij}(t)\Delta}{T_{2,ij}} \right), \quad (3.61)$$

where Δ denotes the duration of one time slot, $T_{2,ij}$ is the coherence time of the quantum memory, and F_{∞} represents the fidelity limit under the selected noise model.

For a two-qubit Werner-state approximation, the fully decohered state approaches the maximally mixed state, whose fidelity is

$$F_{\infty} = \frac{1}{4}. \quad (3.62)$$

Therefore, the storage-induced fidelity degradation model becomes

$$F_{ij}^{\text{eff}}(t) = \frac{1}{4} + \left(F_{ij}^{\text{A},0} - \frac{1}{4}\right) \exp\left(-\frac{A_{ij}(t)\Delta}{T_{2,ij}}\right). \quad (3.63)$$

This formulation ensures that the fidelity gradually approaches the physical lower bound of the Werner-state model rather than incorrectly decaying to zero.

An equivalent representation can be obtained using the Werner parameter:

$$p_{ij}^W(t) = \frac{4F_{ij}^{\text{eff}}(t) - 1}{3}. \quad (3.64)$$

The initial Werner parameter is defined as

$$p_{ij}^{W,0} = \frac{4F_{ij}^{\text{A},0} - 1}{3}. \quad (3.65)$$

Under exponential storage decoherence, the Werner parameter evolves as

$$p_{ij}^W(t) = p_{ij}^{W,0} \exp\left(-\frac{A_{ij}(t)\Delta}{T_{2,ij}}\right). \quad (3.66)$$

The corresponding effective fidelity is recovered by

$$F_{ij}^{\text{eff}}(t) = \frac{1 + 3p_{ij}^W(t)}{4}. \quad (3.67)$$

For an artificial-overlay route

$$R_{s,d}(t) = (v_s, v_{r_1}, \dots, v_{r_h}, v_d), \quad (3.68)$$

the end-to-end fidelity depends on the effective fidelity of all artificial e-links along the selected path and the success probability of intermediate swapping operations. Let the set of artificial e-links contained in route $R_{s,d}(t)$ be denoted by

$$E_A(R_{s,d}(t)). \quad (3.69)$$

Under the Werner-state approximation, the accumulated route-level Werner parameter before considering swapping imperfections is

$$p_{\text{route}}^W(t) = \prod_{(i,j) \in E_A(R_{s,d}(t))} p_{ij}^W(t). \quad (3.70)$$

To model imperfect entanglement-swapping operations, let

$$\lambda_r \in [0, 1] \quad (3.71)$$

denote the fidelity-retention factor of the Bell-state measurement operation performed at intermediate ESP v_r .

The end-to-end Werner parameter after considering swapping imperfections is approximated as

$$p_{s,d}^{W,\text{end}}(t) = \left(\prod_{(i,j) \in E_A(R_{s,d}(t))} p_{ij}^W(t) \right) \left(\prod_{v_r \in R_{s,d}(t) \setminus \{v_s, v_d\}} \lambda_r \right). \quad (3.72)$$

The resulting end-to-end fidelity is obtained as

$$F_{s,d}^{\text{end}}(t) = \frac{1 + 3p_{s,d}^{W,\text{end}}(t)}{4}. \quad (3.73)$$

A selected route is considered fidelity-feasible at the current time only if

$$F_{s,d}^{\text{end}}(t) \geq F_{\min}. \quad (3.74)$$

However, current-time fidelity alone is insufficient for temporal routing decisions. In practice, additional waiting time may occur due to classical communication, resource scheduling, and intermediate swapping operations. During this interval, the stored e-links continue to experience decoherence.

Let

$$\widehat{D}_{s,d}^{\text{route}}(t) \quad (3.75)$$

denote the predicted completion delay of the selected route.

For an artificial e-link (i, j) , the predicted fidelity at consumption time is given by

$$\widehat{F}_{ij}^{\text{cons}}(t) = \frac{1}{4} + \left(F_{ij}^{\text{eff}}(t) - \frac{1}{4} \right) \exp \left(-\frac{\widehat{D}_{ij}^{\text{use}}(t)}{T_{2,ij}} \right), \quad (3.76)$$

where $\widehat{D}_{ij}^{\text{use}}(t)$ represents the predicted waiting time before the e-link is consumed during route establishment.

Using these predicted link fidelities, the end-to-end fidelity at route completion can be estimated as

$$\widehat{F}_{s,d}^{\text{end}}(t + \widehat{D}_{s,d}^{\text{route}}(t)) = \frac{1 + 3\widehat{p}_{s,d}^{W,\text{end}}(t)}{4}. \quad (3.77)$$

A route is considered temporally fidelity-feasible only if

$$\widehat{F}_{s,d}^{\text{end}}(t + \widehat{D}_{s,d}^{\text{route}}(t)) \geq F_{\min}. \quad (3.78)$$

In some cases, purification may be applied when the stored fidelity is insufficient but additional entangled pairs are available. The output fidelity of a purification procedure can be represented generally as

$$F_{ij}^{\text{pur}} = \Phi_{\text{pur}}(F_{ij}^{(1)}, F_{ij}^{(2)}). \quad (3.79)$$

Purification is performed only when the expected fidelity improvement justifies the additional memory consumption, processing delay, and probabilistic success cost.

Therefore, the proposed routing framework distinguishes among:

- current stored-link fidelity;
- predicted fidelity at consumption time;
- end-to-end route fidelity after swapping; and
- optional purification-enhanced fidelity.

This temporal fidelity evaluation prevents the routing algorithm from selecting an artificial e-link that appears feasible at the decision time but becomes unusable before completion of the end-to-end entanglement establishment process.

3.7 Remaining Coherence-Budget Model

The remaining coherence budget estimates whether an artificial e-link can remain usable until it is consumed by an end-to-end entanglement-routing operation. Unlike simple age-based metrics, the coherence budget jointly considers the current storage duration, memory coherence limit, and the expected future delay before utilization.

For artificial e-link (i, j) , let

$$T_{2,ij} \quad (3.80)$$

denote the coherence time of the corresponding quantum memory, and let

$$A_{ij}(t) \quad (3.81)$$

represent the current age of the e-link at time slot t .

A basic estimate of the remaining coherence lifetime is

$$R_{ij}^{\text{coh}}(t) = T_{2,ij} - A_{ij}(t)\Delta, \quad (3.82)$$

where Δ is the duration of one time slot.

However, remaining coherence lifetime alone does not guarantee successful utilization because additional delay may occur before the e-link participates in an end-to-end entanglement operation.

Let

$$\widehat{D}_{ij}^{\text{use}}(t) \quad (3.83)$$

denote the predicted additional delay before artificial e-link (i, j) is consumed.

The remaining coherence budget is therefore defined as

$$B_{ij}(t) = T_{2,ij} - A_{ij}(t)\Delta - \widehat{D}_{ij}^{\text{use}}(t). \quad (3.84)$$

A positive coherence budget indicates that the e-link is expected to remain within its coherence interval until consumption:

$$B_{ij}(t) > 0. \quad (3.85)$$

A non-positive value indicates that the e-link is likely to expire before it can be used:

$$B_{ij}(t) \leq 0. \quad (3.86)$$

The predicted utilization delay can include several components:

$$\widehat{D}_{ij}^{\text{use}}(t) = \widehat{D}_{ij}^{\text{queue}}(t) + \widehat{D}_{ij}^{\text{coord}}(t) + \widehat{D}_{ij}^{\text{pur}}(t) + \widehat{D}_{ij}^{\text{swap}}(t), \quad (3.87)$$

where

- $\widehat{D}_{ij}^{\text{queue}}(t)$ represents scheduling and resource waiting delay;
- $\widehat{D}_{ij}^{\text{coord}}(t)$ represents classical coordination delay;
- $\widehat{D}_{ij}^{\text{pur}}(t)$ represents purification-related delay; and
- $\widehat{D}_{ij}^{\text{swap}}(t)$ represents the expected delay before entanglement swapping.

For a route

$$R_{s,d}(t), \quad (3.88)$$

the route-level coherence budget is determined by the bottleneck e-link:

$$B_{s,d}^{\text{route}}(t) = \min_{(i,j) \in E_A(R_{s,d}(t))} B_{ij}(t). \quad (3.89)$$

This formulation reflects the fact that a single expired artificial e-link can invalidate the complete end-to-end route.

Therefore, a route is coherence-feasible only when

$$B_{s,d}^{\text{route}}(t) > 0. \quad (3.90)$$

To compensate for uncertainty in delay prediction, a safety margin can be introduced:

$$\delta_{\text{safe}} > 0. \quad (3.91)$$

The robust feasibility condition becomes

$$B_{ij}(t) \geq \delta_{\text{safe}}, \quad \forall (i,j) \in E_A(R_{s,d}(t)). \quad (3.92)$$

A normalized coherence-budget ratio is defined as

$$\tilde{B}_{ij}(t) = \frac{B_{ij}(t)}{T_{2,ij}}. \quad (3.93)$$

This normalization enables comparison among e-links with different coherence times.

The normalized budget satisfies

$$\tilde{B}_{ij}(t) \in (-\infty, 1]. \quad (3.94)$$

A larger positive value indicates a temporally reliable e-link, while a value close to zero indicates a high expiration risk.

The coherence-budget risk cost is defined as

$$C_B(i, j, t) = \begin{cases} \frac{1}{\tilde{B}_{ij}(t) + \varepsilon}, & \tilde{B}_{ij}(t) > 0, \\ +\infty, & \tilde{B}_{ij}(t) \leq 0, \end{cases} \quad (3.95)$$

where $\varepsilon > 0$ avoids numerical instability near zero.

This cost increases sharply when an e-link approaches expiration, discouraging selection of temporally unreliable resources.

The coherence-budget state can also be classified using predefined thresholds:

$$\text{State}_{ij}^B(t) = \begin{cases} \text{Safe,} & B_{ij}(t) \geq B_{\text{high}}, \\ \text{Warning,} & B_{\text{low}} \leq B_{ij}(t) < B_{\text{high}}, \\ \text{Critical,} & 0 < B_{ij}(t) < B_{\text{low}}, \\ \text{Expired,} & B_{ij}(t) \leq 0. \end{cases} \quad (3.96)$$

Safe links can be used normally, warning links require monitoring, critical links may trigger rerouting or regeneration, and expired links are removed from the temporally usable overlay.

The regeneration urgency of artificial e-link (i, j) is defined as

$$U_{ij}^{\text{regen}}(t) = \omega_1 C_B(i, j, t) + \omega_2 C_F(i, j, t) + \omega_3 I_{ij}^{\text{struct}}, \quad (3.97)$$

where

- $C_B(i, j, t)$ represents coherence-budget risk;
- $C_F(i, j, t)$ represents fidelity degradation risk; and
- I_{ij}^{struct} represents structural importance of the e-link in the compact overlay.

The structural importance term enables anchor links and frequently used artificial links to receive higher regeneration priority than links with limited routing influence.

Therefore, the coherence-budget model establishes a direct relationship among quantum-memory lifetime, route feasibility, temporal routing decisions, and artificial-overlay maintenance.

3.8 Temporal Route-Feasibility Condition

A route over the artificial entanglement overlay may be structurally valid but temporally infeasible. Structural validity only indicates that all required artificial e-links are currently available in the overlay. Temporal feasibility additionally requires that these e-links remain usable until the completion of the end-to-end entanglement-establishment process.

Let

$$R_{s,d}(t) = (v_s, v_{r_1}, \dots, v_{r_h}, v_d) \quad (3.98)$$

denote a candidate artificial-overlay route between source ESP v_s and destination ESP v_d at time t .

The artificial e-link set contained in the route is denoted by

$$E_A(R_{s,d}(t)). \quad (3.99)$$

The route is structurally feasible if every required artificial e-link exists and is currently stored:

$$x_{ij}(t) = 1, \quad \forall (i, j) \in E_A(R_{s,d}(t)). \quad (3.100)$$

However, structural availability alone does not guarantee successful route completion. Each artificial e-link must satisfy temporal-state, fidelity, delay, and memory constraints.

The first requirement is the remaining coherence-budget constraint:

$$B_{ij}(t) \geq \delta_{\text{safe}}, \quad \forall (i, j) \in E_A(R_{s,d}(t)), \quad (3.101)$$

where δ_{safe} represents the required safety margin against delay-estimation uncertainty.

The second requirement is the predicted link-level fidelity constraint:

$$\hat{F}_{ij}^{\text{cons}}(t) \geq F_{\min}^{\text{link}}, \quad \forall (i, j) \in E_A(R_{s,d}(t)). \quad (3.102)$$

The third requirement considers the final end-to-end fidelity after all swapping operations:

$$\hat{F}_{s,d}^{\text{end}}(t + \widehat{D}_{s,d}^{\text{route}}(t)) \geq F_{\min}^{\text{end}}. \quad (3.103)$$

The fourth requirement limits the route-completion delay:

$$\widehat{D}_{s,d}^{\text{route}}(t) \leq D_{\max}, \quad (3.104)$$

where $D_{\max}$ represents the maximum acceptable entanglement-establishment delay.

The fifth requirement ensures that sufficient quantum-memory resources are available along the route:

$$\sum_{j: e_{ij}^A \in R_{s,d}(t)} n_{ij}^{\text{ebit}}(t) \leq M_i, \quad \forall v_i \in R_{s,d}(t). \quad (3.105)$$

A binary temporal route-feasibility indicator is defined as

$$\chi_{s,d}(R_{s,d}(t)) \in \{0, 1\}. \quad (3.106)$$

For compactness, define the active-edge set of the selected route as $\mathcal{E}_{s,d}(t) = E_A(R_{s,d}(t))$. The indicator takes value one only when all temporal routing requirements are satisfied:

$$\begin{aligned}
\chi_{s,d}(R_{s,d}(t)) = & \prod_{(i,j) \in \mathcal{E}_{s,d}(t)} x_{ij}(t) \\
& \times \prod_{(i,j) \in \mathcal{E}_{s,d}(t)} \mathbf{1}(B_{ij}(t) \geq \delta_{\text{safe}}) \\
& \times \prod_{(i,j) \in \mathcal{E}_{s,d}(t)} \mathbf{1}(\hat{F}_{ij}^{\text{cons}}(t) \geq F_{\min}^{\text{link}}) \\
& \times \mathbf{1}(\hat{F}_{s,d}^{\text{end}}(t + \hat{D}_{s,d}^{\text{route}}(t)) \geq F_{\min}^{\text{end}}) \\
& \times \mathbf{1}(\hat{D}_{s,d}^{\text{route}}(t) \leq D_{\max}).
\end{aligned} \tag{3.107}$$

and the route must be rejected, regenerated, repaired, or replaced by an alternative temporally feasible path. The set of all temporally feasible routes between ESPs v_s and v_d is defined as

$$\mathcal{R}_{s,d}^{\text{TF}}(t) = \{R_{s,d}(t) : \chi_{s,d}(R_{s,d}(t)) = 1\}. \tag{3.108}$$

If

$$\mathcal{R}_{s,d}^{\text{TF}}(t) = \emptyset, \tag{3.109}$$

then no currently available artificial-overlay route satisfies all temporal, fidelity, delay, and memory constraints.

In this situation, the Entanglement-Defined Controller may perform one of the following recovery actions:

- regenerate one or more stale artificial e-links;
- replace critical anchor or tracked links;
- wait for additional entanglement-generation opportunities;
- apply purification when sufficient entangled resources are available; or
- declare the request temporarily infeasible.

Therefore, the proposed framework classifies each candidate route into three temporal states:

$$\text{RouteState} = \begin{cases} \text{Feasible,} & \text{all temporal constraints are satisfied,} \\ \text{Repairable,} & \text{constraints can be restored through maintenance actions,} \\ \text{Infeasible,} & \text{no acceptable recovery action exists.} \end{cases} \quad (3.110)$$

A repairable route contains one or more critical artificial e-links whose regeneration delay and resource requirements remain within predefined limits.

Let

$$\mathcal{E}_{\text{crit}}(R_{s,d}(t)) \quad (3.111)$$

denote the set of critical artificial e-links contained in route $R_{s,d}(t)$:

$$\mathcal{E}_{\text{crit}}(R_{s,d}(t)) = \left\{ (i, j) \in E_A(R_{s,d}(t)) : B_{ij}(t) < \delta_{\text{safe}} \vee \hat{F}_{ij}^{\text{cons}}(t) < F_{\text{min}}^{\text{link}} \right\}. \quad (3.112)$$

The predicted route-repair delay is calculated as

$$\hat{D}_{s,d}^{\text{repair}}(t) = \sum_{(i,j) \in \mathcal{E}_{\text{crit}}(R_{s,d}(t))} D_{ij}^{\text{regen}}. \quad (3.113)$$

A route is considered repairable only if the required regeneration operations satisfy both delay and resource constraints:

$$\hat{D}_{s,d}^{\text{repair}}(t) \leq D_{\text{max}}^{\text{repair}}, \quad (3.114)$$

and

$$\sum_{(i,j) \in \mathcal{E}_{\text{crit}}(R_{s,d}(t))} C_{ij}^{\text{regen}} \leq C_{\text{max}}^{\text{repair}}. \quad (3.115)$$

If both conditions are satisfied, the controller may regenerate the critical links and restore route feasibility. Otherwise, an alternative route must be selected or the request must wait for future resource availability.

This temporal route-feasibility model prevents the compact-routing protocol from selecting artificial e-links solely because they remain structurally present in the routing table. Instead, routing decisions are based on predicted future usability, ensuring that selected routes remain reliable throughout the complete entanglement-establishment process.

3.9 Time-Dependent Entangling-Cost Metric

The time-dependent entangling-cost metric quantifies the cost of selecting an artificial e-link according to its current physical and temporal state. Unlike static routing metrics, where an e-link has a fixed cost, the proposed metric dynamically changes with fidelity degradation, entanglement age, remaining coherence budget, regeneration overhead, and structural characteristics.

For an artificial e-link (i, j) at time t , the proposed time-dependent cost is defined as

$$w_t(i, j) = \alpha_F C_F(i, j, t) + \alpha_A C_A(i, j, t) + \alpha_B C_B(i, j, t) + \alpha_R C_R(i, j, t) + \alpha_S C_S(i, j), \quad (3.116)$$

where

- $C_F(i, j, t)$ represents fidelity degradation cost;
- $C_A(i, j, t)$ represents entanglement-age cost;
- $C_B(i, j, t)$ represents coherence-budget risk;
- $C_R(i, j, t)$ represents regeneration overhead; and
- $C_S(i, j)$ represents structural routing cost.

The weighting coefficients satisfy

$$\alpha_F, \alpha_A, \alpha_B, \alpha_R, \alpha_S \geq 0, \quad (3.117)$$

with

$$\alpha_F + \alpha_A + \alpha_B + \alpha_R + \alpha_S = 1. \quad (3.118)$$

The fidelity-related cost is defined according to the predicted consumption-time fidelity:

$$C_F(i, j, t) = 1 - \hat{F}_{ij}^{\text{cons}}(t). \quad (3.119)$$

Therefore, artificial e-links with higher predicted fidelity obtain lower routing cost.

The age-related cost is defined as

$$C_A(i, j, t) = \min \left\{ \frac{A_{ij}(t)}{A_{ij}^{\max}}, 1 \right\}. \quad (3.120)$$

This cost increases as the stored e-link approaches its maximum allowable age.

The coherence-budget risk cost is defined as

$$C_B(i, j, t) = \begin{cases} \frac{1}{\tilde{B}_{ij}(t) + \varepsilon}, & \tilde{B}_{ij}(t) > 0, \\ +\infty, & \tilde{B}_{ij}(t) \leq 0, \end{cases} \quad (3.121)$$

where

$$\tilde{B}_{ij}(t) = \frac{B_{ij}(t)}{T_{2,ij}}. \quad (3.122)$$

A smaller remaining coherence budget therefore produces a larger routing penalty.

The regeneration-related cost is defined as

$$C_R(i, j, t) = \lambda_D \frac{D_{ij}^{\text{regen}}}{D_{\max}^{\text{regen}}} + \lambda_C \frac{C_{ij}^{\text{regen}}}{C_{\max}^{\text{regen}}}, \quad (3.123)$$

where

$$\lambda_D + \lambda_C = 1. \quad (3.124)$$

The first term represents regeneration delay, while the second term represents regeneration resource consumption.

The structural routing cost models the role of the artificial e-link inside the compact overlay:

$$C_S(i, j) = \begin{cases} c_{\text{local}}, & \text{if } (i, j) \text{ is an e-neighbor link,} \\ c_{\text{anchor}}, & \text{if } (i, j) \text{ is an anchor link,} \\ c_{\text{tracked}}, & \text{if } (i, j) \text{ is a tracked long-range link.} \end{cases} \quad (3.125)$$

The structural cost and structural importance are treated separately because they serve different purposes.

The structural routing cost

$$C_S(i, j) \quad (3.126)$$

is used during route selection, whereas the structural importance

$$I_{ij}^{\text{struct}} \quad (3.127)$$

is used during overlay maintenance and regeneration prioritization.

The structural importance may depend on

- the number of source–destination pairs using the e-link;
- whether the link connects anchor ESPs;
- whether link removal affects overlay connectivity;
- the betweenness of the link in the artificial topology; and
- the availability of alternative temporally feasible paths.

For a complete route

$$R_{s,d}(t), \quad (3.128)$$

the total time-dependent entangling cost is

$$W_t(R_{s,d}(t)) = \sum_{(i,j) \in E_A(R_{s,d}(t))} w_t(i, j). \quad (3.129)$$

The routing objective is therefore formulated as selecting the temporally feasible route with minimum total cost:

$$R_{s,d}^*(t) = \arg \min_{R_{s,d}(t) \in \mathcal{R}_{s,d}^{\text{TF}}(t)} W_t(R_{s,d}(t)). \quad (3.130)$$

This formulation transforms compact quantum routing from a static shortest-path problem into a temporal resource-selection problem where route quality depends on both topology and the evolving physical state of entanglement resources.

A normalized structural-importance score is defined as

$$I_{ij}^{\text{struct}} = \mu_1 I_{ij}^{\text{anchor}} + \mu_2 I_{ij}^{\text{betweenness}} + \mu_3 I_{ij}^{\text{demand}} + \mu_4 I_{ij}^{\text{scarcity}}, \quad (3.131)$$

where

$$\sum_{k=1}^4 \mu_k = 1, \quad (3.132)$$

and the coefficients μ_k determine the relative contribution of each structural factor.

For a candidate artificial-overlay route

$$R_{s,d}(t), \quad (3.133)$$

the accumulated time-dependent entangling cost is represented using the general composition operator:

$$w_t(R_{s,d}(t)) = \bigoplus_{(i,j) \in E_A(R_{s,d}(t))} w_t(i,j). \quad (3.134)$$

For the additive cost case used in this thesis, the route cost becomes

$$w_t(R_{s,d}(t)) = \sum_{(i,j) \in E_A(R_{s,d}(t))} w_t(i,j). \quad (3.135)$$

The optimal temporally feasible route is selected as

$$R_{s,d}^*(t) = \arg \min_{R \in \mathcal{R}_{s,d}^{\text{TF}}(t)} w_t(R), \quad (3.136)$$

where $\mathcal{R}_{s,d}^{\text{TF}}(t)$ denotes the set of routes satisfying all temporal feasibility constraints.

If no fully feasible route exists, the controller evaluates repairable routes. The repair-aware route cost is defined as

$$w_t^{\text{repair}}(R) = w_t(R) + \rho_D \widehat{D}_{s,d}^{\text{repair}}(t) + \rho_C C_{s,d}^{\text{repair}}(t), \quad (3.137)$$

where the additional terms represent regeneration delay and regeneration resource consumption.

The total repair cost is given by

$$C_{s,d}^{\text{repair}}(t) = \sum_{(i,j) \in \mathcal{E}_{\text{crit}}(R)} C_{ij}^{\text{regen}}. \quad (3.138)$$

The minimum-cost repairable route is therefore

$$R_{s,d}^{\text{repair},*}(t) = \arg \min_{R \in \mathcal{R}_{s,d}^{\text{repair}}(t)} w_t^{\text{repair}}(R). \quad (3.139)$$

For routing convergence, the proposed metric preserves the non-negative ordering property required by additive shortest-path algorithms. Specifically, every finite link cost satisfies

$$w_t(i,j) \geq 0. \quad (3.140)$$

Artificial e-links that violate temporal feasibility conditions are excluded by assigning

$$w_t(i,j) = +\infty. \quad (3.141)$$

Therefore, stale or expired artificial e-links cannot be selected during temporally feasible route optimization.

The weighting parameters are not assumed to be universally optimal. Instead, their influence is evaluated through sensitivity analysis to determine how fidelity, age, coherence budget, regeneration overhead, and structural factors affect routing performance.

3.10 Temporal Entangling Stretch

Temporal entangling stretch extends the conventional entangling-stretch metric by evaluating routing performance under the time-varying state of artificial entanglement resources. Unlike static stretch, which assumes that all artificial links remain available, temporal entangling stretch considers fidelity degradation, coherence limitation, and regeneration-related effects.

For source ESP v_s and destination ESP v_d , let

$$R_{QR}(s, d, t) \quad (3.142)$$

denote the route selected by the proposed compact quantum-routing protocol at time t .

Let

$$R_t^*(s, d) \quad (3.143)$$

denote the minimum-cost temporally feasible route between v_s and v_d under the same temporal network state.

The temporal entangling stretch is defined as

$$ES_t(s, d) = \frac{w_t(R_{QR}(s, d, t))}{w_t(R_t^*(s, d))}. \quad (3.144)$$

When both routes are temporally feasible and have finite cost,

$$ES_t(s, d) \geq 1. \quad (3.145)$$

The optimal value is achieved when

$$ES_t(s, d) = 1, \quad (3.146)$$

which indicates that the compact-routing protocol selects the minimum-cost temporally feasible route.

A value greater than one indicates that the selected route has higher temporal entangling cost than the optimal feasible route. This additional cost may result from compact-routing restrictions, limited local visibility, anchor constraints, or avoidance of temporally degraded e-links.

The worst-case temporal entangling stretch at time t is defined as

$$ES_t^{\max} = \max_{v_s, v_d \in V_2, v_s \neq v_d} ES_t(s, d). \quad (3.147)$$

The average temporal entangling stretch over evaluated source–destination pairs is

$$\overline{ES}_t = \frac{1}{|\mathcal{D}(t)|} \sum_{(s,d) \in \mathcal{D}(t)} ES_t(s, d), \quad (3.148)$$

where $\mathcal{D}(t)$ represents the set of source–destination demands with at least one temporally feasible route at time t .

Over a simulation period of T time slots, the time-average temporal stretch is

$$\overline{ES} = \frac{1}{T} \sum_{t=1}^T \overline{ES}_t. \quad (3.149)$$

To evaluate worst-case temporal degradation, the tail stretch is measured using the 95th percentile:

$$ES_{95} = \inf \{x : \Pr [ES_t(s, d) \leq x] \geq 0.95\}. \quad (3.150)$$

The percentile metric captures rare but important cases where temporal degradation causes significant routing overhead.

A structurally reachable route may still be temporally infeasible because one or more artificial e-links may expire before route completion. Therefore, temporal stretch is evaluated only when at least one temporally feasible route exists.

If

$$\mathcal{R}_{s,d}^{\text{TF}}(t) = \emptyset, \quad (3.151)$$

the corresponding demand is classified as temporally disconnected rather than being assigned an infinite stretch value.

This distinction separates routing inefficiency from resource unavailability and provides a more accurate evaluation of temporal-state-aware quantum routing performance. The temporal disconnection rate measures the fraction of source–destination requests for which no temporally feasible route exists during the evaluation period:

$$P_{\text{disc}} = \frac{\sum_{t=1}^T \sum_{(s,d) \in \mathcal{D}(t)} \mathbf{1} [\mathcal{R}_{s,d}^{\text{TF}}(t) = \emptyset]}{\sum_{t=1}^T |\mathcal{D}(t)|}. \quad (3.152)$$

A lower value of P_{disc} indicates that the artificial overlay maintains higher temporal availability under dynamic entanglement conditions.

To quantify the impact of temporal degradation, the proposed temporal stretch is compared with the original static entangling stretch.

Let

$$ES_{\text{static}}(s, d) \quad (3.153)$$

denote the stretch obtained using the original static entangling-cost metric without temporal-state information.

The temporal stretch inflation is defined as

$$\Delta ES_t(s, d) = ES_t(s, d) - ES_{\text{static}}(s, d). \quad (3.154)$$

A positive value indicates that aging, fidelity degradation, coherence limitation, or regeneration constraints increase the effective routing cost compared with the static case.

The normalized temporal-stretch inflation is defined as

$$\Delta ES_t^{\text{rel}}(s, d) = \frac{ES_t(s, d) - ES_{\text{static}}(s, d)}{ES_{\text{static}}(s, d)}. \quad (3.155)$$

The temporal-stretch performance is evaluated separately for Partial-Anchor and Full-Anchor artificial overlays.

For the Partial-Anchor Scheme, the original static compact-routing guarantee is

$$ES_{\text{static}}^{\text{PA}} \leq 5. \quad (3.156)$$

For the Full-Anchor Scheme, the corresponding static guarantee is

$$ES_{\text{static}}^{\text{FA}} \leq 3. \quad (3.157)$$

These structural guarantees assume that the required artificial links remain available and usable. Under temporal degradation, however, the original route may become infeasible due to fidelity loss, coherence expiration, or regeneration delay.

Therefore, the following conditions are not assumed:

$$ES_t^{\text{PA}} \leq 5, \quad (3.158)$$

and

$$ES_t^{\text{FA}} \leq 3. \quad (3.159)$$

Instead, this thesis evaluates the degradation of compact-routing guarantees under realistic temporal conditions by measuring:

- the frequency of static-bound violations;
- the temporal characteristics of links causing violations;
- the recovery capability after e-link regeneration;
- the additional memory and control overhead required for temporal maintenance; and
- the robustness difference between Partial-Anchor and Full-Anchor overlays.

The violation indicators are defined as

$$I_{\text{viol}}^{\text{PA}}(s, d, t) = \mathbf{1} \left[ES_t^{\text{PA}}(s, d) > 5 \right], \quad (3.160)$$

and

$$I_{\text{viol}}^{\text{FA}}(s, d, t) = \mathbf{1} \left[ES_t^{\text{FA}}(s, d) > 3 \right]. \quad (3.161)$$

The average violation probability is then calculated as

$$P_{\text{viol}} = \frac{\sum_{t=1}^T \sum_{(s,d)} I_{\text{viol}}(s, d, t)}{\sum_{t=1}^T |\mathcal{D}(t)|}. \quad (3.162)$$

This metric quantifies how frequently temporal degradation causes the compact-routing stretch guarantees to be violated. The stretch-bound violation rate is defined as

$$P_{\text{viol}} = \frac{\sum_{t=1}^T \sum_{(s,d) \in \mathcal{D}(t)} I_{\text{viol}}(s, d, t)}{\sum_{t=1}^T |\mathcal{D}(t)|}. \quad (3.163)$$

This metric directly quantifies whether temporal degradation weakens the structural guarantees of compact routing.

The proposed temporal entangling-stretch framework therefore evaluates not only route efficiency, but also the robustness of compact-routing guarantees under realistic entanglement aging, decoherence, fidelity degradation, and regeneration delay.

3.11 Problem Formulation

The routing problem is formulated as the joint selection of temporally feasible artificial-overlay routes and regeneration decisions. The objective is to minimize the time-dependent entangling cost while satisfying fidelity, coherence-budget, memory, exclusivity, regeneration, and compact-routing constraints.

The stretch-bound violation rate is defined as

$$P_{\text{viol}} = \frac{\sum_{t=1}^T \sum_{(s,d) \in \mathcal{D}(t)} I_{\text{viol}}(s, d, t)}{\sum_{t=1}^T |\mathcal{D}(t)|}. \quad (3.164)$$

This metric quantifies whether temporal degradation weakens the structural guarantees of compact routing.

The proposed temporal entangling-stretch framework therefore evaluates not only route efficiency but also the robustness of compact-routing guarantees under realistic entanglement aging, decoherence, and regeneration delay.

Let

$$\mathcal{D}(t) \quad (3.165)$$

denote the set of source–destination entanglement requests arriving at time slot t .

For each request

$$k = (s, d) \in \mathcal{D}(t), \quad (3.166)$$

the routing controller selects an artificial-overlay route

$$R_{s,d}(t) \in \mathcal{R}_{s,d}(t), \quad (3.167)$$

where $\mathcal{R}_{s,d}(t)$ is the set of candidate routes between source ESP v_s and destination ESP v_d .

A binary route-selection variable is defined as

$$y_{s,d}^R(t) = \begin{cases} 1, & \text{if route } R \text{ is selected,} \\ 0, & \text{otherwise.} \end{cases} \quad (3.168)$$

At most one route is selected for each demand:

$$\sum_{R \in \mathcal{R}_{s,d}(t)} y_{s,d}^R(t) \leq 1, \quad \forall (s, d) \in \mathcal{D}(t). \quad (3.169)$$

A selected route must satisfy temporal feasibility:

$$y_{s,d}^R(t) \leq \chi_{s,d}(R), \quad (3.170)$$

where $\chi_{s,d}(R)$ is the temporal route-feasibility indicator.
The route-selection objective is

$$\min \sum_{(s,d) \in \mathcal{D}(t)} \sum_{R \in \mathcal{R}_{s,d}^{\text{TF}}(t)} y_{s,d}^R(t) w_t(R). \quad (3.171)$$

To include route repair and rejection decisions, the objective is extended as

$$\min \left[\sum_{(s,d) \in \mathcal{D}(t)} \sum_{R \in \mathcal{R}_{s,d}^{\text{TF}}(t)} y_{s,d}^R(t) w_t(R) + \lambda_{\text{rep}} C_{\text{repair}}(t) + \lambda_{\text{rej}} N_{\text{reject}}(t) \right], \quad (3.172)$$

where

- $C_{\text{repair}}(t)$ represents regeneration and overlay repair cost;
- $N_{\text{reject}}(t)$ represents rejected requests;
- λ_{rep} controls repair-cost importance; and
- λ_{rej} controls rejection penalty.

The regeneration cost is

$$C_{\text{repair}}(t) = \sum_{(i,j) \in E_A(t)} r_{ij}(t) C_{ij}^{\text{regen}}, \quad (3.173)$$

where

$$r_{ij}(t) \in \{0, 1\} \quad (3.174)$$

is the regeneration decision variable:

$$r_{ij}(t) = \begin{cases} 1, & \text{if e-link } (i, j) \text{ is regenerated,} \\ 0, & \text{otherwise.} \end{cases} \quad (3.175)$$

The regeneration budget constraint is

$$\sum_{(i,j) \in E_A(t)} r_{ij}(t) \leq R_{\text{max}}. \quad (3.176)$$

The memory constraint at ESP v_i is

$$\sum_{j: e_{ij}^A(t) \in E_A(t)} n_{ij}^{\text{ebit}}(t) \leq M_i. \quad (3.177)$$

The end-to-end fidelity constraint is

$$\widehat{F}_{s,d}^{\text{end}}(t + \widehat{D}_{s,d}^{\text{route}}(t)) \geq F_{\min}^{\text{end}}. \quad (3.178)$$

The coherence-budget constraint is

$$B_{ij}(t) \geq \delta_{\text{safe}}, \quad \forall (i, j) \in R_{s,d}(t). \quad (3.179)$$

The route completion delay constraint is

$$\widehat{D}_{s,d}^{\text{route}}(t) \leq D_{\max}. \quad (3.180)$$

The e-link exclusivity constraint ensures that available stored entangled resources are not allocated simultaneously to multiple requests:

$$\sum_{(s,d) \in \mathcal{D}(t)} \sum_{R: (i,j) \in R} y_{s,d}^R(t) \leq n_{ij}^{\text{ebit}}(t). \quad (3.181)$$

The compact-routing constraint requires sublinear routing-table size:

$$|\mathcal{T}_i(t)| \leq c_{\text{tab}} \sqrt{n_e} \log n_e, \quad (3.182)$$

where $\mathcal{T}_i(t)$ represents the routing table of ESP v_i .

The complete optimization problem is summarized as

$$\min_{\mathbf{y}(t), \mathbf{r}(t)} C_{\text{route}}(t) + \lambda_{\text{rep}} C_{\text{repair}}(t) + \lambda_{\text{rej}} N_{\text{reject}}(t) \quad (3.183)$$

subject to

$$\text{route-selection constraints,} \quad (3.184)$$

$$\text{temporal-feasibility constraints,} \quad (3.185)$$

$$\text{fidelity constraints,} \quad (3.186)$$

$$\text{coherence-budget constraints,} \quad (3.187)$$

$$\text{memory constraints,} \quad (3.188)$$

$$\text{e-link exclusivity constraints,} \quad (3.189)$$

$$\text{regeneration-budget constraints,} \quad (3.190)$$

$$\text{compact-routing-table constraints.} \quad (3.191)$$

The complete optimization problem becomes computationally expensive for large networks because route selection, regeneration, resource allocation, and temporal-state evolution are strongly coupled.

Therefore, the following chapter develops a practical heuristic algorithm that preserves compact-routing scalability while using temporal e-link information for adaptive path selection and overlay maintenance.

3.12 Chapter Summary

This chapter presented the system model and formal problem formulation for temporal-state-aware compact routing in quantum-native entanglement overlay networks.

The model separated the physical quantum network from the artificial entanglement overlay. The physical layer represents hardware connectivity, elementary-link generation, quantum memories, swapping operations, and generation delay, while the artificial overlay represents pre-established entangled resources maintained among Entanglement Service Providers.

Each artificial e-link was extended with an explicit temporal state containing availability, age, initial fidelity, effective fidelity, coherence time, remaining coherence budget, regeneration delay, and regeneration cost.

A fidelity-degradation model was introduced to estimate both current and predicted future e-link quality. The coherence-budget model determined whether an artificial link could remain usable until route completion.

A temporal route-feasibility condition was developed by combining link availability, fidelity constraints, coherence requirements, memory limitations, and completion delay.

A time-dependent entangling-cost metric was introduced by combining fidelity cost, age cost, coherence-budget risk, regeneration overhead, and structural routing cost.

Temporal entangling stretch, temporal disconnection rate, stretch inflation, and stretch-bound violation rate were defined to evaluate the robustness of compact-routing schemes under dynamic entanglement conditions.

Finally, the routing and overlay-maintenance problem was formulated as a joint optimization problem involving route selection and regeneration decisions under fidelity, coherence, memory, exclusivity, regeneration-budget, and compact-routing constraints.

The next chapter presents the proposed temporal-state-aware compact-routing algorithm, including temporal-state evaluation, adaptive route selection, coherence-budget-aware regeneration, and overlay maintenance procedures.

Chapter 4

Proposed Temporal-State-Aware Compact Routing Method

4.1 Introduction

This chapter presents the proposed temporal-state-aware compact routing method for quantum-native entanglement overlay networks. The method extends compact quantum routing by incorporating the time-dependent state of stored artificial e-links into route selection and overlay-maintenance decisions.

The proposed framework is motivated by the observation that an artificial e-link may remain structurally present in the routing table while its effective routing utility deteriorates because of aging, decoherence, fidelity degradation, memory waiting time, and delayed consumption. Therefore, a routing decision based only on topological reachability or static entangling cost may select a route that is unlikely to satisfy the required end-to-end fidelity at completion time.

The proposed method addresses this limitation through five coordinated operations:

1. monitoring the temporal state of each stored artificial e-link;
2. constructing compact-routing candidate paths;
3. filtering candidates using temporal feasibility conditions;
4. selecting the minimum-cost temporally feasible route; and

5. regenerating or replacing critical artificial e-links when necessary.

The method operates on the artificial entanglement overlay while using physical-network information to estimate generation delay, regeneration cost, fidelity evolution, and memory consumption.

At each time slot, the controller updates the state of every artificial e-link according to

$$\mathcal{L}_{ij}(t) = [x_{ij}(t), A_{ij}(t), F_{ij}^{A,0}, F_{ij}^{\text{eff}}(t), T_{2,ij}, B_{ij}(t), D_{ij}^{\text{regen}}, C_{ij}^{\text{regen}}, I_{ij}^{\text{struct}}]. \quad (4.1)$$

Based on this temporal state, the controller classifies each artificial e-link as

$$\text{State}_{ij}(t) \in \{\text{Safe}, \text{Warning}, \text{Critical}, \text{Expired}\}. \quad (4.2)$$

Only safe or admissible warning-state links are directly considered during route construction. Critical links may trigger route repair or proactive regeneration, whereas expired links are removed from the temporally usable overlay.

The proposed routing process is expressed as

$$\text{Monitor} \rightarrow \text{Construct} \rightarrow \text{Filter} \rightarrow \text{Select} \rightarrow \text{Maintain}. \quad (4.3)$$

The method preserves the compact-routing principle by limiting route discovery to local e-neighbors, anchor nodes, and tracked long-range artificial links. It does not require every ESP to maintain complete global routing information.

The main design challenge is therefore to improve temporal reliability without converting the compact-routing architecture into a fully centralized global-state routing system.

This chapter first defines the design objectives and overall framework. It then describes temporal-state monitoring, candidate-route construction, temporal-feasibility filtering, time-dependent route selection, proactive regeneration, overlay maintenance, algorithmic steps, and computational complexity.

4.2 Design Objectives

The proposed method is designed to improve the reliability of compact quantum routing by incorporating temporal information of artificial e-links while preserving the scalability advantages of compact routing. The method does

not require every Entanglement Service Provider to maintain complete global knowledge of the network state.

The main design objectives are described as follows.

4.2.1 Temporal Reliability

The first objective is to prevent the routing protocol from selecting artificial e-links that are structurally available but expected to become unusable before route completion.

For every selected route $R_{s,d}(t)$, the proposed method aims to satisfy the coherence-budget constraint

$$B_{ij}(t) \geq \delta_{\text{safe}}, \quad \forall (i, j) \in R_{s,d}(t), \quad (4.4)$$

where $B_{ij}(t)$ denotes the remaining coherence budget of artificial e-link (i, j) and δ_{safe} represents the required safety margin.

In addition, the selected route must satisfy the predicted end-to-end fidelity requirement at the expected completion time:

$$\widehat{F}_{s,d}^{\text{end}}(t + \widehat{D}_{s,d}^{\text{route}}(t)) \geq F_{\min}^{\text{end}}. \quad (4.5)$$

These constraints ensure that routing decisions consider future link usability rather than only the current availability state.

4.2.2 Fidelity Preservation

The second objective is to preserve the quality of distributed entanglement by considering both the current fidelity and the predicted fidelity at the expected consumption time.

The routing method therefore avoids selecting an artificial e-link solely because its instantaneous fidelity satisfies the threshold. Instead, it evaluates whether the link quality remains sufficient after additional waiting, coordination, purification, and swapping operations.

The fidelity-preservation objective can be expressed as

$$\max \widehat{F}_{s,d}^{\text{end}}(t + \widehat{D}_{s,d}^{\text{route}}(t)). \quad (4.6)$$

This objective is incorporated into the proposed time-dependent entangling-cost metric rather than optimized independently. Therefore, high-fidelity and temporally stable routes receive lower routing costs during route selection.

4.2.3 Compact Routing-State Preservation

The proposed method must preserve the compact-routing property of the underlying quantum-native architecture.

Let $\mathcal{T}_i(t)$ denote the routing table maintained by ESP v_i . The routing-state constraint is defined as

$$|\mathcal{T}_i(t)| \leq c_{\text{tab}} \sqrt{n_e} \log n_e, \quad (4.7)$$

where $n_e = |V_2|$ represents the number of ESPs.

The method therefore relies on local e-neighbors, anchor information, and selected tracked e-links instead of maintaining complete all-pairs routing information.

4.2.4 Low Route-Selection Cost

The method aims to select a temporally feasible route with minimum time-dependent entangling cost.

For each demand (s, d) , the route-selection objective is

$$R_{s,d}^*(t) = \arg \min_{R \in \mathcal{R}_{s,d}^{\text{TF}}(t)} w_t(R). \quad (4.8)$$

The metric $w_t(R)$ considers fidelity degradation, e-link age, coherence-budget risk, regeneration cost, and structural routing cost.

4.2.5 Controlled Regeneration Overhead

Frequent regeneration may improve temporal freshness but can consume quantum-memory capacity, elementary-link generation resources, classical coordination time, and Bell-state measurement resources.

The proposed method therefore limits the number of regeneration operations in each time slot:

$$\sum_{(i,j) \in E_A(t)} r_{ij}(t) \leq R_{\max}. \quad (4.9)$$

Regeneration is prioritized for artificial e-links with high temporal risk and high structural importance.

4.2.6 Reduced Request Rejection

The routing method aims to reduce entanglement-request rejection caused by stale links, insufficient coherence budget, or fidelity-threshold violations.

Let $N_{\text{reject}}(t)$ denote the number of rejected requests at time t . The long-term request-rejection ratio is

$$P_{\text{reject}} = \frac{\sum_{t=1}^T N_{\text{reject}}(t)}{\sum_{t=1}^T |\mathcal{D}(t)|}. \quad (4.10)$$

The proposed method seeks to minimize this ratio while respecting memory and regeneration constraints.

4.2.7 Robustness to Dynamic E-Link Conditions

The artificial overlay may change because of successful generation, consumption, expiration, memory failure, or regeneration.

The method adapts route selection whenever the state of a relevant e-link changes. However, unnecessary route recomputation should be avoided.

A route update is triggered only when at least one of the following conditions occurs:

- a selected e-link becomes critical or expired;
- the predicted end-to-end fidelity falls below the required threshold;
- a lower-cost temporally feasible route becomes available;
- an anchor or tracked e-link is regenerated; or
- the current route cannot satisfy the completion-delay constraint.

4.2.8 Scalable Decision Making

The proposed method should remain computationally practical as the number of ESPs increases.

Instead of solving the complete joint optimization problem at every time slot, the framework separates the decision process into

$$\text{State Update} \rightarrow \text{Candidate Construction} \rightarrow \text{Feasibility Filtering} \rightarrow \text{Cost Based Selection}. \quad (4.11)$$

This decomposition reduces the number of candidate routes requiring detailed temporal evaluation.

The overall design objective can be summarized as

$$\min \left[\lambda_1 \bar{w} + \lambda_2 P_{\text{reject}} + \lambda_3 C_{\text{regen}} + \lambda_4 \overline{ES} \right], \quad (4.12)$$

subject to

$$\hat{F}_{s,d}^{\text{end}} \geq F_{\min}^{\text{end}}, \quad (4.13)$$

$$B_{ij}(t) \geq \delta_{\text{safe}}, \quad (4.14)$$

$$m_i(t) \leq M_i, \quad (4.15)$$

$$|\mathcal{T}_i(t)| \leq c_{\text{tab}} \sqrt{n_e} \log n_e, \quad (4.16)$$

$$\sum_{(i,j)} r_{ij}(t) \leq R_{\max}. \quad (4.17)$$

Here, $\bar{w}$ is the average selected-route cost, C_{regen} is the total regeneration overhead, and $\overline{ES}$ is the average temporal entangling stretch.

4.3 Overview of the Proposed Framework

The proposed framework integrates compact routing, temporal e-link monitoring, fidelity prediction, coherence-budget evaluation, and selective overlay maintenance.

The framework operates in discrete time slots. At the beginning of each time slot, the controller updates the state of every stored artificial e-link and constructs the temporally usable overlay.

The complete processing sequence is represented as

$$G_A(t) \rightarrow \mathcal{L}(t) \rightarrow G_U(t) \rightarrow \mathcal{R}_{s,d}^{\text{cand}}(t) \rightarrow \mathcal{R}_{s,d}^{\text{TF}}(t) \rightarrow R_{s,d}^*(t). \quad (4.18)$$

Here,

- $G_A(t)$ is the currently stored artificial overlay;
- $\mathcal{L}(t)$ is the collection of temporal states of artificial e-links;
- $G_U(t)$ is the temporally usable overlay;
- $\mathcal{R}_{s,d}^{\text{cand}}(t)$ is the candidate-route set;
- $\mathcal{R}_{s,d}^{\text{TF}}(t)$ is the temporally feasible route set; and
- $R_{s,d}^*(t)$ is the selected minimum-cost route.

The framework contains two interacting control processes:

- Reactive routing;

- Proactive overlay maintenance.

Reactive routing is executed when an entanglement request arrives. It constructs candidate compact routes, evaluates their temporal conditions, removes infeasible routes, and selects the lowest-cost temporally feasible route.

Proactive overlay maintenance is executed periodically or when a critical state change occurs. It identifies artificial e-links that should be regenerated, replaced, or discarded before they cause routing failure.

The framework receives the following inputs:

$$\mathcal{I}(t) = [G_P, G_A(t), \mathcal{D}(t), \mathcal{T}(t), \mathcal{M}(t), \mathcal{Q}(t)], \quad (4.19)$$

where

- G_P is the physical quantum network;
- $G_A(t)$ is the current artificial overlay;
- $\mathcal{D}(t)$ is the set of active entanglement requests;
- $\mathcal{T}(t)$ is the compact-routing-table state;
- $\mathcal{M}(t)$ is the quantum-memory state; and
- $\mathcal{Q}(t)$ is the observed or predicted network-quality state.

The output of the framework is

$$\mathcal{O}(t) = [\mathcal{Y}(t), \mathcal{R}^{\text{regen}}(t), \mathcal{E}^{\text{drop}}(t), \mathcal{T}(t+1)], \quad (4.20)$$

where

- $\mathcal{Y}(t)$ is the set of selected routes;
- $\mathcal{R}^{\text{regen}}(t)$ is the regeneration-action set;
- $\mathcal{E}^{\text{drop}}(t)$ is the set of discarded e-links; and
- $\mathcal{T}(t+1)$ is the updated compact-routing-table state.

The proposed framework is divided into six functional modules.

4.3.1 Temporal-State Monitoring Module

This module updates the age, effective fidelity, predicted consumption fidelity, remaining coherence budget, and stale-link state of every stored artificial e-link.

Its output is the temporally usable artificial overlay

$$G_U(t) = (V_2, E_U(t)). \quad (4.21)$$

4.3.2 Compact Candidate-Construction Module

This module constructs a limited number of candidate routes using only the routing information permitted by the Partial-Anchor or Full-Anchor compact-routing scheme.

4.3.3 Compact Candidate-Construction Module

The candidate-route set generated by this module is represented as

$$\mathcal{R}_{s,d}^{\text{cand}}(t). \quad (4.22)$$

This restriction preserves compact-routing state requirements and avoids unrestricted global path enumeration.

4.3.4 Temporal-Feasibility Module

This module evaluates each candidate route according to the temporal route-feasibility indicator

$$\chi_{s,d}(R_{s,d}(t)). \quad (4.23)$$

Routes that violate fidelity, coherence-budget, memory, delay, or e-link availability constraints are removed.

The remaining route set is

$$\mathcal{R}_{s,d}^{\text{TF}}(t) \subseteq \mathcal{R}_{s,d}^{\text{cand}}(t). \quad (4.24)$$

4.3.5 Time-Dependent Route-Selection Module

For each temporally feasible route, this module calculates the accumulated time-dependent entangling cost:

$$w_t(R) = \sum_{(i,j) \in R} w_t(i,j). \quad (4.25)$$

The selected route is

$$R_{s,d}^*(t) = \arg \min_{R \in \mathcal{R}_{s,d}^{\text{TF}}(t)} w_t(R). \quad (4.26)$$

4.3.6 Route-Repair Module

When no directly feasible route exists, this module identifies routes that can become feasible through limited regeneration or purification operations.

The route-repair process is performed only when

$$\widehat{D}_{s,d}^{\text{repair}}(t) \leq D_{\max}^{\text{repair}} \quad (4.27)$$

and

$$C_{s,d}^{\text{repair}}(t) \leq C_{\max}^{\text{repair}}. \quad (4.28)$$

Otherwise, the request is delayed or rejected.

4.3.7 Overlay-Maintenance Module

This module assigns regeneration priority to critical artificial e-links according to temporal risk and structural importance.

The maintenance priority is defined as

$$\Pi_{ij}(t) = \beta_1 C_B(i,j,t) + \beta_2 C_F(i,j,t) + \beta_3 I_{ij}^{\text{struct}} + \beta_4 H_{ij}^{\text{use}}(t), \quad (4.29)$$

where $H_{ij}^{\text{use}}(t)$ represents the recent or predicted utilization level of the artificial e-link.

The weights satisfy

$$\sum_{k=1}^4 \beta_k = 1. \quad (4.30)$$

The highest-priority links are regenerated subject to quantum-memory availability, generation-attempt budget, and control-overhead limitations.

The framework is intentionally modular. Different fidelity models, delay predictors, compact-routing structures, or regeneration policies can be integrated without changing the overall processing sequence.

The proposed method is a model-based routing and control heuristic. It does not require a trained machine-learning model. Any prediction component used in the implementation is derived from measured averages, analytical estimates, or simulation parameters.

4.4 Temporal E-Link State Monitoring

The temporal e-link state-monitoring module continuously evaluates the condition of every artificial e-link stored in the quantum memories of the Entanglement Service Providers.

Its purpose is to distinguish between an artificial e-link that is merely present in the overlay and an e-link that remains sufficiently reliable for future routing and entanglement-swapping operations.

Let the set of currently stored artificial e-links be

$$E_A(t). \quad (4.31)$$

For each artificial e-link $(i, j) \in E_A(t)$, the controller maintains the temporal state record

$$\mathcal{L}_{ij}(t) = \left[x_{ij}(t), A_{ij}(t), F_{ij}^{A,0}, F_{ij}^{\text{eff}}(t), T_{2,ij}, \hat{F}_{ij}^{\text{cons}}(t), B_{ij}(t), H_{ij}^{\text{use}}(t), S_{ij}^{\text{state}}(t) \right], \quad (4.32)$$

where

- $x_{ij}(t)$ is the current storage and availability indicator;
- $A_{ij}(t)$ is the current e-link age;
- $F_{ij}^{A,0}$ is the initial fidelity after e-link generation;
- $F_{ij}^{\text{eff}}(t)$ is the current effective fidelity;
- $T_{2,ij}$ is the coherence time of the stored entanglement;
- $\hat{F}_{ij}^{\text{cons}}(t)$ is the predicted fidelity at consumption time;
- $B_{ij}(t)$ is the remaining coherence budget;
- $H_{ij}^{\text{use}}(t)$ is the recent or predicted utilization score; and
- $S_{ij}^{\text{state}}(t)$ is the temporal-state classification.

At the end of each time slot, the age of an active e-link is updated as

$$A_{ij}(t+1) = A_{ij}(t) + 1, \quad (4.33)$$

provided that the link remains stored and is neither consumed nor regenerated.

When regeneration succeeds, the age is reset:

$$A_{ij}(t+1) = 0. \quad (4.34)$$

The current effective fidelity is estimated using the selected decoherence model:

$$F_{ij}^{\text{eff}}(t) = f_{\text{dec}} \left(F_{ij}^{\text{A},0}, A_{ij}(t), T_{2,ij} \right). \quad (4.35)$$

The controller also estimates the additional delay before the link is expected to be consumed:

$$\widehat{D}_{ij}^{\text{use}}(t). \quad (4.36)$$

This delay may be estimated from recent queueing observations, route depth, expected swapping order, and classical coordination delay.

The predicted fidelity at consumption time is calculated as

$$\widehat{F}_{ij}^{\text{cons}}(t) = f_{\text{dec}} \left(F_{ij}^{\text{eff}}(t), \widehat{D}_{ij}^{\text{use}}(t), T_{2,ij} \right). \quad (4.37)$$

The remaining coherence budget is updated according to

$$B_{ij}(t) = T_{2,ij} - A_{ij}(t)\Delta - \widehat{D}_{ij}^{\text{use}}(t). \quad (4.38)$$

4.4.1 Temporal-State Classification and Link Availability

To reduce sensitivity to short-term fluctuations, the utilization score is updated using an exponential moving average:

$$H_{ij}^{\text{use}}(t) = \gamma_H H_{ij}^{\text{use}}(t-1) + (1 - \gamma_H) I_{ij}^{\text{used}}(t), \quad (4.39)$$

where

$$0 \leq \gamma_H < 1 \quad (4.40)$$

and

$$I_{ij}^{\text{used}}(t) = \begin{cases} 1, & \text{if the e-link is selected or consumed at time } t, \\ 0, & \text{otherwise.} \end{cases} \quad (4.41)$$

The temporal state of an artificial e-link is classified according to predicted consumption fidelity and remaining coherence budget.

A link is classified as safe when

$$B_{ij}(t) \geq B_{\text{high}} \quad (4.42)$$

and

$$\hat{F}_{ij}^{\text{cons}}(t) \geq F_{\text{high}}. \quad (4.43)$$

A link is classified as warning when it remains usable but approaches temporal degradation:

$$B_{\text{low}} \leq B_{ij}(t) < B_{\text{high}} \quad (4.44)$$

or

$$F_{\text{min}}^{\text{link}} \leq \hat{F}_{ij}^{\text{cons}}(t) < F_{\text{high}}. \quad (4.45)$$

A link is classified as critical when it is still potentially recoverable but has a high risk of becoming unusable:

$$0 < B_{ij}(t) < B_{\text{low}} \quad (4.46)$$

or

$$\hat{F}_{ij}^{\text{cons}}(t) \approx F_{\text{min}}^{\text{link}}. \quad (4.47)$$

Critical links are not directly preferred for route selection and may trigger proactive regeneration or purification.

A link is classified as expired when

$$B_{ij}(t) \leq 0 \quad (4.48)$$

or

$$x_{ij}(t) = 0. \quad (4.49)$$

The resulting temporal-state classification is

$$S_{ij}^{\text{state}}(t) = \begin{cases} \text{Safe,} & \text{if high-quality conditions are satisfied,} \\ \text{Warning,} & \text{if the link remains usable with degradation risk,} \\ \text{Critical,} & \text{if recovery actions may restore usability,} \\ \text{Expired,} & \text{if the link is unusable.} \end{cases} \quad (4.50)$$

The direct-use indicator is defined as

$$u_{ij}^{\text{direct}}(t) = \begin{cases} 1, & S_{ij}^{\text{state}}(t) \in \{\text{Safe, Warning}\}, \\ 0, & \text{otherwise.} \end{cases} \quad (4.51)$$

The regeneration-candidate indicator is defined as

$$u_{ij}^{\text{regen}}(t) = \begin{cases} 1, & S_{ij}^{\text{state}}(t) = \text{Critical}, \\ 0, & \text{otherwise.} \end{cases} \quad (4.52)$$

Expired links are removed from route construction and may be discarded from memory:

$$u_{ij}^{\text{drop}}(t) = \mathbf{1} [S_{ij}^{\text{state}}(t) = \text{Expired}]. \quad (4.53)$$

The temporally usable edge set is therefore

$$E_U(t) = \{(i, j) \in E_A(t) : u_{ij}^{\text{direct}}(t) = 1\}. \quad (4.54)$$

The temporal-state monitor operates periodically. Let

$$\tau_{\text{mon}} \quad (4.55)$$

denote the monitoring interval.

A shorter monitoring interval improves responsiveness but increases control and state-update overhead. A longer interval reduces overhead but may allow rapidly degrading links to remain incorrectly classified.

Therefore, the monitoring interval is evaluated experimentally under different coherence times, request rates, and overlay sizes.

The output of this module is a synchronized temporal-state table used by candidate-route construction, temporal-feasibility filtering, route selection, and proactive regeneration.

4.5 Candidate Route Construction

The candidate-route construction module generates a limited set of compact-routing paths between a source ESP and a destination ESP. Its purpose is to avoid unrestricted enumeration of all possible paths while preserving the structural properties of the underlying Partial-Anchor or Full-Anchor routing scheme.

For an entanglement request between source ESP v_s and destination ESP v_d , the candidate-route set at time t is denoted by

$$\mathcal{R}_{s,d}^{\text{cand}}(t). \quad (4.56)$$

Candidate routes are constructed over the currently available compact-overlay information. The module uses the temporally usable overlay

$$G_U(t) = (V_2, E_U(t)), \quad (4.57)$$

where $E_U(t)$ contains artificial e-links classified as safe or warning.

The candidate-generation process first checks whether the source and destination are directly connected by a usable artificial e-link. If

$$(v_s, v_d) \in E_U(t), \quad (4.58)$$

then the direct route

$$R_{s,d}^{\text{dir}}(t) = (v_s, v_d) \quad (4.59)$$

is inserted into the candidate set.

If no direct usable e-link exists, the controller searches through local e-neighbors, anchor ESPs, and tracked long-range links according to the selected compact-routing architecture.

Let

$$\mathcal{N}_U(v_i, t) \quad (4.60)$$

denote the usable e-neighborhood of ESP v_i :

$$\mathcal{N}_U(v_i, t) = \{v_j \in V_2 : (v_i, v_j) \in E_U(t)\}. \quad (4.61)$$

A one-intermediate-node candidate route has the form

$$R_{s,d}^{(1)}(t) = (v_s, v_k, v_d), \quad (4.62)$$

where

$$v_k \in \mathcal{N}_U(v_s, t) \cap \mathcal{N}_U(v_d, t). \quad (4.63)$$

For longer compact routes, the controller follows the routing structure of the selected overlay scheme. In the Partial-Anchor scheme, route construction is restricted by local e-neighborhood information and selected anchor links. In the Full-Anchor scheme, additional tracked long-range links are considered during candidate generation.

4.5.1 Candidate Construction in the Partial-Anchor Scheme

In the Partial-Anchor Scheme, each ESP maintains local e-neighbor information, while a subset of ESPs acts as anchors.

Let

$$\mathcal{T} \subseteq V_2 \quad (4.64)$$

denote the anchor set.

The usable source-anchor set is

$$\mathcal{T}_s^U(t) = \{v_a \in \mathcal{T} : v_a \in \mathcal{N}_U(v_s, t)\}. \quad (4.65)$$

Similarly, the usable destination-anchor set is

$$\mathcal{T}_d^U(t) = \{v_b \in \mathcal{T} : v_b \in \mathcal{N}_U(v_d, t)\}. \quad (4.66)$$

A Partial-Anchor candidate route can be written as

$$R_{s,d}^{\text{PA}}(t) = (v_s, v_a, \dots, v_b, v_d), \quad (4.67)$$

where

$$v_a \in \mathcal{T}_s^U(t) \quad (4.68)$$

and

$$v_b \in \mathcal{T}_d^U(t). \quad (4.69)$$

The internal anchor segment is obtained from the currently usable anchor overlay.

Let

$$G_{\text{anchor}}^U(t) \quad (4.70)$$

denote the usable subgraph formed by anchor ESPs and usable anchor e-links.

A bounded number of low-cost anchor paths is generated between v_a and v_b .

4.5.2 Candidate Construction in the Full-Anchor Scheme

In the Full-Anchor Scheme, every ESP may maintain local e-links and selected tracked long-range e-links.

Let

$$E_{\text{tracked}}^{\text{U}}(t) \quad (4.71)$$

denote the set of currently usable tracked long-range artificial e-links.

The Full-Anchor candidate set is constructed using

$$E_{\text{local}}^{\text{U}}(t) \cup E_{\text{tracked}}^{\text{U}}(t). \quad (4.72)$$

A candidate route is constructed by combining local source segments, tracked long-range segments, and local destination segments.

A general candidate route is represented by

$$R_{s,d}^{\text{FA}}(t) = (v_s, v_{r_1}, \dots, v_{r_h}, v_d), \quad (4.73)$$

where every consecutive node pair corresponds to a usable local or tracked artificial e-link.

4.5.3 Bounded Candidate-Set Size

To control computational overhead, the controller retains at most

$$K_{\text{max}} \quad (4.74)$$

candidate routes for each source–destination request.

Therefore,

$$|\mathcal{R}_{s,d}^{\text{cand}}(t)| \leq K_{\text{max}}. \quad (4.75)$$

Candidate routes are initially ranked using a lightweight preliminary score

$$\tilde{w}_t(R) = \sum_{(i,j) \in R} \tilde{w}_t(i,j), \quad (4.76)$$

where $\tilde{w}_t(i,j)$ includes only readily available local information such as hop count, current fidelity, normalized age, and direct-use state.

A possible preliminary link score is

$$\tilde{w}_t(i, j) = \eta_h + \eta_F \left(1 - F_{ij}^{\text{eff}}(t)\right) + \eta_A \frac{A_{ij}(t)}{A_{ij}^{\text{max}}}. \quad (4.77)$$

The coefficients satisfy

$$\eta_h + \eta_F + \eta_A = 1. \quad (4.78)$$

This preliminary score is used only for candidate pruning. Final route selection uses the complete time-dependent entangling-cost metric.

To avoid routing loops, every candidate route must contain unique nodes:

$$v_m \neq v_n, \quad \forall m \neq n. \quad (4.79)$$

A maximum overlay-hop constraint is also applied:

$$|R| - 1 \leq H_{\text{max}}, \quad (4.80)$$

where H_{max} is the maximum number of artificial e-links allowed in a candidate route.

The final candidate set is

$$\mathcal{R}_{s,d}^{\text{cand}}(t) = \text{TopK}_{K_{\text{max}}} [\tilde{w}_t(R)], \quad (4.81)$$

subject to compact-routing, loop-free, direct-use, and maximum-hop constraints.

The output of this module is a small set of structurally valid candidate routes. These routes are passed to the temporal-feasibility module, where predicted consumption fidelity, remaining coherence budget, delay, memory availability, and end-to-end fidelity are evaluated in detail.

4.5.4 Availability Test

The first test verifies that every required artificial e-link is currently available for direct use:

$$u_{ij}^{\text{direct}}(t) = 1, \quad \forall (i, j) \in E(R). \quad (4.82)$$

The route-level availability indicator is

$$\chi_{\text{avail}}(R, t) = \prod_{(i,j) \in E(R)} u_{ij}^{\text{direct}}(t). \quad (4.83)$$

If

$$\chi_{\text{avail}}(R, t) = 0, \quad (4.84)$$

the route is removed from the directly feasible set.

4.5.5 Predicted Consumption-Time Calculation

Therefore, the temporal feasibility evaluation is performed individually for each e-link based on its predicted consumption time. Therefore, the controller estimates the consumption time of each e-link according to the expected swapping schedule.

Let

$$\widehat{\tau}_{ij}^{\text{cons}}(R, t) \quad (4.85)$$

denote the estimated additional time before e-link (i, j) is consumed as part of route R .

A general estimate is

$$\widehat{\tau}_{ij}^{\text{cons}}(R, t) = \widehat{D}_{\text{queue}}(R, t) + \widehat{D}_{\text{coord}}(R, t) + \widehat{D}_{ij}^{\text{sched}}(R, t), \quad (4.86)$$

where

- $\widehat{D}_{\text{queue}}(R, t)$ is the expected request waiting delay;
- $\widehat{D}_{\text{coord}}(R, t)$ is the classical coordination delay; and
- $\widehat{D}_{ij}^{\text{sched}}(R, t)$ is the delay introduced by the selected swapping schedule.

For a simple sequential swapping schedule, the estimated scheduling delay can be approximated as

$$\widehat{D}_{ij}^{\text{sched}}(R, t) = \ell_{ij}(R) D_{\text{swap}}, \quad (4.87)$$

where $\ell_{ij}(R)$ is the number of swapping stages completed before e-link (i, j) is consumed.

For a parallel or nested swapping schedule, this value may be smaller and is determined by the swapping depth of the route.

4.5.6 Link-Level Fidelity Test

The predicted fidelity of e-link (i, j) at its estimated consumption time is

$$\hat{F}_{ij}^{\text{cons}}(R, t) = f_{\text{dec}} \left(F_{ij}^{\text{eff}}(t), \hat{\tau}_{ij}^{\text{cons}}(R, t), T_{2,ij} \right). \quad (4.88)$$

The route passes the link-level fidelity test only if

$$\hat{F}_{ij}^{\text{cons}}(R, t) \geq F_{\min}^{\text{link}}, \quad \forall (i, j) \in E(R). \quad (4.89)$$

The corresponding indicator is

$$\chi_{\text{linkF}}(R, t) = \prod_{(i,j) \in E(R)} \mathbf{1} \left[\hat{F}_{ij}^{\text{cons}}(R, t) \geq F_{\min}^{\text{link}} \right]. \quad (4.90)$$

4.5.7 End-to-End Fidelity Test

Using the predicted consumption-time fidelities, the controller estimates the end-to-end route fidelity.

Under the Werner-state approximation, the predicted link parameter is

$$\hat{p}_{ij}^W(R, t) = \frac{4\hat{F}_{ij}^{\text{cons}}(R, t) - 1}{3}. \quad (4.91)$$

The predicted end-to-end Werner parameter is

$$\hat{p}_{s,d}^{W,\text{end}}(R, t) = \left(\prod_{(i,j) \in E(R)} \hat{p}_{ij}^W(R, t) \right) \left(\prod_{v_r \in \text{Int}(R)} \lambda_r \right), \quad (4.92)$$

where $\text{Int}(R)$ denotes the set of intermediate ESPs participating in entanglement swapping along route R .

The predicted end-to-end fidelity is

$$\hat{F}_{s,d}^{\text{end}}(R, t) = \frac{1 + 3\hat{p}_{s,d}^{W,\text{end}}(R, t)}{4}. \quad (4.93)$$

The route passes the end-to-end fidelity test when

$$\hat{F}_{s,d}^{\text{end}}(R, t) \geq F_{\min}^{\text{end}}. \quad (4.94)$$

The associated indicator is

$$\chi_{\text{endF}}(R, t) = \mathbf{1} \left[\hat{F}_{s,d}^{\text{end}}(R, t) \geq F_{\min}^{\text{end}} \right]. \quad (4.95)$$

4.5.8 Route-Delay Test

The predicted route-completion delay is denoted by

$$\widehat{D}_{s,d}^{\text{route}}(R, t). \quad (4.96)$$

A general decomposition is

$$\widehat{D}_{s,d}^{\text{route}}(R, t) = \widehat{D}_{\text{queue}}(R, t) + \widehat{D}_{\text{coord}}(R, t) + \widehat{D}_{\text{pur}}(R, t) + \widehat{D}_{\text{swap}}(R, t). \quad (4.97)$$

The delay estimation is route-dependent because different candidate routes may contain different numbers of swapping operations, purification requirements, and queueing conditions.

The route satisfies the delay constraint only if

$$\widehat{D}_{s,d}^{\text{route}}(R, t) \leq D_{\text{max}}. \quad (4.98)$$

The route-delay indicator is

$$\chi_{\text{delay}}(R, t) = \mathbf{1} \left[\widehat{D}_{s,d}^{\text{route}}(R, t) \leq D_{\text{max}} \right]. \quad (4.99)$$

4.5.9 Memory and E-Bit Availability Test

The number of stored e-bits on each artificial e-link must be sufficient to satisfy the entanglement request.

For a single-pair request, the link-level resource condition is

$$n_{ij}^{\text{ebit}}(t) \geq 1, \quad \forall (i, j) \in E(R). \quad (4.100)$$

For a request requiring $b_{s,d}$ entangled pairs, where $b_{s,d}$ denotes the required number of entangled pairs between source and destination, the condition becomes

$$n_{ij}^{\text{ebit}}(t) \geq b_{s,d}, \quad \forall (i, j) \in E(R). \quad (4.101)$$

The route-level resource indicator is

$$\chi_{\text{mem}}(R, t) = \prod_{(i,j) \in E(R)} \mathbf{1} \left[n_{ij}^{\text{ebit}}(t) \geq b_{s,d} \right]. \quad (4.102)$$

4.5.10 Overall Temporal-Feasibility Decision

The overall temporal-feasibility indicator is defined as

$$\chi_{s,d}(R, t) = \chi_{\text{avail}}(R, t) \chi_{\text{linkF}}(R, t) \chi_{\text{coh}}(R, t) \chi_{\text{endF}}(R, t) \chi_{\text{delay}}(R, t) \chi_{\text{mem}}(R, t). \quad (4.103)$$

A candidate route is directly feasible when

$$\chi_{s,d}(R, t) = 1. \quad (4.104)$$

The temporally feasible route set is therefore

$$\mathcal{R}_{s,d}^{\text{TF}}(t) = \{R \in \mathcal{R}_{s,d}^{\text{cand}}(t) : \chi_{s,d}(R, t) = 1\}. \quad (4.105)$$

Candidate routes that fail the direct-feasibility test are not immediately discarded if their infeasibility can be corrected through limited regeneration or repair operations.

Such routes are transferred to the route-repair set

$$\mathcal{R}_{s,d}^{\text{repair}}(t). \quad (4.106)$$

A route enters this set only if its predicted repair delay and repair cost satisfy

$$\widehat{D}_{s,d}^{\text{repair}}(R, t) \leq D_{\text{max}}^{\text{repair}} \quad (4.107)$$

and

$$C_{s,d}^{\text{repair}}(R, t) \leq C_{\text{max}}^{\text{repair}}. \quad (4.108)$$

This two-stage filtering process distinguishes among immediately feasible, repairable, and infeasible routes. It prevents the route-selection module from assigning low cost to a path that is topologically available but temporally incapable of satisfying the entanglement request.

4.6 Time-Dependent Route Selection

The time-dependent route-selection module chooses the best route from the set of temporally feasible candidates.

For an entanglement request between source ESP v_s and destination ESP v_d , the feasible candidate set is

$$\mathcal{R}_{s,d}^{\text{TF}}(t). \quad (4.109)$$

For each candidate route

$$R \in \mathcal{R}_{s,d}^{\text{TF}}(t), \quad (4.110)$$

the controller evaluates the complete time-dependent route cost.

The route cost is defined as

$$w_t(R) = \sum_{(i,j) \in E(R)} w_t(i,j) + C_{\text{route}}^{\text{swap}}(R,t) + C_{\text{route}}^{\text{delay}}(R,t), \quad (4.111)$$

where

- $w_t(i,j)$ is the time-dependent cost of artificial e-link (i,j) ;
- $C_{\text{route}}^{\text{swap}}(R,t)$ is the route-level swapping cost; and
- $C_{\text{route}}^{\text{delay}}(R,t)$ is the normalized completion-delay cost.

The link-level time-dependent cost is

$$w_t(i,j) = \alpha_F C_F(i,j,R,t) + \alpha_A C_A(i,j,t) + \alpha_B C_B(i,j,R,t) + \alpha_R C_R(i,j,t) + \alpha_S C_S(i,j). \quad (4.112)$$

The coefficients satisfy

$$\alpha_F + \alpha_A + \alpha_B + \alpha_R + \alpha_S = 1. \quad (4.113)$$

The fidelity cost is defined as

$$C_F(i,j,R,t) = 1 - \hat{F}_{ij}^{\text{cons}}(R,t). \quad (4.114)$$

This term penalizes e-links that are expected to have low fidelity when they are consumed.

The age cost is

$$C_A(i,j,t) = \min \left\{ \frac{A_{ij}(t)}{A_{ij}^{\text{max}}}, 1 \right\}. \quad (4.115)$$

The normalized coherence-budget cost is defined as

$$C_B(i,j,R,t) = \frac{1}{\tilde{B}_{ij}(R,t) + \varepsilon}, \quad (4.116)$$

where

$$\tilde{B}_{ij}(R,t) = \frac{B_{ij}(R,t)}{T_{2,ij}}. \quad (4.117)$$

Since only temporally feasible routes reach this module, links with non-positive coherence budget are already removed before route selection:

$$\tilde{B}_{ij}(R, t) > 0. \quad (4.118)$$

The normalized regeneration-risk cost is

$$C_R(i, j, t) = \lambda_D \frac{D_{ij}^{\text{regen}}}{D_{\max}^{\text{regen}}} + \lambda_C \frac{C_{ij}^{\text{regen}}}{C_{\max}^{\text{regen}}}, \quad (4.119)$$

where

$$\lambda_D + \lambda_C = 1. \quad (4.120)$$

The structural routing cost is represented by

$$C_S(i, j). \quad (4.121)$$

This component distinguishes local e-neighbor links, anchor links, and tracked long-range links according to their resource and maintenance requirements.

4.6.1 Route-Level Cost Components

The swapping cost of route R is defined as

$$C_{\text{route}}^{\text{swap}}(R, t) = \alpha_Q \sum_{v_r \in \text{Int}(R)} (1 - q_r), \quad (4.122)$$

where q_r is the swapping-success probability at intermediate ESP v_r .

The normalized route-delay cost is

$$C_{\text{route}}^{\text{delay}}(R, t) = \alpha_D \frac{\widehat{D}_{s,d}^{\text{route}}(R, t)}{D_{\max}}. \quad (4.123)$$

The selected directly feasible route is

$$R_{s,d}^*(t) = \arg \min_{R \in \mathcal{R}_{s,d}^{\text{TF}}(t)} w_t(R). \quad (4.124)$$

When multiple routes have the same minimum cost, the controller applies the following tie-breaking order:

1. select the route with the highest predicted end-to-end fidelity;
2. select the route with the largest minimum coherence budget;

3. select the route with the smallest predicted completion delay;
4. select the route with the fewest artificial e-links.

The minimum route-level coherence budget is

$$B_{\min}(R, t) = \min_{(i,j) \in E(R)} B_{ij}(R, t). \quad (4.125)$$

The route-selection decision can therefore be represented lexicographically as

$$\min \left[w_t(R), -\widehat{F}_{s,d}^{\text{end}}(R, t), -B_{\min}(R, t), \widehat{D}_{s,d}^{\text{route}}(R, t), |E(R)| \right]. \quad (4.126)$$

After selecting a route, the required e-bits are temporarily reserved before swapping begins.

For a request requiring $b_{s,d}$ entangled pairs, the reservation variable is

$$z_{ij}^{s,d}(t) = \begin{cases} b_{s,d}, & \text{if } (i, j) \text{ belongs to the selected route,} \\ 0, & \text{otherwise.} \end{cases} \quad (4.127)$$

The reservation constraint is

$$\sum_{(s,d) \in \mathcal{D}(t)} z_{ij}^{s,d}(t) \leq n_{ij}^{\text{ebit}}(t). \quad (4.128)$$

This reservation prevents the same stored e-bit from being assigned to multiple requests.

After successful end-to-end swapping, the consumed e-bits are removed:

$$n_{ij}^{\text{ebit}}(t+1) = n_{ij}^{\text{ebit}}(t) - z_{ij}^{s,d}(t). \quad (4.129)$$

If a swapping operation fails, all e-bits already consumed by the failed route are treated as lost.

The route outcome is represented by

$$Y_{s,d}(t) = \begin{cases} 1, & \text{if end-to-end entanglement is established,} \\ 0, & \text{otherwise.} \end{cases} \quad (4.130)$$

The realized route success probability is estimated as

$$P_{\text{succ}}(R, t) = \left(\prod_{(i,j) \in E(R)} p_{ij}^{\text{avail}}(t) \right) \left(\prod_{v_r \in \text{Int}(R)} q_r \right), \quad (4.131)$$

where $p_{ij}^{\text{avail}}(t)$ represents the probability that artificial e-link (i, j) remains usable until its scheduled consumption time.

A reliability-aware extension of the route cost is

$$w_t^{\text{rel}}(R) = w_t(R) + \alpha_P [-\log P_{\text{succ}}(R, t)]. \quad (4.132)$$

This logarithmic transformation converts the multiplicative success probability into an additive routing penalty.

The route selected under the reliability-aware metric is

$$R_{s,d}^{*,\text{rel}}(t) = \arg \min_{R \in \mathcal{R}_{s,d}^{\text{TF}}(t)} w_t^{\text{rel}}(R). \quad (4.133)$$

The reliability-aware metric provides an optional extension of the basic temporal entangling-cost metric. Its effectiveness will be evaluated through comparative experiments.

When

$$\mathcal{R}_{s,d}^{\text{TF}}(t) = \emptyset, \quad (4.134)$$

the module does not select a directly feasible route.

Instead, it forwards the request and the repairable candidate set

$$\mathcal{R}_{s,d}^{\text{repair}}(t) \quad (4.135)$$

to the proactive regeneration and route-repair process described in the following section.

The normalized fidelity risk is defined as

$$\hat{C}_F(i, j, t) = \min \left\{ \frac{F_{\text{high}} - \hat{F}_{ij}^{\text{cons}}(t)}{F_{\text{high}} - F_{\text{min}}^{\text{link}} + \varepsilon}, 1 \right\}. \quad (4.136)$$

The predicted demand relevance may be estimated from the number of active or expected routes that can use the e-link:

$$Q_{ij}^{\text{demand}}(t) = \frac{N_{ij}^{\text{demand}}(t)}{\max_{(u,v)} N_{uv}^{\text{demand}}(t) + \varepsilon}. \quad (4.137)$$

The normalized regeneration cost is

$$\hat{C}_R(i, j, t) = \lambda_D \frac{D_{ij}^{\text{regen}}}{D_{\text{max}}^{\text{regen}}} + \lambda_C \frac{C_{ij}^{\text{regen}}}{C_{\text{max}}^{\text{regen}}}, \quad (4.138)$$

where

$$\lambda_D + \lambda_C = 1. \quad (4.139)$$

A link becomes a proactive-regeneration candidate when

$$\Pi_{ij}^{\text{regen}}(t) \geq \Pi_{\min} \quad (4.140)$$

and sufficient regeneration resources are available.

The candidate set is

$$\mathcal{E}_{\text{regen}}^{\text{cand}}(t) = \{(i, j) \in \mathcal{E}_{\text{crit}}(t) : \Pi_{ij}^{\text{regen}}(t) \geq \Pi_{\min}\}. \quad (4.141)$$

The regeneration-decision variable is

$$r_{ij}(t) \in \{0, 1\}. \quad (4.142)$$

It is defined as

$$r_{ij}(t) = \begin{cases} 1, & \text{if e-link } (i, j) \text{ is selected for regeneration,} \\ 0, & \text{otherwise.} \end{cases} \quad (4.143)$$

The controller selects at most $R_{\max}$ highest-priority regeneration candidates:

$$\mathcal{E}_{\text{regen}}^*(t) = \text{TopK}(\mathcal{E}_{\text{regen}}^{\text{cand}}(t), R_{\max}). \quad (4.144)$$

The selected regeneration actions must satisfy the regeneration budget constraint:

$$|\mathcal{E}_{\text{regen}}^*(t)| \leq R_{\max}. \quad (4.145)$$

Let $m_i(t)$ denote the currently occupied quantum-memory capacity at ESP v_i . The regeneration decision must satisfy

$$m_i(t) + \sum_{j:(i,j) \in \mathcal{E}_{\text{regen}}^*(t)} r_{ij}(t) \leq M_i, \quad \forall v_i \in V_2. \quad (4.146)$$

They must also satisfy the physical-generation capacity constraint:

$$\sum_{(i,j) \in E_A(t)} r_{ij}(t) g_{ij}^{\text{req}} \leq G_{\max}, \quad (4.147)$$

where g_{ij}^{req} represents the expected number of elementary-link generation attempts required for regenerating artificial e-link (i, j) , and $G_{\max}$ denotes the available generation-attempt budget.

For an artificial e-link generated through physical path P_{ij}^{gen} , the expected regeneration delay is

$$D_{ij}^{\text{regen}} = D^{\text{phys}}(P_{ij}^{\text{gen}}) + D_{ij}^{\text{swap}} + D_{ij}^{\text{coord}}. \quad (4.148)$$

When purification is required, the regeneration delay becomes

$$D_{ij}^{\text{regen,pur}} = D_{ij}^{\text{regen}} + D_{ij}^{\text{pur}}. \quad (4.149)$$

The regeneration process may follow either a replace-after-success strategy or a discard-before-generation strategy.

Under the replace-after-success strategy, the old artificial e-link remains available until the new e-link is successfully generated. After successful regeneration,

$$F_{ij}^{\text{eff}}(t+1) = F_{ij}^{\text{A},0}, \quad (4.150)$$

$$A_{ij}(t+1) = 0, \quad (4.151)$$

and

$$B_{ij}(t+1) = T_{2,ij} - \widehat{D}_{ij}^{\text{use}}(t+1). \quad (4.152)$$

This strategy reduces temporary disconnection risk but requires additional memory capacity because both old and newly generated e-links may coexist during the transition period.

Under the discard-before-generation strategy, the old e-link is released before regeneration. This reduces memory consumption but temporarily removes the link from the usable overlay.

The replacement-policy variable is defined as

$$\rho_{ij}^{\text{replace}} \in \{0, 1\}, \quad (4.153)$$

where

$$\rho_{ij}^{\text{replace}} = 1 \quad (4.154)$$

indicates that the replace-after-success strategy is selected.

The policy is determined according to available memory capacity:

$$\rho_{ij}^{\text{replace}} = \mathbf{1} [M_i - m_i(t) \geq 1 \wedge M_j - m_j(t) \geq 1]. \quad (4.155)$$

Regeneration is considered successful only when the required elementary-link generation, entanglement swapping, and coordination operations are completed successfully.

Let

$$Z_{ij}^{\text{regen}}(t) \in \{0, 1\} \quad (4.156)$$

denote the regeneration outcome of artificial e-link (i, j) at time slot t . The regeneration success probability is approximated as

$$P[Z_{ij}^{\text{regen}}(t) = 1] = P_{ij}^{\text{regen}} = P_{\text{succ}}(P_{ij}^{\text{gen}}). \quad (4.157)$$

If regeneration fails, the e-link remains in its previous state under the replace-after-success strategy. Under the discard-before-generation strategy, the e-link remains unavailable until a subsequent regeneration attempt succeeds.

Reactive route repair uses the same regeneration mechanism but assigns additional priority according to the blocked request.

For a repairable route R , the repair priority of a critical e-link is increased by

$$\Pi_{ij}^{\text{repair}}(R, t) = \Pi_{ij}^{\text{regen}}(t) + \beta_{\text{req}} I_{ij}^R, \quad (4.158)$$

where

$$I_{ij}^R = \mathbf{1}[(i, j) \in \mathcal{E}_{\text{crit}}(R)]. \quad (4.159)$$

This request-specific term provides higher priority to links whose regeneration can restore a currently infeasible route.

A regeneration action is approved only when its expected benefit exceeds its expected cost.

The expected regeneration utility is defined as

$$U_{ij}^{\text{regen}}(t) = P_{ij}^{\text{regen}} G_{ij}^{\text{future}}(t) - \kappa_D D_{ij}^{\text{regen}} - \kappa_C C_{ij}^{\text{regen}}, \quad (4.160)$$

where $G_{ij}^{\text{future}}(t)$ represents the predicted future routing benefit obtained by refreshing artificial e-link (i, j) .

Regeneration is executed only if

$$U_{ij}^{\text{regen}}(t) > 0. \quad (4.161)$$

This condition prevents unnecessary regeneration of low-impact e-links whose expected routing benefit does not justify the required generation resources and delay.

The proposed proactive-regeneration mechanism therefore balances temporal freshness, fidelity preservation, structural importance, future demand, memory availability, physical-generation capacity, and regeneration overhead.

4.7 Overlay Maintenance Strategy

The overlay-maintenance strategy updates the artificial entanglement overlay according to temporal degradation, routing demand, memory availability, and compact-routing constraints.

Its purpose is not to keep every artificial e-link permanently fresh. Instead, the strategy maintains a bounded set of useful e-links whose expected routing benefit justifies their generation and storage cost.

At time slot t , the stored artificial overlay is

$$G_A(t) = (V_2, E_A(t)). \quad (4.162)$$

The overlay is updated through four principal actions:

$$\text{Retain,} \quad \text{Regenerate,} \quad \text{Replace,} \quad \text{Discard.} \quad (4.163)$$

For each artificial e-link (i, j) , the maintenance action is denoted by

$$a_{ij}^{\text{maint}}(t) \in \{\text{retain, regenerate, replace, discard}\}. \quad (4.164)$$

4.7.1 Retention Decision

An e-link is retained without regeneration when it remains temporally reliable and sufficiently useful.

The retention indicator is

$$k_{ij}^{\text{retain}}(t) = \mathbf{1} [S_{ij}^{\text{state}}(t) \in \{\text{Safe, Warning}\}]. \quad (4.165)$$

A warning-state link may be retained only if its predicted consumption fidelity and remaining coherence budget satisfy the minimum operational margins:

$$\hat{F}_{ij}^{\text{cons}}(t) \geq F_{\min}^{\text{link}} \quad (4.166)$$

and

$$B_{ij}(t) \geq \delta_{\text{safe}}. \quad (4.167)$$

4.7.2 Regeneration Decision

A critical e-link is considered for regeneration when its expected future routing benefit is high and its regeneration cost is acceptable.

The regeneration decision follows the previously defined variable:

$$r_{ij}(t) = \mathbf{1} \left[\Pi_{ij}^{\text{regen}}(t) \geq \Pi_{\min} \right], \quad (4.168)$$

subject to memory, physical-generation, and per-slot regeneration constraints.

The total number of regeneration actions satisfies

$$\sum_{(i,j) \in E_A(t)} r_{ij}(t) \leq R_{\max}. \quad (4.169)$$

4.7.3 Replacement Decision

Replacement changes the endpoint pair or physical realization of an artificial e-link when repeated regeneration becomes inefficient or when the existing link has limited routing value.

Let

$$h_{ij}^{\text{fail}}(t) \quad (4.170)$$

denote the recent number of failed regeneration attempts for e-link (i, j) . The link becomes a replacement candidate when

$$h_{ij}^{\text{fail}}(t) \geq H_{\max}^{\text{fail}}, \quad (4.171)$$

or when

$$U_{ij}^{\text{regen}}(t) \leq 0. \quad (4.172)$$

A replacement candidate may also be identified when its structural importance is low while another non-stored e-link provides higher predicted benefit.

Let

$$\mathcal{E}_{\text{new}}^{\text{cand}}(t) \quad (4.173)$$

denote the set of candidate artificial e-links that are not currently stored.

For a candidate new e-link (u, v) , the predicted placement utility is defined as

$$U_{uv}^{\text{place}}(t) = \theta_D Q_{uv}^{\text{demand}}(t) + \theta_S I_{uv}^{\text{struct}} + \theta_C G_{uv}^{\text{connect}}(t) - \theta_R \hat{C}_R(u, v, t). \quad (4.174)$$

The weights satisfy

$$\theta_D + \theta_S + \theta_C + \theta_R = 1. \quad (4.175)$$

A stored e-link (i, j) may be replaced by candidate (u, v) when

$$U_{uv}^{\text{place}}(t) - U_{ij}^{\text{retain}}(t) \geq \Delta U_{\min}, \quad (4.176)$$

where $U_{ij}^{\text{retain}}(t)$ represents the predicted utility of keeping the existing artificial e-link without replacement.

The threshold $\Delta U_{\min}$ prevents frequent overlay changes caused by small utility fluctuations.

4.7.4 Discard Decision

An artificial e-link is discarded when it becomes expired, repeatedly unused, structurally redundant, or too expensive to maintain.

The discard indicator is

$$d_{ij}(t) \in \{0, 1\}. \quad (4.177)$$

It is defined as

$$d_{ij}(t) = \mathbf{1} \left[S_{ij}^{\text{state}}(t) = \text{Expired} \right]. \quad (4.178)$$

A non-expired link may also be discarded when the following redundancy conditions are simultaneously satisfied:

$$H_{ij}^{\text{use}}(t) < H_{\min}^{\text{use}}, \quad (4.179)$$

$$I_{ij}^{\text{struct}} < I_{\min}^{\text{struct}}, \quad (4.180)$$

and

$$U_{ij}^{\text{regen}}(t) \leq 0. \quad (4.181)$$

An e-link is not discarded when doing so would violate a required compact-overlay connectivity condition.

4.7.5 Compact-Overlay Constraints

For every ESP v_i , the number of maintained routing-table entries is bounded by

$$|\mathcal{T}_i(t)| \leq T_i^{\max}, \quad (4.182)$$

where

$$T_i^{\max} = c_{\text{tab}} \sqrt{n_e} \log n_e. \quad (4.183)$$

This constraint preserves the compact-routing property by preventing each ESP from maintaining complete global routing information.

In the Partial-Anchor Scheme, the required local and anchor connectivity structures must be preserved:

$$E_{\text{local}}^{\text{req}} \subseteq E_A^{\text{PA}}(t). \quad (4.184)$$

In addition, the anchor overlay subgraph should remain connected whenever sufficient physical entanglement resources are available.

In the Full-Anchor Scheme, each ESP maintains a bounded tracked-link set:

$$|\mathcal{L}_i^{\text{track}}(t)| \leq L_{\text{max}}^{\text{track}}. \quad (4.185)$$

New tracked links are admitted only when available routing-table capacity exists or when an existing tracked link is removed.

4.7.6 Memory-Aware Maintenance

The overlay-maintenance strategy must satisfy the quantum-memory constraint

$$\sum_{j:(i,j) \in E_A(t)} n_{ij}^{\text{ebit}}(t) \leq M_i, \quad \forall v_i \in V_2. \quad (4.186)$$

When memory capacity becomes insufficient, the controller releases artificial e-links according to a discard-priority score.

The discard-priority score is defined as

$$\Pi_{ij}^{\text{drop}}(t) = \phi_1 (1 - H_{ij}^{\text{use}}(t)) + \phi_2 (1 - I_{ij}^{\text{struct}}) + \phi_3 \hat{C}_R(i, j, t) + \phi_4 I_{ij}^{\text{expired}}(t), \quad (4.187)$$

where

$$I_{ij}^{\text{expired}}(t) = \mathbf{1} [S_{ij}^{\text{state}}(t) = \text{Expired}]. \quad (4.188)$$

Links with higher discard priority are released first while preserving required compact-overlay connectivity.

4.7.7 Maintenance Triggering

Overlay maintenance may be executed periodically or through event-driven triggers.

Periodic maintenance is performed every

$$\tau_{\text{maint}} \quad (4.189)$$

time slots.

Event-driven maintenance is triggered when one of the following conditions occurs:

$$S_{ij}^{\text{state}}(t) = \text{Critical} \quad (4.190)$$

for a structurally important e-link,

$$S_{ij}^{\text{state}}(t) = \text{Expired}, \quad (4.191)$$

$$m_i(t) \geq M_i, \quad (4.192)$$

or

$$P_{\text{disc}}(t) \geq P_{\text{disc}}^{\text{max}}. \quad (4.193)$$

To prevent excessive overlay reconfiguration, a minimum holding interval

$$\tau_{\text{hold}} \quad (4.194)$$

is enforced between two structural changes affecting the same routing-table entry.

4.7.8 Overlay-State Update

Let

$$E_{\text{retain}}(t), \quad E_{\text{regen}}^{\text{succ}}(t), \quad E_{\text{new}}^{\text{succ}}(t), \quad E_{\text{drop}}(t) \quad (4.195)$$

denote the sets of retained, successfully regenerated, newly established, and discarded artificial e-links, respectively.

The updated artificial overlay edge set is defined as

$$E_A(t+1) = \left(E_{\text{retain}}(t) \cup E_{\text{regen}}^{\text{succ}}(t) \cup E_{\text{new}}^{\text{succ}}(t) \right) \setminus E_{\text{drop}}(t). \quad (4.196)$$

The corresponding routing tables are updated only after successful entanglement generation or regeneration is confirmed through classical heralding information.

Therefore, a routing entry is not advertised as usable until the corresponding artificial e-link has been successfully established and its initial temporal state has been recorded.

The proposed maintenance strategy balances temporal reliability and resource efficiency by preserving useful links, selectively regenerating critical links, replacing inefficient links, and removing stale or low-value links while maintaining the compact structure of the artificial overlay.

4.8 Proposed Routing Algorithm

The proposed routing algorithm integrates temporal-state monitoring, compact candidate construction, temporal-feasibility filtering, time-dependent route selection, and selective artificial e-link regeneration.

The algorithm is executed at each time slot t and processes the active entanglement-request set

$$\mathcal{D}(t). \quad (4.197)$$

The main inputs are

$$G_P, \quad G_A(t), \quad \mathcal{D}(t), \quad \mathcal{T}(t), \quad \mathcal{M}(t), \quad (4.198)$$

together with fidelity, coherence, delay, memory, and regeneration parameters.

The main outputs are

$$\mathcal{Y}(t), \quad \mathcal{E}_{\text{regen}}^*(t), \quad \mathcal{E}_{\text{drop}}(t), \quad G_A(t+1), \quad (4.199)$$

where $\mathcal{Y}(t)$ contains selected routes and their execution outcomes.

The complete procedure is described as follows.

4.8.1 Algorithmic Procedure

1. Update temporal e-link states.

For every artificial e-link $(i, j) \in E_A(t)$, update

$$A_{ij}(t), \quad F_{ij}^{\text{eff}}(t), \quad \widehat{F}_{ij}^{\text{cons}}(t), \quad B_{ij}(t), \quad H_{ij}^{\text{use}}(t). \quad (4.200)$$

Each e-link is classified as safe, warning, critical, or expired according to the temporal-state classification rules.

2. Construct the temporally usable overlay.

The usable edge set is constructed as

$$E_U(t) = \left\{ (i, j) \in E_A(t) : S_{ij}^{\text{state}}(t) \in \{\text{Safe}, \text{Warning}\} \right\}. \quad (4.201)$$

The temporally usable overlay is then defined as

$$G_U(t) = (V_2, E_U(t)). \quad (4.202)$$

3. Process each entanglement request.

For every request

$$(s, d) \in \mathcal{D}(t), \quad (4.203)$$

the controller generates a compact candidate-route set

$$\mathcal{R}_{s,d}^{\text{cand}}(t). \quad (4.204)$$

The candidate set contains no more than $K_{\max}$ routes.

4. Evaluate direct temporal feasibility.

For every candidate route R , calculate

$$\chi_{\text{avail}}(R, t), \quad \chi_{\text{linkF}}(R, t), \quad \chi_{\text{coh}}(R, t), \quad (4.205)$$

and

$$\chi_{\text{endF}}(R, t), \quad \chi_{\text{delay}}(R, t), \quad \chi_{\text{mem}}(R, t). \quad (4.206)$$

The overall feasibility indicator is

$$\chi_{s,d}(R, t) = \chi_{\text{avail}}(R, t) \chi_{\text{linkF}}(R, t) \chi_{\text{coh}}(R, t) \chi_{\text{endF}}(R, t) \chi_{\text{delay}}(R, t) \chi_{\text{mem}}(R, t). \quad (4.207)$$

5. Build the directly feasible route set.

The temporally feasible route set is defined as

$$\mathcal{R}_{s,d}^{\text{TF}}(t) = \left\{ R \in \mathcal{R}_{s,d}^{\text{cand}}(t) : \chi_{s,d}(R, t) = 1 \right\}. \quad (4.208)$$

6. Select a directly feasible route.

If

$$\mathcal{R}_{s,d}^{\text{TF}}(t) \neq \emptyset, \quad (4.209)$$

the controller calculates the complete time-dependent cost

$$W_t(R) \quad (4.210)$$

for every feasible route.

The selected route is

$$R_{s,d}^*(t) = \arg \min_{R \in \mathcal{R}_{s,d}^{\text{TF}}(t)} W_t(R). \quad (4.211)$$

The required e-bits are reserved and the planned swapping operations are executed.

7. Construct the repairable route set.

If no directly feasible route exists, candidate routes containing a bounded number of regenerable critical links are considered.

The repairable route set is defined as

$$\mathcal{R}_{s,d}^{\text{repair}}(t) = \left\{ R \in \mathcal{R}_{s,d}^{\text{cand}}(t) : \widehat{D}_{s,d}^{\text{repair}}(R, t) \leq D_{\max}^{\text{repair}} \wedge C_{s,d}^{\text{repair}}(R, t) \leq C_{\max}^{\text{repair}} \right\}. \quad (4.212)$$

8. Select a repairable route.

If

$$\mathcal{R}_{s,d}^{\text{repair}}(t) \neq \emptyset, \quad (4.213)$$

the repair-aware cost is calculated as

$$W_t^{\text{repair}}(R) = W_t(R) + \rho_D \widehat{D}_{s,d}^{\text{repair}}(R, t) + \rho_C C_{s,d}^{\text{repair}}(R, t). \quad (4.214)$$

The selected repairable route is

$$R_{s,d}^{\text{repair},*}(t) = \arg \min_{R \in \mathcal{R}_{s,d}^{\text{repair}}(t)} W_t^{\text{repair}}(R). \quad (4.215)$$

The critical e-links of this route are added to the request-specific regeneration queue.

9. Reject or postpone infeasible requests.

If both

$$\mathcal{R}_{s,d}^{\text{TF}}(t) = \emptyset \quad (4.216)$$

and

$$\mathcal{R}_{s,d}^{\text{repair}}(t) = \emptyset, \quad (4.217)$$

the request is classified as temporarily infeasible.

Depending on the request deadline, the controller either postpones or rejects the request.

10. Calculate proactive regeneration priorities.

For every critical artificial e-link, calculate

$$\Pi_{ij}^{\text{regen}}(t). \quad (4.218)$$

Request-specific critical links receive an additional repair-priority term.

11. Select regeneration actions.

Regeneration candidates are ranked according to priority and expected utility.

The selected regeneration set is

$$\mathcal{E}_{\text{regen}}^*(t), \quad (4.219)$$

subject to

$$|\mathcal{E}_{\text{regen}}^*(t)| \leq R_{\text{max}}, \quad (4.220)$$

memory constraints, and physical-generation capacity constraints.

12. Execute overlay-maintenance actions.

Useful links are retained, selected critical links are regenerated, inefficient links are replaced, and expired or low-value links are discarded.

13. Update the artificial overlay.

After receiving generation and swapping outcomes, update

$$E_A(t+1) = \left(E_{\text{retain}}(t) \cup E_{\text{regen}}^{\text{succ}}(t) \cup E_{\text{new}}^{\text{succ}}(t) \right) \setminus E_{\text{drop}}(t). \quad (4.221)$$

14. Update compact routing tables.

A routing-table entry is added only after the corresponding artificial e-link is successfully established and confirmed through classical communication.

Entries associated with consumed, discarded, or permanently unavailable e-links are removed.

4.8.2 Algorithm Characteristics

The proposed algorithm is heuristic rather than globally optimal. It does not solve the complete joint optimization problem involving route selection, regeneration scheduling, and overlay maintenance at every time slot.

Instead, the computational cost is reduced through

- compact routing-state restriction;
- bounded candidate-route generation;
- early temporal-feasibility filtering;
- additive time-dependent route-cost evaluation;
- bounded regeneration operations; and
- threshold-based overlay maintenance.

For a fixed network state, parameter configuration, candidate ordering, and tie-breaking rule, the algorithm produces deterministic routing decisions.

Randomness appears only from physical entanglement generation processes, decoherence behavior, and probabilistic swapping operations.

The performance of the proposed method is evaluated by comparison with static compact routing, fidelity-only routing, age-only routing, shortest-path routing, and regeneration-disabled variants.

4.9 Computational Complexity Analysis

The computational complexity of the proposed method consists of temporal-state monitoring, candidate-route construction, temporal-feasibility evaluation, route-cost calculation, and regeneration-priority computation.

Let

$$n = |V_2| \quad (4.222)$$

denote the number of Entanglement Service Providers (ESPs), and let

$$m_A(t) = |E_A(t)| \quad (4.223)$$

denote the number of stored artificial e-links at time slot t .

The number of active entanglement requests at time t is

$$D_t = |\mathcal{D}(t)|. \quad (4.224)$$

The maximum number of candidate routes generated for each request is denoted by

$$K_{\max}, \quad (4.225)$$

while the maximum number of artificial e-links contained in one candidate route is denoted by

$$H_{\max}. \quad (4.226)$$

4.9.1 Temporal-State Monitoring Complexity

The temporal-state monitoring module updates the age, effective fidelity, predicted consumption fidelity, remaining coherence budget, utilization score, and state classification of every stored artificial e-link.

Since each update requires a constant number of operations, the monitoring complexity is

$$O(m_A(t)). \quad (4.227)$$

The memory requirement for storing temporal-state information is also

$$O(m_A(t)), \quad (4.228)$$

because one temporal-state record is maintained for each artificial e-link.

4.9.2 Candidate-Route Construction Complexity

Candidate-route construction is performed on the temporally usable compact overlay rather than the complete physical network.

When Dijkstra's algorithm or a bounded K -shortest-path method is applied to the usable overlay, the shortest-path computation using adjacency lists and a binary heap has complexity

$$O(m_U(t) \log n), \quad (4.229)$$

where

$$m_U(t) = |E_U(t)|. \quad (4.230)$$

For generating at most $K_{\max}$ candidate routes, a conservative upper bound is

$$O(K_{\max} m_U(t) \log n). \quad (4.231)$$

In practice, the computational cost is lower because candidate generation is restricted by compact routing tables, anchor structures, tracked long-range links, and maximum-hop constraints.

If every ESP maintains at most

$$T_{\max} = O\left(\sqrt{n} \log n\right) \quad (4.232)$$

routing entries, the local search space remains significantly smaller than unrestricted global path enumeration.

4.9.3 Temporal-Feasibility Filtering Complexity

For each candidate route, the temporal-feasibility module evaluates every artificial e-link contained in the route.

Since a candidate route contains at most $H_{\max}$ artificial e-links, the availability, fidelity, coherence-budget, and resource-availability checks require

$$O(H_{\max}) \quad (4.233)$$

computational complexity per candidate route.

The end-to-end fidelity estimation also requires a product operation over at most $H_{\max}$ link parameters and swapping-retention factors. Therefore, its complexity is

$$O(H_{\max}). \quad (4.234)$$

For at most $K_{\max}$ candidate routes, the total feasibility-filtering complexity for one request is

$$O(K_{\max} H_{\max}). \quad (4.235)$$

For all entanglement requests in time slot t , the total complexity becomes

$$O(D_t K_{\max} H_{\max}). \quad (4.236)$$

4.9.4 Time-Dependent Route-Selection Complexity

After feasibility filtering, let

$$K_{\text{TF}} \leq K_{\text{max}} \quad (4.237)$$

denote the number of temporally feasible candidate routes.

The complete time-dependent cost evaluation of one route requires examining at most H_{max} artificial e-links. Therefore, evaluating all feasible routes requires

$$O(K_{\text{TF}} H_{\text{max}}) \quad (4.238)$$

computational complexity per request.

Selecting the minimum-cost route requires

$$O(K_{\text{TF}}) \quad (4.239)$$

additional comparisons.

Because

$$K_{\text{TF}} \leq K_{\text{max}}, \quad (4.240)$$

the overall route-selection complexity is bounded by

$$O(K_{\text{max}} H_{\text{max}}). \quad (4.241)$$

4.9.5 Route-Repair Complexity

When no directly feasible route is available, the algorithm evaluates the repairable candidate-route set.

For each candidate route, the controller identifies critical artificial e-links and estimates the corresponding repair delay and regeneration cost.

Since each route contains at most H_{max} artificial e-links, the repair evaluation requires

$$O(H_{\max}) \quad (4.242)$$

computational complexity per candidate route.

Therefore, the total repair-evaluation complexity per request is

$$O(K_{\max} H_{\max}). \quad (4.243)$$

This repair procedure is activated only when the directly feasible route set is empty, and therefore does not introduce additional overhead during normal routing operation.

4.9.6 Regeneration-Priority Complexity

The proactive regeneration module evaluates all currently critical artificial e-links.

Let

$$m_{\text{crit}}(t) = |\mathcal{E}_{\text{crit}}(t)| \quad (4.244)$$

denote the number of critical e-links at time t .

The calculation of regeneration-priority and utility scores requires constant computation per critical e-link. Therefore, the scoring complexity is

$$O(m_{\text{crit}}(t)). \quad (4.245)$$

Selecting the highest-priority $R_{\max}$ regeneration candidates through complete sorting requires

$$O(m_{\text{crit}}(t) \log m_{\text{crit}}(t)). \quad (4.246)$$

Alternatively, maintaining a bounded priority queue with maximum size $R_{\max}$ reduces the selection complexity to

$$O(m_{\text{crit}}(t) \log R_{\max}). \quad (4.247)$$

Since

$$R_{\max} \ll m_{\text{crit}}(t), \quad (4.248)$$

the bounded-priority-queue implementation is preferred for large-scale quantum networks.

4.9.7 Overlay-Maintenance Complexity

The retain, regenerate, replace, and discard decisions are evaluated over the currently stored artificial e-link set.

Therefore, the basic maintenance-state evaluation requires

$$O(m_A(t)). \quad (4.249)$$

If replacement candidates are selected from a bounded candidate set of size

$$m_{\text{new}}(t), \quad (4.250)$$

the ranking complexity becomes

$$O(m_{\text{new}}(t) \log m_{\text{new}}(t)). \quad (4.251)$$

In the proposed implementation, replacement evaluation is restricted to a bounded candidate set instead of considering all possible ESP pairs. This limitation preserves the scalability of the compact-overlay architecture.

4.9.8 Total Per-Slot Complexity

Combining temporal-state monitoring, candidate construction, feasibility filtering, route selection, and regeneration operations, the overall computational complexity in time slot t is

$$O(m_A(t) + D_t K_{\max} m_U(t) \log n + D_t K_{\max} H_{\max} + m_{\text{crit}}(t) \log R_{\max}). \quad (4.252)$$

When candidate routes are generated directly from pre-established compact-routing structures rather than performing a complete graph search for every request, the route-construction component is reduced.

In this case, the dominant computational complexity becomes

$$O(m_A(t) + D_t K_{\max} H_{\max} + m_{\text{crit}}(t) \log R_{\max}). \quad (4.253)$$

Since $K_{\max}$ and $H_{\max}$ are bounded design parameters, the request-processing complexity increases approximately linearly with the number of active entanglement requests.

4.9.9 Space Complexity

The artificial-overlay graph requires

$$O(n + m_A(t)) \quad (4.254)$$

storage space.

The temporal-state information maintained for each artificial e-link requires additional

$$O(m_A(t)) \quad (4.255)$$

storage.

The temporary storage required for candidate routes of all active requests is bounded by

$$O(D_t K_{\max} H_{\max}). \quad (4.256)$$

Therefore, the overall space complexity is

$$O(n + m_A(t) + D_t K_{\max} H_{\max}). \quad (4.257)$$

The compact-routing-table constraint prevents each ESP from maintaining complete global routing information. Instead of requiring $O(n)$ routing state, the proposed architecture maintains the target per-ESP routing-state complexity of

$$O(\sqrt{n} \log n). \quad (4.258)$$

Therefore, the proposed algorithm sacrifices global optimality in exchange for bounded candidate evaluation, compact routing-state requirements, and scalable temporal-state processing.

4.10 Chapter Summary

This chapter presented the proposed temporal-state-aware compact routing and overlay-maintenance framework for quantum-native entanglement networks.

The proposed method extends compact quantum routing by incorporating the temporal evolution of stored artificial e-links into routing decisions. The main objectives were to improve temporal reliability, preserve entanglement fidelity, maintain compact routing-state requirements, control regeneration overhead, reduce request rejection, and achieve scalable decision making.

The proposed framework consists of two interacting processes: reactive routing and proactive overlay maintenance. Reactive routing handles incoming entanglement requests by constructing a bounded candidate-route set, evaluating temporal feasibility, and selecting the minimum-cost feasible route. Proactive maintenance continuously evaluates stored artificial e-links and determines whether they should be retained, regenerated, replaced, or discarded.

A temporal-state monitoring mechanism was introduced to update e-link age, effective fidelity, predicted consumption fidelity, remaining coherence budget, utilization history, and temporal classification. Based on these factors, artificial e-links are classified into safe, warning, critical, and expired states.

The candidate-route construction mechanism preserves compact routing by limiting route discovery to local e-neighbors, anchor information, and tracked long-range links. The number of candidate routes and maximum route length are explicitly bounded to avoid unrestricted path exploration.

The temporal-feasibility filtering module evaluates candidate routes using link availability, predicted fidelity, coherence budget, end-to-end fidelity, completion delay, and e-bit resource availability. This process separates directly feasible routes from repairable and infeasible routes.

A time-dependent route-selection metric was developed by combining temporal e-link quality, fidelity degradation, age, coherence risk, regeneration cost, swapping reliability, and route delay. Additional reliability-aware extensions were introduced using predicted route-success probability.

The proactive regeneration mechanism prioritizes critical artificial e-links according to temporal risk, structural importance, utilization history, pre-

dicted demand, and regeneration overhead. Regeneration decisions are constrained by quantum-memory capacity, physical-generation capability, and per-slot regeneration limits.

The overlay-maintenance strategy provides a controlled mechanism for preserving useful links, refreshing degraded links, replacing inefficient links, and removing obsolete links while maintaining compact routing constraints.

The complete routing procedure was summarized as a model-based heuristic algorithm. It does not attempt to solve the globally optimal joint routing and maintenance problem at every time slot. Instead, it achieves practical scalability through bounded candidate generation, temporal filtering, and limited regeneration operations.

Finally, the computational complexity analysis demonstrated that the proposed framework depends mainly on the number of stored artificial e-links, active requests, bounded candidate routes, maximum route length, and critical-link population. With fixed candidate and route-length limits, the request-processing complexity scales approximately linearly with the number of active requests.

The next chapter presents the simulation environment, implementation details, parameter configuration, baseline algorithms, evaluation metrics, and experimental scenarios used to validate the proposed method.

Chapter 5

Simulation Methodology and Experimental Setup

5.1 Introduction

This chapter presents the simulation methodology and experimental setup used to evaluate the proposed temporal-state-aware compact routing method.

The evaluation is designed to examine whether explicit temporal e-link information can improve routing reliability, preserve end-to-end fidelity, reduce request rejection, and limit unnecessary regeneration overhead.

The simulation compares the proposed method with several benchmark routing strategies under different network sizes, traffic loads, memory coherence times, fidelity thresholds, and artificial-overlay configurations.

The experimental methodology contains the following components:

- simulation environment and implementation tools;
- physical quantum-network topology generation;
- artificial entanglement-overlay construction;
- temporal e-link and decoherence parameters;
- traffic and entanglement-request generation;
- benchmark routing methods;
- evaluation metrics;
- experimental scenarios; and
- statistical result collection.

The complete experimental workflow is organized as

$$\begin{aligned} \text{Topology Generation} \rightarrow \text{Overlay Construction} \rightarrow \text{Traffic Generation} \\ \rightarrow \text{Routing} \rightarrow \text{Regeneration} \rightarrow \text{Performance Evaluation.} \end{aligned} \quad (5.1)$$

Figure 5.1 illustrates the overall simulation workflow used in this thesis.

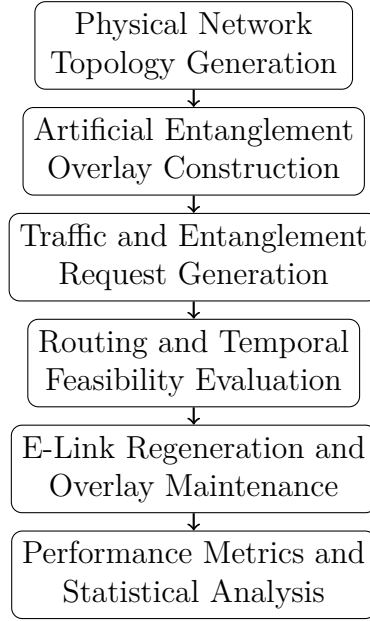


Figure 5.1: Overall simulation and experimental-evaluation workflow.

The proposed method is evaluated in discrete time slots. During each slot, new requests may arrive, stored artificial e-links may age, fidelity may decrease, swapping may succeed or fail, and selected e-links may be regenerated or discarded.

The simulation therefore maintains both network state and temporal e-link state over the complete experimental horizon.

To ensure a fair comparison, all benchmark methods use the same physical topology, artificial-overlay capacity, request sequence, memory limits, generation probabilities, swapping probabilities, and random seeds.

The primary evaluation questions are:

1. Does temporal-state-aware routing improve end-to-end fidelity?
2. Does coherence-budget filtering reduce failed route selections?

3. How much regeneration overhead is introduced?
4. How does performance change with network size and traffic load?
5. Does the method preserve compact-routing efficiency?
6. How robust are the results to different parameter weights?

The numerical values used in the experiments are not treated as universal hardware constants. They are configurable simulation parameters selected from representative ranges reported in quantum-networking literature and varied through sensitivity analysis.

The following sections describe the simulation environment, topology generation, parameter configuration, benchmark methods, evaluation metrics, and experimental scenarios in detail.

5.2 Simulation Environment

The proposed routing framework is implemented as a discrete-event and time-slotted simulation in Python. The simulator models physical quantum links, artificial e-links, quantum-memory occupation, entanglement generation, decoherence, swapping, routing requests, and overlay-maintenance actions.

The implementation uses Python because it provides mature graph-processing, numerical-computation, and data-analysis libraries. The principal software components are:

- Python for the simulation engine;
- NetworkX for graph representation and route computation;
- NumPy for numerical operations and random sampling;
- pandas for data storage and experiment aggregation;
- Matplotlib for result visualization; and
- Jupyter Notebook or Visual Studio Code for implementation and analysis.

The physical quantum network is represented as a graph

$$G_P = (V_2, E_P), \quad (5.2)$$

while the artificial entanglement overlay is represented as a time-dependent graph

$$G_A(t) = (V_2, E_A(t)). \quad (5.3)$$

Each physical edge stores the following attributes:

- physical distance;
- elementary-entanglement generation probability;
- initial fidelity;
- coherence time;
- expected generation delay; and
- operational success probability.

Each artificial e-link stores:

- creation time;
- current age;
- current and predicted fidelity;
- remaining coherence budget;
- number of stored e-bits;
- structural type;
- utilization history; and
- regeneration state.

The simulator advances through discrete time slots of duration

$$\Delta. \quad (5.4)$$

At each time slot, the simulator performs the following operations:

1. generates new entanglement requests;
2. updates artificial e-link ages;

3. applies the selected decoherence model;
4. classifies e-links as safe, warning, critical, or expired;
5. constructs candidate routes;
6. evaluates temporal feasibility;
7. selects routes and executes swapping;
8. updates memory occupation;
9. performs regeneration and overlay maintenance; and
10. records performance metrics.

The general simulation-loop structure is shown in Algorithm 1.

Algorithm 1 Main Simulation Loop

- 1: Initialize physical topology G_P
 - 2: Construct initial artificial overlay $G_A(0)$
 - 3: Initialize quantum-memory and e-link states
 - 4: **for** $t = 0$ to $T_{\text{sim}} - 1$ **do**
 - 5: Generate new requests $\mathcal{D}(t)$
 - 6: Update age, fidelity, and coherence budget
 - 7: Classify all stored artificial e-links
 - 8: Construct temporally usable overlay $G_U(t)$
 - 9: **for** each request $(s, d) \in \mathcal{D}(t)$ **do**
 - 10: Construct compact candidate routes
 - 11: Apply temporal-feasibility filtering
 - 12: Select or repair a route
 - 13: Execute swapping and update consumed resources
 - 14: **end for**
 - 15: Rank and execute regeneration actions
 - 16: Update artificial overlay and routing tables
 - 17: Record metrics for time slot t
 - 18: **end for**
 - 19: Aggregate results over all simulation runs
-

Random events such as entanglement-generation success and swapping success are sampled using pseudorandom number generation. For reproducibility, every experiment uses a recorded random seed.

Each experimental configuration is repeated over multiple independent runs. Let

$$N_{\text{run}} \quad (5.5)$$

denote the number of independent simulation runs.

For a measured quantity X , the reported sample mean is

$$\overline{X} = \frac{1}{N_{\text{run}}} \sum_{r=1}^{N_{\text{run}}} X_r. \quad (5.6)$$

The sample standard deviation is

$$s_X = \sqrt{\frac{1}{N_{\text{run}} - 1} \sum_{r=1}^{N_{\text{run}}} (X_r - \overline{X})^2}. \quad (5.7)$$

Where appropriate, results are reported with confidence intervals rather than as single-run values.

The simulation code, configuration files, random seeds, and processed result files will be organized in separate folders to support reproducibility and later publication.

5.3 Network Topology Generation

The physical quantum network topology determines the available elementary quantum channels and the physical paths used to create artificial e-links.

The physical network is represented by

$$G_P = (V_2, E_P), \quad (5.8)$$

where V_2 is the set of Entanglement Service Providers and E_P is the set of physical quantum links.

The simulations consider three topology categories:

1. random connected topology;
2. grid-based topology; and
3. small-world topology.

Using multiple topology types prevents the evaluation from depending on a single network structure.

5.3.1 Random Connected Topology

In the random topology, ESP positions are generated inside a two-dimensional square region.

The coordinate of ESP v_i is

$$\mathbf{x}_i = (x_i, y_i). \quad (5.9)$$

The Euclidean distance between ESPs v_i and v_j is

$$L_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}. \quad (5.10)$$

A physical link is created when

$$L_{ij} \leq L_{\max}^{\text{phys}}, \quad (5.11)$$

where $L_{\max}^{\text{phys}}$ is the maximum physical-link distance.

Disconnected topology samples are rejected or repaired by connecting the nearest disconnected components.

5.3.2 Grid Topology

The grid topology arranges ESPs in a regular two-dimensional structure.

Each node is connected to its horizontal and vertical neighbors. Optional diagonal links may be added to increase path diversity.

The grid topology provides a controlled environment for evaluating route length, anchor placement, and temporal-stretch behavior.

5.3.3 Small-World Topology

The small-world topology is generated by starting from a locally connected graph and rewiring selected edges with probability

$$p_{\text{rewire}}. \quad (5.12)$$

This structure combines local clustering with a limited number of long-range physical links.

It is useful for evaluating whether compact routing benefits from short topological paths while remaining robust to aging and fidelity degradation.

5.3.4 Network Sizes

The proposed method is evaluated using several network sizes:

$$n \in \{20, 40, 60, 80, 100\}. \quad (5.13)$$

These values may be adjusted according to simulation runtime and available computing resources.

For every topology configuration, the following conditions are enforced:

- the physical graph is connected;
- every ESP has at least one physical neighbor;
- physical-link distances remain within the configured range;
- the same topology is used by all compared routing methods; and
- multiple independent topology instances are evaluated.

Figure 5.2 shows an example physical quantum-network topology.

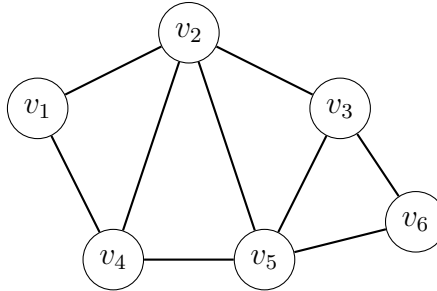


Figure 5.2: Example physical quantum-network topology used to illustrate topology generation.

The illustrated topology is conceptual. The actual experimental topologies are generated automatically by the simulation code and stored with their random seeds.

For reproducibility, every generated topology is exported with

- node coordinates;
- physical-edge list;
- link distances;
- topology-generation parameters; and

- random seed.

This topology-generation procedure provides controlled and repeatable physical substrates for constructing the artificial entanglement overlays evaluated in the following sections.

5.4 Physical and Quantum-Link Parameters

The physical and quantum-link parameters determine the success probability, delay, fidelity, and temporal lifetime of elementary entanglement resources.

Each physical link $(i, j) \in E_P$ is assigned the parameter set

$$\Theta_{ij}^P = [L_{ij}, p_{ij}^{\text{gen}}, F_{ij}^0, T_{2,ij}, q_{ij}, D_{ij}^{\text{gen}}, S_{ij}], \quad (5.14)$$

where L_{ij} is the physical-link distance, p_{ij}^{gen} is the elementary-entanglement generation probability, F_{ij}^0 is the initial fidelity, $T_{2,ij}$ is the coherence time, q_{ij} is the swapping-success probability, D_{ij}^{gen} is the expected generation delay, and S_{ij} is the number of parallel generation modes.

The parameter values used in the simulations are treated as configurable experimental assumptions rather than universal hardware constants. Each important parameter is varied through sensitivity experiments.

The elementary-entanglement generation probability is modeled as a decreasing function of physical distance. A simplified attenuation-based model is

$$p_{ij}^{\text{gen}} = p_0 \exp(-\alpha L_{ij}), \quad (5.15)$$

where p_0 is the short-distance generation factor and α is the effective attenuation coefficient.

This expression is used as a simulation abstraction. It does not represent every optical component, detector inefficiency, coupling loss, or protocol-specific heralding process separately.

When S_{ij} independent parallel modes are available, the probability of at least one successful generation event in a time slot is

$$p_{ij}^{\text{link}} = 1 - (1 - p_{ij}^{\text{gen}})^{S_{ij}}. \quad (5.16)$$

The expected number of attempts required for one elementary-pair success is

$$\mathbb{E}[N_{ij}^{\text{attempt}}] = \frac{1}{p_{ij}^{\text{link}}}. \quad (5.17)$$

Table 5.1: Baseline physical and quantum-link simulation parameters.

Parameter	Symbol	Baseline setting
Time-slot duration	Δ	Configurable
Physical-link distance	L_{ij}	Topology dependent
Generation probability	p_{ij}^{gen}	Distance dependent
Initial elementary fidelity	F_{ij}^0	Configurable range
Memory coherence time	$T_{2,ij}$	Configurable range
Swapping-success probability	q_{ij}	Configurable range
Number of parallel modes	S_{ij}	Hardware dependent
Classical coordination delay	D_{ij}^{coord}	Distance dependent
Elementary-generation delay	D_{ij}^{gen}	Probability dependent
Link-level fidelity threshold	$F_{\min}^{\text{link}}$	Scenario dependent
End-to-end fidelity threshold	$F_{\min}^{\text{end}}$	Scenario dependent

The approximate expected generation delay is therefore

$$D_{ij}^{\text{gen}} = \frac{\Delta_{\text{attempt}}}{p_{ij}^{\text{link}}}, \quad (5.18)$$

where Δ_{attempt} is the duration of one generation attempt, including heralding.

The initial fidelity of a successful elementary pair may be modeled as

$$F_{ij}^0 = F_0 - \xi_L L_{ij} - \epsilon_{ij}^{\text{op}}, \quad (5.19)$$

where F_0 is the reference short-distance fidelity, ξ_L is a distance-related degradation coefficient, and $\epsilon_{ij}^{\text{op}}$ represents operational error.

For the baseline implementation, fidelity values are clipped to the physically meaningful interval

$$\frac{1}{4} \leq F_{ij}^0 \leq 1. \quad (5.20)$$

Memory coherence times may be homogeneous or heterogeneous.

For a homogeneous experiment,

$$T_{2,ij} = T_2, \quad \forall(i, j). \quad (5.21)$$

For a heterogeneous experiment,

$$T_{2,ij} \sim \mathcal{U}(T_2^{\min}, T_2^{\max}), \quad (5.22)$$

where $\mathcal{U}$ denotes a uniform distribution.

The stored-fidelity evolution follows the selected Werner-state approximation:

$$F_{ij}^{\text{eff}}(t) = \frac{1}{4} + \left(F_{ij}^0 - \frac{1}{4}\right) \exp\left(-\frac{A_{ij}(t)\Delta}{T_{2,ij}}\right). \quad (5.23)$$

The swapping-success probability at intermediate ESP v_r is denoted by

$$q_r \in [0, 1]. \quad (5.24)$$

The fidelity-retention factor of the swapping operation is represented by

$$\lambda_r \in [0, 1]. \quad (5.25)$$

For a route containing multiple artificial e-links, the end-to-end success probability includes the product of the required swapping-success probabilities.

Classical coordination delay is approximated as

$$D_{ij}^{\text{coord}} = \frac{L_{ij}}{v_{\text{signal}}}, \quad (5.26)$$

where v_{signal} is the effective propagation speed of the classical control signal.

The baseline parameters will later be replaced by verified numerical values supported by quantum-networking literature. Every reported experiment will clearly distinguish between literature-derived values, calibrated values, and simulation assumptions.

5.5 Artificial Overlay Configuration

The artificial entanglement overlay is constructed on top of the physical quantum network. It contains pre-established entangled connections among Entanglement Service Providers and is used as the logical substrate for compact routing.

At time slot t , the overlay is represented by

$$G_A(t) = (V_2, E_A(t)). \quad (5.27)$$

An artificial e-link may connect two ESPs that are not directly adjacent in the physical network. Such a link is generated through a feasible physical path and is stored in quantum memories at its endpoint ESPs.

The experiments consider two compact-overlay structures:

1. Partial-Anchor overlay; and
2. Full-Anchor overlay.

5.5.1 Partial-Anchor Overlay Configuration

In the Partial-Anchor configuration, only a subset of ESPs is selected as anchors.

Let

$$\mathcal{T} \subseteq V_2 \quad (5.28)$$

denote the anchor set.

The number of anchors is defined as

$$|\mathcal{T}| = \lceil c_{\text{anchor}} \sqrt{n} \rceil, \quad (5.29)$$

where c_{anchor} is a configurable scaling coefficient.

Anchors may be selected using one of the following strategies:

- random selection;
- highest-degree selection;
- highest-betweenness selection; or
- geographically distributed selection.

The baseline configuration uses degree-based or betweenness-based anchor selection because structurally central ESPs are more likely to provide efficient long-range connectivity.

Each non-anchor ESP maintains local e-neighbor links and one or more links to nearby or assigned anchors.

The Partial-Anchor overlay edge set is

$$E_A^{\text{PA}}(t) = E_{\text{local}}(t) \cup E_{\text{anchor}}(t). \quad (5.30)$$

5.5.2 Full-Anchor Overlay Configuration

In the Full-Anchor configuration, every ESP may maintain local e-neighbor links and selected tracked long-range artificial e-links.

The edge set is

$$E_A^{\text{FA}}(t) = E_{\text{local}}(t) \cup E_{\text{tracked}}(t). \quad (5.31)$$

Each ESP maintains at most

$$L_{\text{max}}^{\text{track}} \quad (5.32)$$

tracked long-range e-links.

Tracked links are selected according to structural utility, expected traffic relevance, and physical generation cost.

5.5.3 Local E-Neighbor Construction

A local artificial e-link between ESPs v_i and v_j is created when the corresponding physical route satisfies a maximum path-length constraint:

$$d_{\text{phys}}(v_i, v_j) \leq H_{\text{local}}^{\text{max}}, \quad (5.33)$$

where $d_{\text{phys}}(v_i, v_j)$ is the physical-hop distance and $H_{\text{local}}^{\text{max}}$ is the maximum physical-hop count for a local e-link.

The local-neighbor set of ESP v_i is

$$\mathcal{N}_{\text{local}}(v_i) = \{v_j : d_{\text{phys}}(v_i, v_j) \leq H_{\text{local}}^{\text{max}}\}. \quad (5.34)$$

5.5.4 Artificial E-Link Generation

For each selected endpoint pair (i, j) , the simulator identifies a physical generation path

$$P_{ij}^{\text{gen}}. \quad (5.35)$$

The path is selected using a physical-layer cost based on generation delay, success probability, and expected fidelity.

A simplified physical generation-path cost is

$$C_{ij}^{\text{phys}} = \omega_D D^{\text{phys}}(P_{ij}^{\text{gen}}) + \omega_P [-\log P_{\text{succ}}(P_{ij}^{\text{gen}})] + \omega_F (1 - F_{ij}^{\text{A},0}). \quad (5.36)$$

The selected generation path is

$$P_{ij}^{\text{gen},*} = \arg \min_P C_{ij}^{\text{phys}}(P). \quad (5.37)$$

5.5.5 Initial Overlay Density

The initial overlay density is controlled by

$$\rho_A = \frac{2|E_A(0)|}{n(n-1)}. \quad (5.38)$$

The experiments consider sparse, medium-density, and relatively dense overlay configurations.

A sparse overlay reduces memory and maintenance overhead but may increase route length and temporal disconnection.

A denser overlay provides more route alternatives but consumes more memory and generation resources.

5.5.6 Memory Allocation

Each ESP v_i has memory capacity

$$M_i. \quad (5.39)$$

The number of stored artificial e-links incident to v_i must satisfy

$$\sum_{j:(i,j) \in E_A(t)} n_{ij}^{\text{ebit}}(t) \leq M_i. \quad (5.40)$$

A baseline overlay assigns one stored e-bit per artificial e-link:

$$n_{ij}^{\text{ebit}}(0) = 1. \quad (5.41)$$

Additional experiments may allocate multiple e-bits to high-demand or structurally important e-links.

5.5.7 Initial E-Link Age

To avoid an unrealistically synchronized overlay in which every e-link is newly created at time zero, initial ages may be sampled from

$$A_{ij}(0) \sim \mathcal{U}(0, A_{\text{init}}^{\text{max}}). \quad (5.42)$$

This produces a mixed initial condition containing fresh and partially aged artificial e-links.

A separate cold-start scenario is also evaluated, where

$$A_{ij}(0) = 0 \quad (5.43)$$

for all initial artificial e-links.

5.5.8 Overlay Initialization Procedure

The initial overlay is constructed through the following steps:

1. generate the physical quantum-network topology;
2. select anchor ESPs or tracked-link candidates;
3. create required local artificial e-links;
4. identify feasible physical generation paths;
5. verify memory-capacity constraints;
6. assign initial fidelity and coherence parameters;
7. assign initial ages; and
8. initialize compact routing-table entries.

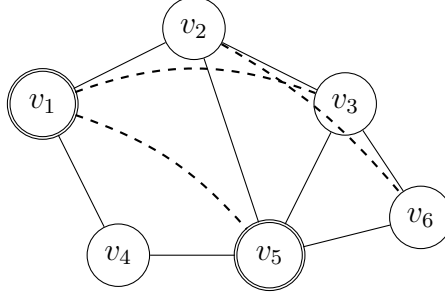


Figure 5.3: Conceptual artificial entanglement overlay. Solid lines represent physical quantum links, dashed lines represent stored artificial e-links, and double-circle nodes represent anchor ESPs.

Figure 5.3 illustrates a conceptual artificial entanglement overlay constructed over a physical topology.

The actual overlays used in the experiments are generated automatically and exported with their anchor sets, tracked-link sets, physical generation paths, routing-table entries, memory occupation, and random seeds.

5.6 Traffic and Entanglement-Request Model

The traffic model determines when entanglement requests arrive, which ESP pairs generate requests, how many entangled pairs are required, and how long each request can wait before it is considered unsuccessful.

At time slot t , the active request set is denoted by

$$\mathcal{D}(t). \quad (5.44)$$

Each entanglement request is represented by

$$d_k = (s_k, d_k, b_k, F_k^{\text{req}}, D_k^{\text{max}}, c_k), \quad (5.45)$$

where

- s_k is the source ESP;
- d_k is the destination ESP;
- b_k is the number of required entangled pairs;
- F_k^{req} is the minimum required end-to-end fidelity;
- D_k^{max} is the maximum acceptable completion delay; and
- c_k is the service class of the request.

5.6.1 Request-Arrival Process

The number of new requests arriving during time slot t is denoted by

$$N_{\text{arr}}(t). \quad (5.46)$$

For the baseline experiments, request arrivals are modeled using a Poisson process:

$$N_{\text{arr}}(t) \sim \text{Poisson}(\lambda_{\text{req}}), \quad (5.47)$$

where λ_{req} is the average number of requests per time slot. Three traffic-load levels are considered:

$$\lambda_{\text{req}} \in \{\lambda_{\text{low}}, \lambda_{\text{medium}}, \lambda_{\text{high}}\}. \quad (5.48)$$

This enables evaluation under lightly loaded, moderately loaded, and congested network conditions.

5.6.2 Source–Destination Selection

For uniform traffic, source and destination ESPs are selected randomly from V_2 , subject to

$$s_k \neq d_k. \quad (5.49)$$

A non-uniform demand model is also considered to represent traffic concentration around popular ESPs or anchor regions.

Under hotspot traffic, a subset

$$\mathcal{H} \subseteq V_2 \quad (5.50)$$

is selected as the hotspot set.

Requests involving hotspot ESPs are generated with probability

$$p_{\text{hot}}. \quad (5.51)$$

The remaining requests follow uniform source–destination selection.

5.6.3 Entanglement Demand Size

The number of requested entangled pairs is denoted by

$$b_k. \quad (5.52)$$

The baseline experiments use single-pair requests:

$$b_k = 1. \quad (5.53)$$

Additional experiments consider multi-pair requests selected from

$$b_k \in \{1, 2, 4, 8\}. \quad (5.54)$$

Multi-pair requests increase memory pressure and may require repeated generation and swapping operations.

5.6.4 Fidelity Requirement

Each request specifies a minimum end-to-end fidelity requirement:

$$F_k^{\text{req}}. \quad (5.55)$$

The experiments consider several service levels, for example:

$$F_k^{\text{req}} \in \{F_{\text{low}}, F_{\text{medium}}, F_{\text{high}}\}. \quad (5.56)$$

Higher fidelity requirements reduce the number of feasible routes and may increase purification or regeneration demand.

5.6.5 Request Deadline

Each request is assigned a maximum completion delay:

$$D_k^{\text{max}}. \quad (5.57)$$

A request is successfully served only if end-to-end entanglement is established before this deadline.

The request waiting time at slot t is

$$W_k(t) = t - t_k^{\text{arr}}, \quad (5.58)$$

where t_k^{arr} is the arrival time of request k .

A request is expired when

$$W_k(t)\Delta > D_k^{\text{max}}. \quad (5.59)$$

Expired requests are counted as rejected or deadline-violating requests.

5.6.6 Service Classes

Three request classes are considered:

1. best-effort requests;
2. delay-sensitive requests; and
3. fidelity-sensitive requests.

Best-effort requests use moderate fidelity and delay requirements.

Delay-sensitive requests use smaller values of $D_k^{\max}$.

Fidelity-sensitive requests use larger values of F_k^{req} .

The request class influences route-selection priority and regeneration urgency.

Table 5.2: Entanglement-request service classes used in the simulation.

Service class	Main requirement	Routing preference
Best effort	Moderate fidelity and delay	Minimum total route cost
Delay sensitive	Short completion deadline	Low predicted route delay
Fidelity sensitive	High end-to-end fidelity	High predicted consumption fidelity

5.6.7 Request Queue

Requests that cannot be served immediately may be placed in a waiting queue.

The queue at time t is denoted by

$$\mathcal{Q}_{\text{req}}(t). \quad (5.60)$$

A request remains in the queue when

- no directly feasible route is available;
- a repairable route is being regenerated;
- required memory resources are temporarily unavailable; or

- the request deadline has not yet expired.

Requests may be ordered using first-in-first-out scheduling or priority-based scheduling.

The baseline implementation uses a priority score

$$\Pi_k^{\text{req}}(t) = \zeta_D \frac{W_k(t)}{D_k^{\text{max}}/\Delta} + \zeta_F F_k^{\text{req}} + \zeta_C c_k, \quad (5.61)$$

where the coefficients control the relative importance of urgency, fidelity requirement, and service class.

5.6.8 Request Outcome

The outcome of request k is represented by

$$Y_k = \begin{cases} 1, & \text{if the request is successfully served,} \\ 0, & \text{otherwise.} \end{cases} \quad (5.62)$$

A successful request must satisfy

$$F_k^{\text{end}} \geq F_k^{\text{req}} \quad (5.63)$$

and

$$D_k^{\text{completion}} \leq D_k^{\text{max}}. \quad (5.64)$$

Figure 5.4 illustrates the request-processing lifecycle.

The same generated request sequence is provided to every routing method in a given experimental run. This ensures that differences in performance are caused by routing and maintenance decisions rather than by different traffic realizations.

5.7 Benchmark Routing Methods

The proposed method is compared with multiple benchmark routing strategies. All methods operate on the same physical topology, initial artificial overlay, request sequence, memory capacity, generation outcomes, swapping outcomes, and random seeds.

The purpose of the comparison is to determine whether the performance improvement is caused specifically by temporal-state awareness rather than by differences in available network resources.

The following benchmark methods are considered.

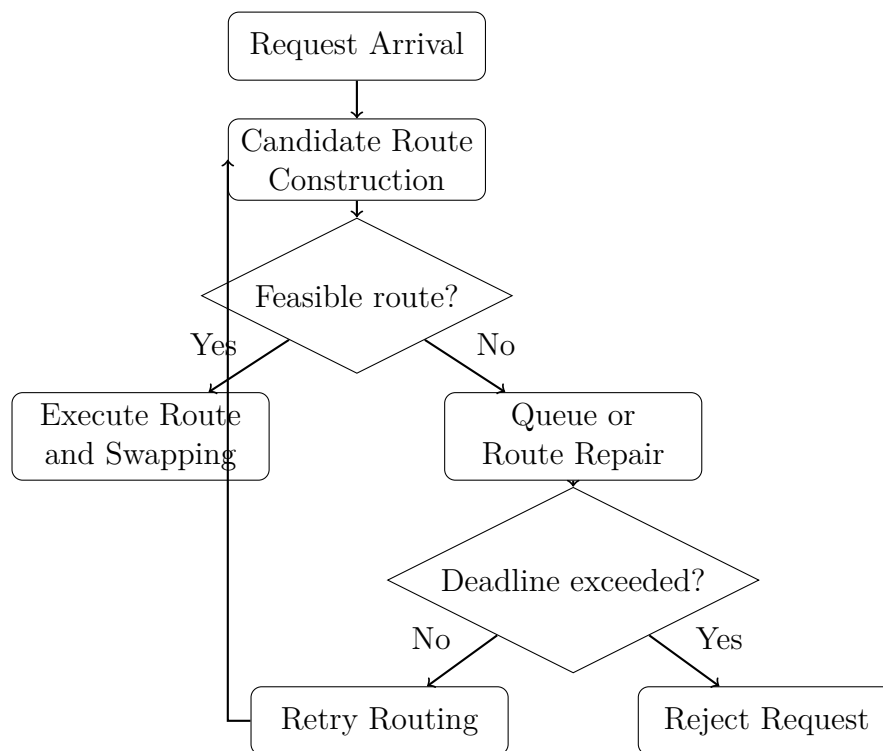


Figure 5.4: Lifecycle of an entanglement-distribution request in the simulation.

5.7.1 Shortest-Hop Routing

Shortest-Hop Routing selects the candidate route with the minimum number of artificial e-links:

$$R_{s,d}^{\text{SH}}(t) = \arg \min_{R \in \mathcal{R}_{s,d}^{\text{cand}}(t)} |E(R)|. \quad (5.65)$$

This method does not explicitly consider fidelity, age, coherence budget, regeneration cost, or predicted route-completion delay.

It represents a topology-oriented routing baseline.

5.7.2 Static Entangling-Cost Routing

Static Entangling-Cost Routing assigns each artificial e-link a fixed cost:

$$w_{\text{static}}(i, j). \quad (5.66)$$

The selected route is

$$R_{s,d}^{\text{static}}(t) = \arg \min_{R \in \mathcal{R}_{s,d}^{\text{cand}}(t)} \sum_{(i,j) \in E(R)} w_{\text{static}}(i, j). \quad (5.67)$$

The link cost may include physical generation distance, hop count, or expected establishment cost, but it is not updated according to the current e-link age or coherence budget.

This baseline represents the original compact-routing behavior without explicit temporal-state awareness.

5.7.3 Fidelity-Only Routing

Fidelity-Only Routing selects the candidate route with the highest predicted end-to-end fidelity:

$$R_{s,d}^{\text{FO}}(t) = \arg \max_{R \in \mathcal{R}_{s,d}^{\text{cand}}(t)} \hat{F}_{s,d}^{\text{end}}(R, t). \quad (5.68)$$

Routes below the required fidelity threshold are rejected.

This method considers fidelity but does not explicitly optimize coherence-budget margin, regeneration overhead, structural cost, or request delay.

5.7.4 Age-Only Routing

Age-Only Routing prefers routes containing the freshest artificial e-links.

The route-level age cost is

$$C_A(R, t) = \sum_{(i,j) \in E(R)} \frac{A_{ij}(t)}{A_{ij}^{\max}}. \quad (5.69)$$

The selected route is

$$R_{s,d}^{\text{AO}}(t) = \arg \min_{R \in \mathcal{R}_{s,d}^{\text{cand}}(t)} C_A(R, t). \quad (5.70)$$

This baseline evaluates whether age alone is sufficient for reliable routing.

5.7.5 Current-Fidelity-Aware Routing

Current-Fidelity-Aware Routing uses the fidelity measured at route-selection time but does not predict additional degradation before consumption.

The selected route is

$$R_{s,d}^{\text{CF}}(t) = \arg \max_{R \in \mathcal{R}_{s,d}^{\text{cand}}(t)} F_{s,d}^{\text{current}}(R, t). \quad (5.71)$$

This method helps isolate the benefit of consumption-time fidelity prediction.

5.7.6 Reactive-Regeneration Routing

Reactive-Regeneration Routing uses static or fidelity-aware route selection and regenerates an e-link only after the selected route becomes infeasible.

No proactive regeneration is performed.

The regeneration decision satisfies

$$r_{ij}^{\text{reactive}}(t) = 1 \quad (5.72)$$

only when e-link (i, j) directly blocks an active request.

This benchmark evaluates whether proactive maintenance provides measurable benefit over failure-triggered repair.

5.7.7 Proposed Method without Regeneration

This ablation variant uses temporal monitoring, coherence-budget filtering, and time-dependent route selection but disables all regeneration actions:

$$R_{\max} = 0. \quad (5.73)$$

It isolates the contribution of proactive and reactive regeneration.

5.7.8 Proposed Method without Coherence-Budget Filtering

This ablation variant removes the coherence-budget feasibility condition.

A route may be selected when its predicted fidelity is acceptable even if

$$B_{ij}(R, t) < \delta_{\text{safe}}. \quad (5.74)$$

This comparison measures the specific contribution of the remaining coherence-budget model.

5.7.9 Proposed Method without Consumption-Time Prediction

This variant uses current e-link fidelity instead of predicted consumption-time fidelity:

$$\hat{F}_{ij}^{\text{cons}}(R, t) = F_{ij}^{\text{eff}}(t). \quad (5.75)$$

It evaluates whether predicting additional waiting and swapping delay improves route reliability.

For fairness, every method is constrained by the same maximum candidate-route count

$$K_{\max}, \quad (5.76)$$

maximum overlay-hop count

$$H_{\max}, \quad (5.77)$$

memory capacities, fidelity thresholds, and request deadlines.

The benchmark set supports both direct performance comparison and ablation analysis. Direct comparison evaluates the proposed method against conventional routing strategies, while ablation analysis determines which proposed components contribute most to the final performance.

Table 5.3: Comparison of the proposed method and benchmark routing strategies.

Method	Age	Fidelity	Coherence budget	Regeneration
Shortest-Hop	No	No	No	No
Static Entangling Cost	No	Optional static	No	No
Fidelity-Only	No	Yes	No	No
Age-Only	Yes	No	No	No
Current-Fidelity-Aware	Indirectly	Current only	No	No
Reactive Regeneration	Optional	Optional	No	Reactive
Proposed without Regeneration	Yes	Predicted	Yes	No
Proposed without Budget Filtering	Yes	Predicted	No	Yes
Proposed without Prediction	Yes	Current only	Yes	Yes
Proposed Full Method	Yes	Predicted	Yes	Proactive and reactive

5.8 Evaluation Metrics

The performance of the proposed method is evaluated using routing reliability, fidelity, delay, resource-consumption, regeneration, and scalability metrics.

No single metric is sufficient because a routing method may improve fidelity while increasing delay, memory usage, or regeneration overhead. Therefore, the evaluation considers multiple complementary measures.

5.8.1 Request Success Ratio

The request success ratio measures the fraction of submitted requests that successfully establish end-to-end entanglement within their fidelity and delay requirements.

Let

$$N_{\text{success}} \quad (5.78)$$

denote the number of successfully completed requests, and let

$$N_{\text{total}} \quad (5.79)$$

denote the total number of generated requests.

The request success ratio is

$$P_{\text{success}} = \frac{N_{\text{success}}}{N_{\text{total}}}. \quad (5.80)$$

A request is counted as successful only when

$$F_k^{\text{end}} \geq F_k^{\text{req}} \quad (5.81)$$

and

$$D_k^{\text{completion}} \leq D_k^{\text{max}}. \quad (5.82)$$

5.8.2 Request Rejection Ratio

The request rejection ratio is

$$P_{\text{reject}} = \frac{N_{\text{reject}}}{N_{\text{total}}}. \quad (5.83)$$

Rejected requests include requests for which no feasible or repairable route exists and requests whose deadlines expire before completion.

5.8.3 Temporal Disconnection Ratio

A source–destination pair is temporally disconnected when a structural path exists but no route satisfies the current temporal constraints.

The temporal disconnection ratio is

$$P_{\text{disc}} = \frac{N_{\text{temporally disconnected}}}{N_{\text{route evaluations}}}. \quad (5.84)$$

This metric directly measures the effect of stale or low-coherence artificial e-links.

5.8.4 Average End-to-End Fidelity

The average realized end-to-end fidelity is calculated over successfully completed requests:

$$\bar{F}_{\text{end}} = \frac{1}{N_{\text{success}}} \sum_{k:Y_k=1} F_k^{\text{end}}. \quad (5.85)$$

The minimum observed successful fidelity is also recorded:

$$F_{\text{end}}^{\min} = \min_{k:Y_k=1} F_k^{\text{end}}. \quad (5.86)$$

5.8.5 Fidelity-Violation Ratio

A fidelity violation occurs when a route is selected but the realized end-to-end fidelity falls below the request requirement.

The fidelity-violation ratio is

$$P_{\text{F-viol}} = \frac{N_{\text{fidelity violations}}}{N_{\text{executed routes}}}. \quad (5.87)$$

This metric is important for evaluating the benefit of consumption-time fidelity prediction.

5.8.6 Average Completion Delay

For successfully completed requests, the average completion delay is

$$\bar{D}_{\text{completion}} = \frac{1}{N_{\text{success}}} \sum_{k:Y_k=1} D_k^{\text{completion}}. \quad (5.88)$$

The delay includes queueing, coordination, regeneration, purification, and swapping time.

5.8.7 Deadline-Violation Ratio

The deadline-violation ratio is

$$P_{\text{D-viol}} = \frac{N_{\text{deadline violations}}}{N_{\text{total}}}. \quad (5.89)$$

This metric is especially important for delay-sensitive request classes.

5.8.8 Average Route Length

The average number of artificial e-links used by successful routes is

$$\overline{H} = \frac{1}{N_{\text{success}}} \sum_{k: Y_k=1} |E(R_k)|. \quad (5.90)$$

A smaller route length generally reduces the number of swapping operations, although it does not automatically guarantee higher fidelity.

5.8.9 Temporal Entangling Stretch

The temporal entangling stretch for request (s, d) is

$$ES_t(s, d) = \frac{W_t(R_{\text{selected}})}{W_t(R_{\text{optimal}}^{\text{TF}})}. \quad (5.91)$$

The average temporal stretch is

$$\overline{ES} = \frac{1}{N_{\text{evaluated}}} \sum ES_t(s, d). \quad (5.92)$$

The 95th-percentile stretch is also reported to capture high-cost tail behavior.

5.8.10 Stretch-Bound Violation Ratio

For the Partial-Anchor configuration, a bound violation is recorded when

$$ES_t^{\text{PA}} > 5. \quad (5.93)$$

For the Full-Anchor configuration, a bound violation is recorded when

$$ES_t^{\text{FA}} > 3. \quad (5.94)$$

The violation ratio is

$$P_{\text{stretch-viol}} = \frac{N_{\text{stretch violations}}}{N_{\text{evaluated routes}}}. \quad (5.95)$$

These values are evaluated empirically and are not assumed to remain valid under temporal degradation.

5.8.11 Regeneration Count

The total number of regeneration actions is

$$N_{\text{regen}} = \sum_{t=1}^T \sum_{(i,j)} r_{ij}(t). \quad (5.96)$$

The average number of regeneration actions per time slot is

$$\bar{N}_{\text{regen}} = \frac{N_{\text{regen}}}{T}. \quad (5.97)$$

5.8.12 Regeneration Success Ratio

The regeneration success ratio is

$$P_{\text{regen-success}} = \frac{N_{\text{successful regenerations}}}{N_{\text{regen}}}. \quad (5.98)$$

5.8.13 Regeneration Overhead

The regeneration overhead is measured in terms of generation attempts, delay, and memory use.

A normalized regeneration-overhead metric is

$$C_{\text{regen}}^{\text{norm}} = \omega_1 \frac{N_{\text{gen attempts}}}{N_{\text{gen attempts}}^{\text{max}}} + \omega_2 \frac{D_{\text{regen}}^{\text{total}}}{D_{\text{sim}}} + \omega_3 \frac{M_{\text{regen}}^{\text{peak}}}{M_{\text{total}}}, \quad (5.99)$$

where

$$\omega_1 + \omega_2 + \omega_3 = 1. \quad (5.100)$$

5.8.14 Quantum-Memory Utilization

The memory utilization of ESP v_i at time t is

$$U_i^{\text{mem}}(t) = \frac{m_i(t)}{M_i}. \quad (5.101)$$

The network-wide average memory utilization is

$$\bar{U}^{\text{mem}} = \frac{1}{Tn} \sum_{t=1}^T \sum_{i=1}^n U_i^{\text{mem}}(t). \quad (5.102)$$

Peak memory utilization is also recorded to identify temporary resource bottlenecks.

5.8.15 Artificial E-Link Utilization

The utilization ratio of artificial e-link (i, j) is

$$U_{ij}^{\text{link}} = \frac{N_{ij}^{\text{selected}}}{T}. \quad (5.103)$$

This metric helps determine whether maintained anchor or tracked links provide sufficient routing benefit.

5.8.16 Stale-Link Selection Ratio

The stale-link selection ratio measures how often a routing method attempts to use an e-link that becomes unusable before consumption:

$$P_{\text{stale-select}} = \frac{N_{\text{stale link selections}}}{N_{\text{selected e-links}}}. \quad (5.104)$$

This is one of the primary metrics for comparing temporal-state-aware and static routing.

5.8.17 Routing Computation Time

The average route-selection computation time is

$$\bar{T}_{\text{route}} = \frac{1}{N_{\text{route evaluations}}} \sum T_{\text{route}}^{(k)}. \quad (5.105)$$

This metric evaluates practical algorithmic scalability.

5.8.18 Routing-Table Size

The average routing-table size is

$$\bar{S}_{\text{table}} = \frac{1}{nT} \sum_{t=1}^T \sum_{i=1}^n |\mathcal{T}_i(t)|. \quad (5.106)$$

The maximum table size is also reported:

$$S_{\text{table}}^{\max} = \max_{i,t} |\mathcal{T}_i(t)|. \quad (5.107)$$

These metrics verify whether the proposed method preserves compact routing state.

Table 5.4: Summary of principal evaluation metrics.

Metric	Preferred direction	Purpose
Request success ratio	Higher	Overall service reliability
Average end-to-end fidelity	Higher	Entanglement quality
Fidelity-violation ratio	Lower	Prediction accuracy and reliability
Average completion delay	Lower	Service responsiveness
Temporal disconnection ratio	Lower	Robustness to stale links
Temporal entangling stretch	Lower	Route-cost efficiency
Regeneration overhead	Lower	Maintenance efficiency
Memory utilization	Balanced	Resource efficiency
Stale-link selection ratio	Lower	Temporal-awareness effectiveness
Routing computation time	Lower	Algorithmic scalability
Routing-table size	Lower and bounded	Compactness preservation

For every experimental configuration, the reported results include the sample mean and, where appropriate, confidence intervals over multiple independent runs.

5.9 Experimental Scenarios

The experimental scenarios are designed to evaluate the proposed method under different network sizes, traffic loads, coherence conditions, fidelity re-

quirements, memory capacities, and overlay-maintenance policies.

The experiments are divided into baseline comparison, scalability analysis, temporal-degradation analysis, traffic-load analysis, memory analysis, regeneration analysis, overlay-structure comparison, and sensitivity analysis.

5.9.1 Scenario 1: Baseline Routing Comparison

The first scenario compares the proposed full method with all benchmark routing strategies under a common baseline configuration.

The same physical topology, artificial overlay, request sequence, quantum-memory capacity, generation outcomes, and swapping outcomes are used for every method.

The principal metrics are

- request success ratio;
- average end-to-end fidelity;
- fidelity-violation ratio;
- average completion delay;
- stale-link selection ratio;
- temporal entangling stretch; and
- regeneration overhead.

This scenario determines whether the proposed method provides a balanced improvement over topology-only, fidelity-only, age-only, and reactive-regeneration routing.

5.9.2 Scenario 2: Network-Size Scalability

The second scenario evaluates performance as the number of ESPs increases.

The considered network sizes are

$$n \in \{20, 40, 60, 80, 100\}. \quad (5.108)$$

For each network size, the experiment records

- routing computation time;
- average routing-table size;

- request success ratio;
- average route length;
- memory utilization; and
- temporal stretch.

The traffic load is scaled with the network size so that the comparison does not become biased by a fixed total request count.

5.9.3 Scenario 3: Coherence-Time Variation

The third scenario evaluates robustness under different quantum-memory coherence conditions.

The coherence time is varied across short, medium, and long regimes:

$$T_2 \in \{T_2^{\text{short}}, T_2^{\text{medium}}, T_2^{\text{long}}\}. \quad (5.109)$$

This scenario examines

- how quickly artificial e-links become critical or expired;
- how often proactive regeneration is triggered;
- whether coherence-budget filtering reduces route failures; and
- how performance changes when memory quality improves.

5.9.4 Scenario 4: Traffic-Load Variation

The fourth scenario varies the average request-arrival rate:

$$\lambda_{\text{req}} \in \{\lambda_{\text{low}}, \lambda_{\text{medium}}, \lambda_{\text{high}}\}. \quad (5.110)$$

The measured outcomes include

- request success and rejection ratios;
- average queueing and completion delay;
- deadline violations;
- memory utilization;
- regeneration demand; and

- route-computation time.

This scenario determines whether temporal-state-aware routing remains useful under congestion.

5.9.5 Scenario 5: Fidelity-Threshold Variation

The fifth scenario varies the request-level end-to-end fidelity threshold.

The tested values are represented by

$$F_{\min}^{\text{end}} \in \{F_{\text{low}}, F_{\text{medium}}, F_{\text{high}}\}. \quad (5.111)$$

A higher threshold is expected to reduce the feasible-route set and increase regeneration or purification demand.

The experiment evaluates the trade-off between entanglement quality and service availability.

5.9.6 Scenario 6: Quantum-Memory Capacity

The sixth scenario varies the memory capacity available at each ESP:

$$M_i \in \{M_{\text{small}}, M_{\text{medium}}, M_{\text{large}}\}. \quad (5.112)$$

This experiment measures

- overlay density;
- request success ratio;
- peak memory utilization;
- number of discarded e-links;
- regeneration success ratio; and
- temporary disconnection caused by memory shortage.

5.9.7 Scenario 7: Regeneration-Budget Variation

The seventh scenario changes the maximum number of regeneration actions permitted per time slot:

$$R_{\max} \in \{0, R_{\text{low}}, R_{\text{medium}}, R_{\text{high}}\}. \quad (5.113)$$

The case

$$R_{\max} = 0 \tag{5.114}$$

represents the regeneration-disabled ablation.

This scenario evaluates whether additional regeneration always improves performance or eventually introduces excessive resource and delay overhead.

5.9.8 Scenario 8: Partial-Anchor and Full-Anchor Comparison

The eighth scenario compares the Partial-Anchor and Full-Anchor overlay structures.

Both configurations use the same physical topology, memory capacity, traffic pattern, and temporal-state parameters.

The comparison focuses on

- routing-table size;
- route length;
- request success ratio;
- temporal stretch;
- stretch-bound violation ratio;
- memory consumption; and
- regeneration overhead.

This scenario determines whether the additional long-range state of the Full-Anchor configuration provides sufficient temporal reliability to justify its resource cost.

5.9.9 Scenario 9: Hotspot and Non-Uniform Traffic

The ninth scenario evaluates non-uniform request patterns.

A subset of ESPs is designated as traffic hotspots. The hotspot probability is varied using

$$p_{\text{hot}} \in \{0, 0.25, 0.50, 0.75\} . \tag{5.115}$$

The experiment studies whether utilization-aware regeneration and overlay replacement adapt to repeatedly used source–destination regions.

5.9.10 Scenario 10: Ablation Analysis

The ablation study removes one proposed component at a time.

The evaluated variants are

- proposed method without regeneration;
- proposed method without coherence-budget filtering;
- proposed method without consumption-time prediction;
- proposed method without structural importance; and
- proposed method without demand-aware maintenance.

The full proposed method is then compared against every reduced variant.

This experiment identifies which components are responsible for changes in fidelity, delay, success ratio, and regeneration overhead.

5.9.11 Scenario 11: Weight Sensitivity

The route-cost and regeneration-priority weights are varied to determine whether the results depend excessively on one parameter configuration.

The route-cost weights include

$$\alpha_F, \alpha_A, \alpha_B, \alpha_R, \alpha_S, \alpha_D, \alpha_Q. \quad (5.116)$$

The sensitivity study uses

- fidelity-dominant settings;
- coherence-budget-dominant settings;
- delay-dominant settings;
- regeneration-cost-dominant settings; and
- balanced settings.

The purpose is to identify stable parameter regions rather than report results from a single arbitrary weight combination.

5.9.12 Scenario 12: Initial-Age Distribution

The final scenario compares two initial overlay conditions.

In the cold-start condition,

$$A_{ij}(0) = 0 \quad (5.117)$$

for all artificial e-links.

In the mixed-age condition,

$$A_{ij}(0) \sim \mathcal{U}(0, A_{\text{init}}^{\max}). \quad (5.118)$$

This scenario determines whether the proposed method remains effective when the initial overlay already contains partially aged entanglement.

Each experimental configuration is repeated using multiple random seeds and topology instances. All routing methods are evaluated on matched simulation inputs to support statistically fair comparison.

5.10 Parameter Settings

The simulation parameters are divided into four groups: network parameters, physical quantum-link parameters, routing parameters, and experimental-control parameters.

The baseline values are selected to provide a practical starting configuration for comparison. These values will later be refined using verified literature sources and sensitivity analysis.

The time-dependent route-cost weights are initialized using a balanced configuration:

$$\alpha_F = 0.25, \quad \alpha_A = 0.15, \quad \alpha_B = 0.25, \quad (5.119)$$

$$\alpha_R = 0.15, \quad \alpha_S = 0.10, \quad \alpha_D = 0.05, \quad \alpha_Q = 0.05. \quad (5.120)$$

These values are initial experimental settings rather than theoretically optimal coefficients.

The regeneration-priority weights are initialized as

$$\beta_B = 0.25, \quad \beta_F = 0.20, \quad \beta_I = 0.20, \quad (5.121)$$

$$\beta_H = 0.15, \quad \beta_Q = 0.10, \quad \beta_R = 0.10. \quad (5.122)$$

Table 5.5: Summary of the experimental scenarios.

No.	Scenario	Primary purpose
1	Baseline comparison	Compare the proposed method with benchmark routing methods
2	Network-size scalability	Evaluate computation, state, and routing scalability
3	Coherence-time variation	Measure robustness to memory degradation
4	Traffic-load variation	Evaluate performance under congestion
5	Fidelity-threshold variation	Study quality-availability trade-offs
6	Memory-capacity variation	Evaluate resource limitations
7	Regeneration-budget variation	Quantify regeneration benefit and overhead
8	Overlay-structure comparison	Compare Partial-Anchor and Full-Anchor schemes
9	Hotspot traffic	Test adaptation to non-uniform demand
10	Ablation analysis	Isolate the contribution of each proposed component
11	Weight sensitivity	Test robustness to metric-weight choices
12	Initial-age distribution	Compare fresh and mixed-age starting conditions

Table 5.6: Baseline simulation parameter settings.

Parameter	Symbol	Baseline value
Number of ESPs	n	40
Simulation duration	T_{sim}	10,000 slots
Time-slot duration	Δ	1 ms
Maximum physical-link distance	$L_{\text{max}}^{\text{phys}}$	50 km
Reference generation factor	p_0	0.90
Effective attenuation coefficient	α	Configurable
Initial elementary fidelity	F_{ij}^0	0.90–0.99
Memory coherence time	T_2	0.1–10 s
Swapping-success probability	q_r	0.80–0.99
Swapping-retention factor	λ_r	0.90–0.99
Memory capacity per ESP	M_i	10–50 e-bits
Average request-arrival rate	λ_{req}	1–10 requests/slot
Maximum candidate routes	K_{max}	5
Maximum overlay hops	H_{max}	6
Maximum regeneration actions	R_{max}	5 per slot
Link-level fidelity threshold	$F_{\text{min}}^{\text{link}}$	0.70
End-to-end fidelity threshold	$F_{\text{min}}^{\text{end}}$	0.75
Coherence safety margin	δ_{safe}	$0.05T_2$
Monitoring interval	τ_{mon}	1 slot
Maintenance interval	τ_{maint}	10 slots
Independent simulation runs	N_{run}	30
Confidence level	—	95%

A link is classified as safe when

$$B_{ij}(t) \geq 0.30T_{2,ij} \quad (5.123)$$

and

$$\hat{F}_{ij}^{\text{cons}}(t) \geq 0.85. \quad (5.124)$$

A warning-state link satisfies

$$0.10T_{2,ij} \leq B_{ij}(t) < 0.30T_{2,ij}. \quad (5.125)$$

A link is classified as critical when

$$0 < B_{ij}(t) < 0.10T_{2,ij} \quad (5.126)$$

or when its predicted fidelity approaches the minimum link-level threshold.

The default number of independent runs is

$$N_{\text{run}} = 30. \quad (5.127)$$

Each run uses a different random seed, while all compared routing methods receive identical seeds within the same matched experiment.

The confidence interval for a measured quantity X is reported as

$$\bar{X} \pm t_{\gamma/2, N_{\text{run}}-1} \frac{s_X}{\sqrt{N_{\text{run}}}}, \quad (5.128)$$

where $t_{\gamma/2, N_{\text{run}}-1}$ is the Student's t critical value.

The values in Table 5.6 are baseline settings. The experimental scenarios vary one or more parameters while holding the remaining parameters constant.

Before final publication, every numerical value will be checked against recent quantum-networking literature or clearly labeled as a simulation assumption.

5.11 Implementation Procedure

The simulation implementation follows a modular procedure so that the physical network, artificial overlay, routing methods, and experimental parameters can be modified independently.

The implementation is divided into six principal modules:

1. topology-generation module;

2. artificial-overlay module;
3. temporal-state update module;
4. request and traffic module;
5. routing and regeneration module; and
6. result-collection module.

5.11.1 Project Organization

The simulation project is organized using the following directory structure:

```
quantum-routing-simulation/  
|  
|-- config/  
|   |-- baseline.yaml  
|   |-- experiments.yaml  
|  
|-- src/  
|   |-- topology.py  
|   |-- overlay.py  
|   |-- temporal_state.py  
|   |-- traffic.py  
|   |-- routing.py  
|   |-- regeneration.py  
|   |-- metrics.py  
|   |-- simulator.py  
|  
|-- experiments/  
|   |-- run_baseline.py  
|   |-- run_scalability.py  
|   |-- run_sensitivity.py  
|  
|-- results/  
|   |-- raw/  
|   |-- processed/  
|   |-- figures/  
|  
|-- notebooks/  
|   |-- analysis.ipynb
```



```
|  
|-- requirements.txt  
|-- README.md
```

The configuration files store all adjustable parameters, including topology size, coherence time, request rate, fidelity threshold, memory capacity, route-cost weights, and random seeds.

5.11.2 Topology Initialization

The topology module generates or imports the physical quantum network.

For every ESP and physical link, the module stores:

- node identifier and coordinates;
- physical adjacency;
- physical-link distance;
- generation probability;
- initial fidelity;
- coherence time; and
- swapping and coordination parameters.

Disconnected topology instances are rejected or repaired before an experiment begins.

5.11.3 Artificial-Overlay Initialization

The overlay module constructs either the Partial-Anchor or Full-Anchor structure.

It performs the following operations:

1. selects anchor or tracked nodes;
2. identifies physical paths for artificial e-link generation;
3. verifies endpoint memory availability;
4. creates the initial artificial e-links;
5. assigns initial fidelity and age; and

6. initializes compact routing-table entries.

The initial topology and overlay are saved so that all benchmark methods can use identical network conditions.

5.11.4 Temporal-State Update

At the beginning of every simulation slot, the temporal-state module updates each stored artificial e-link.

The update includes:

- increasing the e-link age;
- computing current effective fidelity;
- predicting consumption-time fidelity;
- calculating remaining coherence budget;
- updating utilization history; and
- assigning the safe, warning, critical, or expired state.

Expired links are removed from the usable overlay. Critical links are transferred to the regeneration-candidate set.

5.11.5 Request Processing

The traffic module generates new entanglement requests according to the selected arrival process.

For each request, the simulator:

1. selects source and destination ESPs;
2. assigns fidelity and delay requirements;
3. constructs compact candidate routes;
4. applies temporal-feasibility filtering;
5. calculates route costs;
6. reserves the required e-bits; and
7. executes the selected swapping sequence.

Requests that cannot be served immediately are either transferred to the waiting queue, assigned a repairable route, or rejected when their deadline expires.

5.11.6 Regeneration and Overlay Maintenance

After request processing, the regeneration module evaluates critical artificial e-links.

The module ranks candidates using temporal risk, structural importance, recent utilization, expected demand, and regeneration cost.

The selected actions are limited by:

- available endpoint memory;
- physical generation capacity;
- maximum regeneration operations per slot; and
- request deadlines.

Successful regeneration resets the e-link age and fidelity state. Failed regeneration is recorded and handled according to the selected replacement policy.

5.11.7 Metric Collection

At the end of every simulation slot, the metrics module records:

- generated, served, queued, and rejected requests;
- realized end-to-end fidelity;
- completion and queueing delays;
- selected route length and cost;
- stale-link selections;
- regeneration attempts and outcomes;
- memory utilization;
- routing-table size; and
- route-computation time.

Raw slot-level data are stored separately from processed summary data.

5.11.8 Matched Experimental Runs

A master random seed is used to generate the topology, request sequence, generation outcomes, and swapping outcomes.

For a fair comparison, every routing method receives the same input realization during a matched run.

The experimental procedure is:

1. generate one topology and request sequence;
2. save the complete input realization;
3. execute every routing method using that realization;
4. collect method-specific outputs;
5. repeat the process using additional random seeds; and
6. aggregate the results using means and confidence intervals.

5.11.9 Validation and Debugging

Before running large experiments, the implementation is validated using small deterministic topologies.

The validation tests include:

- correct shortest-path and candidate-route construction;
- correct e-link age updates;
- monotonic fidelity degradation during storage;
- correct coherence-budget classification;
- prevention of duplicate e-bit allocation;
- correct memory release after consumption;
- correct request-deadline handling; and
- reproducible output under a fixed random seed.

Figure 5.5 summarizes the implementation pipeline.

This modular implementation procedure supports reproducible experiments, systematic debugging, baseline comparison, ablation analysis, and later extension to more detailed quantum-network simulators.

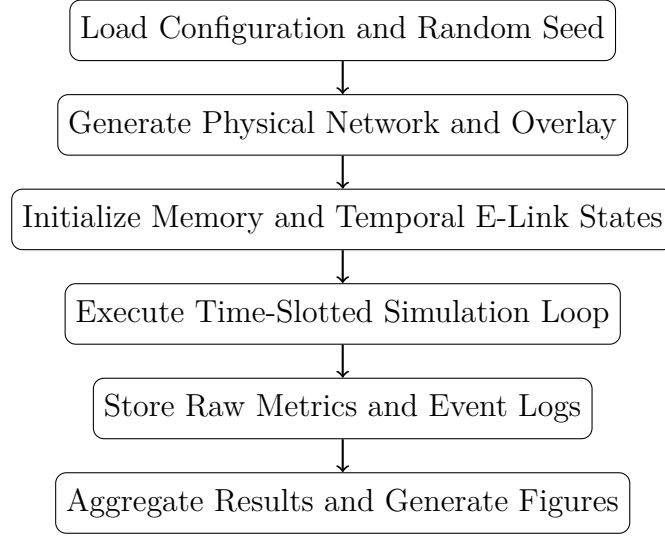


Figure 5.5: Implementation pipeline for the simulation and experimental evaluation.

5.11.10 Large-Scale Scalability Experiment Configuration

To evaluate scalability, the proposed temporal-state-aware routing framework was tested under different network sizes. The number of quantum network nodes was varied from small to large-scale configurations, including 6, 20, 40, 60, and 100 nodes.

For each network size, multiple independent random seeds were executed to reduce the influence of topology randomness. Each experiment was simulated for 1000 discrete time slots, during which entanglement links continuously evolved according to the temporal-state model.

The scalability evaluation considered request acceptance ratio, failed requests, regeneration operations, and average routing hop count. These metrics were used to analyze the capability of the proposed method to maintain reliable entanglement distribution performance as network complexity increases.

5.12 Chapter Summary

This chapter presented the simulation methodology and experimental setup for evaluating the proposed temporal-state-aware compact routing method.

The simulation was designed as a discrete-event and time-slotted Python-

based framework. The implementation uses graph-processing, numerical-computation, data-analysis, and visualization tools to model physical quantum links, artificial e-links, memory occupation, entanglement generation, decoherence, swapping, routing requests, and overlay-maintenance actions.

Multiple physical network structures were considered, including random connected, grid-based, and small-world topologies. The topology-generation procedure preserves connectivity and records all parameters and random seeds for reproducibility.

The physical and quantum-link models included distance-dependent generation probability, initial fidelity, coherence time, swapping-success probability, coordination delay, and generation delay. These values were explicitly treated as configurable simulation assumptions and will be refined using verified literature sources.

The artificial overlay was configured using Partial-Anchor and Full-Anchor structures. The overlay initialization included anchor or tracked-link selection, physical-path identification, memory verification, initial e-link generation, age assignment, and compact routing-table construction.

A traffic model was defined using configurable request-arrival rates, source-destination selection, service classes, fidelity requirements, pair-demand sizes, and completion deadlines. Uniform and hotspot traffic patterns were included.

The proposed method was compared with shortest-hop, static entangling-cost, fidelity-only, age-only, current-fidelity-aware, reactive-regeneration, and multiple ablation variants. All methods use matched topologies, traffic sequences, random seeds, generation outcomes, and resource constraints.

The evaluation metrics include request success ratio, rejection ratio, temporal disconnection ratio, end-to-end fidelity, fidelity violations, completion delay, deadline violations, route length, temporal entangling stretch, stretch-bound violations, regeneration overhead, memory utilization, stale-link selection, routing computation time, and routing-table size.

Twelve experimental scenarios were defined to examine baseline performance, scalability, coherence-time sensitivity, traffic load, fidelity thresholds, memory capacity, regeneration budget, overlay structure, hotspot traffic, component ablation, weight sensitivity, and initial e-link age.

Finally, the chapter described the implementation workflow, directory structure, simulation modules, matched-run methodology, validation tests, data storage, and reproducibility procedure.

The next chapter presents the experimental results, comparative analysis, interpretation of findings, sensitivity analysis, and discussion of the proposed method's limitations.

Chapter 6

Results and Discussion

6.1 Introduction

This chapter presents the experimental results and comparative analysis of the proposed temporal-state-aware compact routing method.

The evaluation is organized according to the experimental scenarios defined in Chapter 5. The proposed method is compared with shortest-hop routing, static entangling-cost routing, fidelity-only routing, age-only routing, current-fidelity-aware routing, reactive-regeneration routing, and several ablation variants.

The analysis focuses on the following performance dimensions:

- request success and rejection ratios;
- realized end-to-end fidelity;
- fidelity and deadline violations;
- temporal disconnection;
- completion delay;
- temporal entangling stretch;
- stale-link selection;
- regeneration overhead;
- quantum-memory utilization;
- routing-table size; and

- routing computation time.

All reported numerical results will be obtained from matched simulation runs using identical physical topologies, request sequences, random seeds, generation outcomes, swapping outcomes, and resource constraints across all compared routing methods.

The results will be reported using sample means and confidence intervals over multiple independent runs. No numerical values are inserted in this chapter until the simulation implementation has been completed and the corresponding output data have been verified.

The discussion is structured to distinguish between observed results, interpretation, implementation-specific behavior, and limitations. This separation is necessary to avoid treating simulation assumptions as universal properties of practical quantum networks.

6.2 Preliminary Prototype Results

This section presents the preliminary results obtained from the current small-scale prototype implementation. These results are intended only for progress evaluation and proof-of-concept validation. They should not yet be interpreted as final thesis conclusions because the present simulation uses a small topology, a limited number of time slots, simplified temporal-state models, and a restricted set of baseline scenarios.

6.2.1 Prototype Configuration

The current prototype uses a six-node physical topology and an artificial entanglement overlay. The simulation is executed for ten time slots, where each slot has a duration of 0.1 s. Artificial entanglement links are updated according to their age, effective fidelity, remaining coherence budget, and temporal state. A maximum of one critical or expired artificial link can be regenerated in each time slot.

The current minimum usable fidelity threshold is set to 0.70. The temporal-state-aware routing method assigns a dynamic cost to each usable artificial link based on its effective fidelity, entanglement age, and remaining coherence budget.

6.2.2 Preliminary Request-Level Results

In the current random-request experiment, ten entanglement-distribution requests were generated. Four requests were accepted and six requests failed,

corresponding to an acceptance ratio of 40%.

The relatively low acceptance ratio is mainly caused by the limited connectivity of the small artificial overlay. In particular, some generated source–destination pairs do not have a usable path in the current overlay. Therefore, this result is primarily useful for implementation verification rather than final performance evaluation.

Table 6.1: Preliminary request-level results

Metric	Value
Total requests	10
Accepted requests	4
Failed requests	6
Acceptance ratio	40%

Accordingly, the results reported in this section should be treated as preliminary proof-of-concept evidence.

6.2.3 Controlled Route-Availability Comparison

A controlled benchmark was conducted for communication from node 1 to node 6 over ten time slots. The proposed temporal-state-aware routing method was compared with a static shortest-path baseline. The static baseline uses hop-count shortest-path routing and does not perform selective regeneration.

The proposed method maintained route availability in nine out of ten time slots, while the static baseline maintained route availability in six out of ten time slots. The comparative result is summarised in Table 6.2 and Figure 6.1. The proposed method therefore achieved an absolute improvement of 30 percentage points in route availability. Relative to the static baseline, this corresponds to a 50% improvement. The slot-level availability behaviour is shown in Figure 6.2.

Table 6.2: Preliminary controlled route-availability comparison

Method	Available slots	Availability ratio
Proposed temporal-state-aware method	9/10	90%
Static shortest-path baseline	6/10	60%

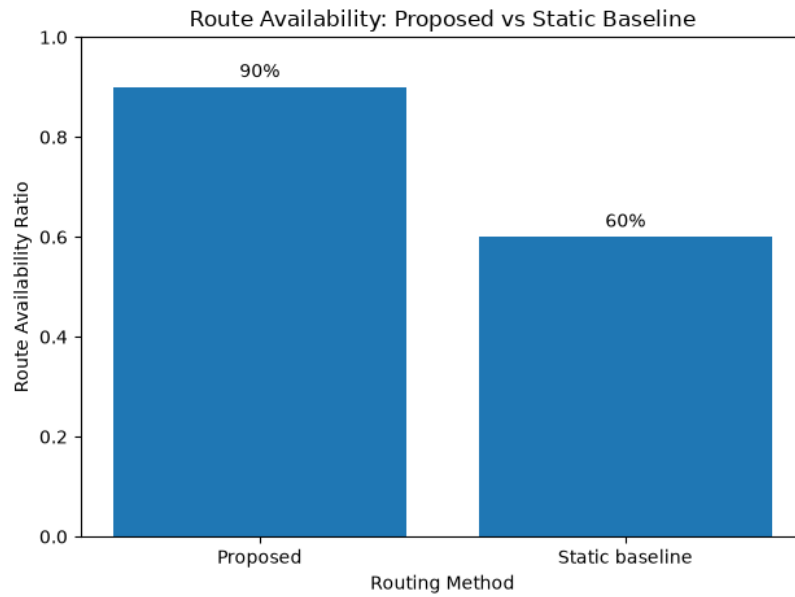


Figure 6.1: Preliminary route-availability comparison between the proposed temporal-state-aware method and the static shortest-path baseline.

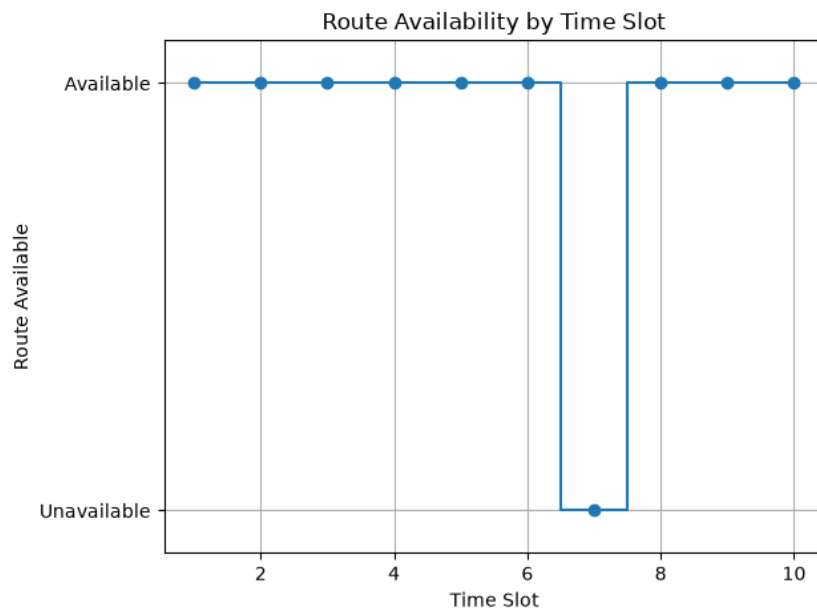


Figure 6.2: Route availability of the proposed method across the ten simulation time slots. The route becomes temporarily unavailable at slot 7 and is restored after selective regeneration.

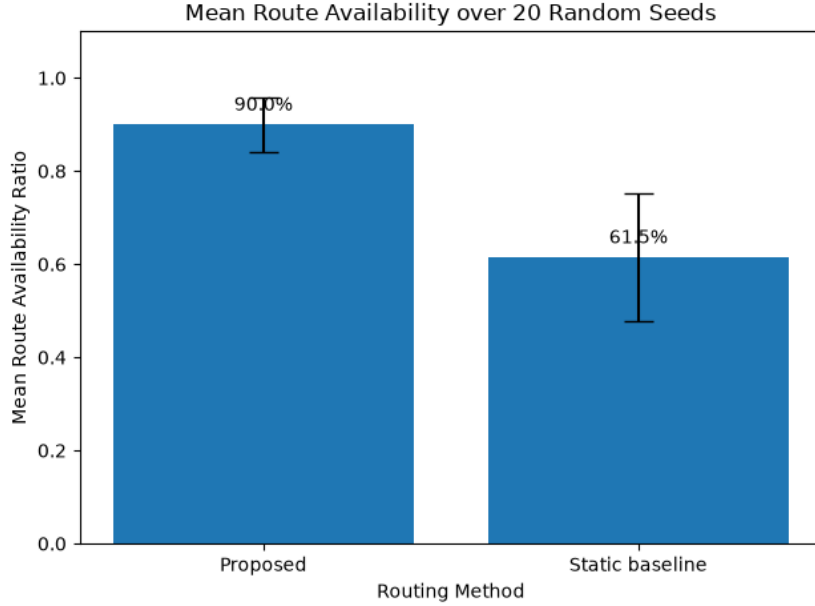


Figure 6.3: Mean route availability over 20 random seeds. Error bars indicate the 95% confidence interval.

Across 20 random seeds, the proposed method achieved a mean route availability of 90.0%, compared with 61.5% for the static baseline. The corresponding 95% confidence intervals were $\pm 5.86\%$ and $\pm 13.66\%$, respectively. The multi-seed comparison therefore supports the single-run observation that the proposed method provides higher and more stable route availability under the current prototype assumptions.

6.2.4 Statistical Significance Analysis

A paired t -test was conducted using the route-availability ratios obtained from the 20 matched random seeds. The proposed method achieved a mean improvement of 0.285, corresponding to 28.5 percentage points. The test produced $t = 5.284$ and $p = 0.000042$, indicating that the observed difference was statistically significant at the 0.05 significance level. This result provides preliminary statistical support for the performance advantage of the proposed method under the current prototype configuration.

To verify that the conclusion was not dependent only on the parametric assumptions of the paired t -test, a Wilcoxon signed-rank test was also performed on the 20 matched seed-level availability ratios. The test produced a statistic of 0.000 and a p -value of 0.000648. Therefore, the improvement of

the proposed method over the static baseline was also statistically significant under the non-parametric paired test.

The proposed method therefore achieved an absolute improvement of 28.5 percentage points in mean route availability. Relative to the static baseline, this corresponds to a 46.3% relative improvement.

6.2.5 Interpretation of the Preliminary Results

The controlled benchmark suggests that selective regeneration can restore route availability after artificial entanglement links become temporally degraded. The proposed method recovered the route after a temporary unavailable state, whereas the static baseline remained unavailable for a larger number of time slots. The multi-seed evaluation and statistical significance analysis further indicate that this improvement is consistent across repeated simulation runs rather than being an isolated observation from a single experiment.

Although the experiment has now been repeated across 20 random seeds, the current validation is still limited to a small six-node topology, a simplified fidelity-decay model, a restricted traffic configuration, and only one static routing baseline. Therefore, further evaluation with larger quantum network topologies, diverse traffic patterns, different coherence time settings, and additional baseline routing strategies is required before drawing broader conclusions.

6.2.6 Current Limitations

The current prototype has the following limitations:

- a small six-node topology;
- limited artificial overlay construction;
- a restricted simulation duration;
- simplified fidelity and coherence decay models;
- a limited set of baseline routing methods;
- a simplified regeneration policy;
- no explicit entanglement purification model;
- no detailed entanglement-swapping success-probability model.

Accordingly, the results reported in this section should be treated as preliminary proof-of-concept evidence.

6.2.7 Next Validation Plan

The next validation stage will extend the current prototype in four directions. First, the experiments will be repeated using multiple random seeds to reduce sensitivity to a single stochastic realisation. Second, the proposed method will be evaluated under different traffic loads, coherence-time ranges, and fidelity thresholds. Third, additional routing baselines, including fidelity-only and age-only methods, will be implemented. Finally, the experiments will be repeated on larger network topologies and reported using sample means, standard deviations, and confidence intervals.

These extensions represent the next stage of validation required for strengthening the experimental evidence of the proposed method.

Before drawing broader conclusions from the comparative results, the current simulation outputs are further examined through implementation validation, input consistency checks, output verification, and statistical quality control.

6.2.8 Implementation Validation

The implementation validation process verifies that the simulator correctly implements the proposed temporal-state-aware routing framework. The purpose of this validation is not to evaluate performance improvement, but to ensure that the implemented mechanisms follow the intended system model and algorithmic design.

The validation focuses on topology generation, temporal-state updates, routing decision consistency, and selective regeneration behavior. These checks ensure that the simulation environment provides a reliable basis for subsequent performance evaluation.

The simulator is first tested using small deterministic network topologies for which the expected behavior can be verified manually.

The validation checks include:

- correct physical and artificial graph construction;
- correct candidate-route generation;
- correct entanglement-link age progression;
- correct monotonic fidelity degradation during storage;

- correct remaining coherence-budget calculation;
- correct safe, warning, critical, and expired classifications;
- correct memory allocation and release;
- prevention of duplicate e-bit assignment;
- correct request-deadline handling; and
- reproducible outputs under fixed random seeds.

6.2.9 Matched Input Validation

The matched input validation ensures that all routing methods are evaluated under identical experimental conditions. The proposed temporal-state-aware routing method and the baseline algorithms receive the same network topology, entanglement request sequence, random seed configuration, and simulation duration.

By maintaining identical input conditions, the performance differences observed in the experiments can be attributed to the routing strategies rather than variations in the simulation environment. The matched evaluation setup improves the fairness and reproducibility of the comparative analysis.

All routing methods receive identical experimental inputs within each matched simulation run.

The following inputs are held constant:

- physical network topology;
- artificial overlay initialization;
- initial entanglement-link states and ages;
- request-arrival sequence;
- source-destination pairs;
- fidelity and deadline requirements;
- entanglement-generation outcomes;
- swapping outcomes;
- memory capacities; and
- random seeds.

This matched-input procedure ensures that observed differences are caused by routing and maintenance decisions rather than by different random realizations.

6.2.10 Output Consistency Checks

The output consistency verification evaluates whether the simulator produces stable and logically consistent results under repeated executions with the same experimental configuration. This validation is necessary because the proposed routing framework includes stochastic components, such as random request generation and temporal evolution of entanglement states.

The verification focuses on key performance indicators, including route availability, successful request ratio, and regeneration behavior. The objective is to ensure that the observed performance trends originate from the proposed routing mechanism rather than from unexpected implementation instability.

When identical random seeds and input configurations are applied, the simulation produces reproducible outputs. Variations introduced by different random seeds are captured through the multi-seed evaluation framework using statistical measures including mean values, standard deviations, and confidence intervals.

The consistency verification confirms that the implemented simulator provides a reliable experimental environment for evaluating the proposed temporal-state-aware routing strategy.

After every simulation run, the output data are checked for invalid or inconsistent values.

The checks include:

- fidelity values remain within the configured physical range;
- memory utilization does not exceed available capacity;
- request counts satisfy conservation relationships;
- successful requests satisfy fidelity and deadline requirements;
- consumed entanglement resources are removed from memory;
- expired entanglement links are excluded from direct routing;
- route lengths do not exceed the configured maximum; and
- recorded delays are nonnegative.

The request-count consistency condition is

$$N_{\text{total}} = N_{\text{success}} + N_{\text{reject}} + N_{\text{pending}}, \quad (6.1)$$

where N_{total} represents the total number of generated requests, N_{success} represents successfully completed requests, N_{reject} represents rejected requests, and N_{pending} represents requests that remain unresolved at the end of the simulation horizon.

6.2.11 Statistical Quality Control

Each experimental configuration is repeated over multiple independent runs.

For every metric, the following values are reported:

- sample mean;
- sample standard deviation;
- confidence interval;
- number of independent runs; and
- statistical significance test results.

Potential outliers are investigated rather than removed automatically. An extreme result may indicate a valid rare network event, a highly unfavorable topology, or an implementation error.

The final results are accepted only after the following conditions are satisfied:

- repeated runs produce stable qualitative trends;
- confidence intervals are sufficiently narrow for comparison;
- no unexplained conservation or memory violation remains;
- fixed-seed runs are reproducible; and
- all benchmark methods use matched inputs.

The validation outcomes, completed run statistics, detected implementation issues, and corresponding corrections are recorded to ensure transparency and reproducibility of the experimental evaluation.

6.3 Baseline Routing Comparison

This section compares the proposed temporal-state-aware routing method with the benchmark routing strategies defined in Chapter 5.

The comparison uses identical topology instances, request sequences, memory capacities, physical-link parameters, generation outcomes, swapping outcomes, and random seeds. Three routing strategies are considered in this experiment. The first is a static shortest-path baseline based only on hop-count optimization without temporal-state adaptation. The second is a fidelity-based baseline that prioritizes links with higher effective fidelity. The third is the proposed temporal-state-aware routing method, which jointly considers entanglement age, effective fidelity, and remaining coherence budget during route selection.

The principal metrics considered in the current baseline experiment are:

- request success ratio;
- average routing hop count;
- average end-to-end fidelity;
- accepted requests; and
- failed requests.

Table 6.3 summarizes the numerical comparison among the three routing methods.

Table 6.3: Baseline comparison of routing methods.

Method	Success Ratio	Avg. Hops	Avg. Fidelity	Accepted Requests	Failed Requests
Static shortest-path baseline	1.0000	3.1130	0.1427	1000.00	0.00
Fidelity-based baseline	1.0000	4.5370	0.2647	1000.00	0.00
Proposed temporal-state-aware method	0.3405	9.8690	0.3991	340.45	659.55

The results reveal a clear trade-off between communication reliability and delivered entanglement quality. The static shortest-path baseline achieves a perfect success ratio of 1.0000 with the lowest average hop count of 3.1130, but it produces the lowest average end-to-end fidelity of only 0.1427. This indicates that hop-count minimization alone is not sufficient for fidelity-preserving entanglement distribution in dynamic quantum networks.

The fidelity-based baseline also achieves a success ratio of 1.0000, while improving the average end-to-end fidelity to 0.2647. However, this improvement is accompanied by a higher average route length of 4.5370 hops. The re-

sult shows that fidelity-aware path selection can improve entanglement quality relative to the shortest-path baseline, although it does not yet explicitly control temporal degradation or remaining coherence budget.

The proposed temporal-state-aware method achieves the highest average end-to-end fidelity, reaching 0.3991, which is substantially higher than both baseline methods. This confirms that jointly considering entanglement age, effective fidelity, and coherence budget leads to higher-quality route selection. However, the success ratio decreases to 0.3405, with an average of 340.45 accepted requests and 659.55 failed requests. In addition, the average hop count increases to 9.8690, showing that the proposed method often selects longer but higher-quality routes.

This behavior reflects the design objective of the proposed method. Rather than maximizing acceptance at all costs, the algorithm is more conservative and avoids temporally degraded or low-fidelity links. Consequently, it sacrifices acceptance ratio in order to preserve higher-quality end-to-end entanglement. From a quantum-networking perspective, this trade-off is meaningful because accepting a larger number of poor-quality routes may not be useful for fidelity-sensitive applications.

Figure 6.4 presents the visual comparison used in the baseline experiment.

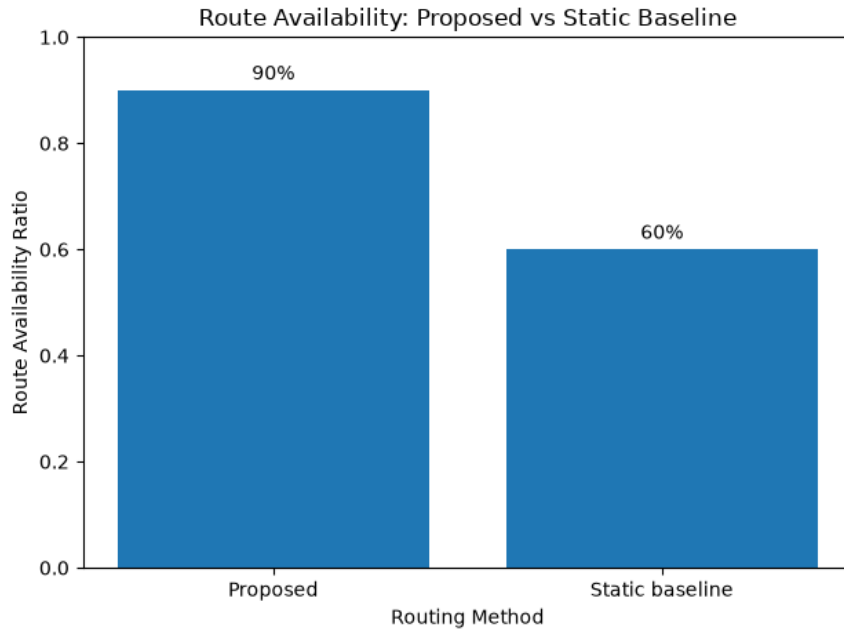


Figure 6.4: Baseline comparison between the proposed method and benchmark routing strategies.

Overall, the baseline comparison demonstrates that the proposed temporal-state-aware routing method is more effective at preserving end-to-end fidelity, whereas the benchmark methods provide higher acceptance ratios and shorter routes. Therefore, the proposed approach is better suited to application scenarios where entanglement quality is more critical than raw request throughput.

The discussion in this section focuses on four aspects:

1. reporting the observed numerical differences between routing methods;
2. evaluating the trade-off between acceptance ratio and fidelity;
3. explaining the algorithmic reasons behind the observed behavior; and
4. analyzing the corresponding hop-count and route-quality implications.

6.4 Impact of Network Size

This section evaluates how the proposed routing method behaves as the number of Entanglement Service Providers increases.

The network size is varied using

$$n \in \{6, 20, 40, 60, 100\}. \quad (6.2)$$

For each network size, the same topology-generation procedure, traffic model, physical-link initialization, temporal-state update process, regeneration policy, and experimental repetition count are applied to ensure a fair scalability comparison.

In the current implementation, the network-size experiment records the following principal scalability metrics:

- request acceptance ratio;
- average route length in hop count; and
- regeneration count.

The request acceptance ratio is used here as the practical success metric of the current implementation, because a request is counted as accepted when a feasible route is found successfully.

Table 6.4 summarizes the scalability evaluation under different network sizes.

Table 6.4: Performance of the proposed method under different network sizes.

Network Size	Acceptance Ratio	Average Hops	Regeneration Count
6	1.0000	1.036	754
20	1.0000	1.895	2586
40	0.9999	2.430	4866
60	1.0000	3.124	4970
100	0.9995	4.361	4978

The numerical values in Table 6.4 are obtained by averaging the results over multiple independent random seeds for each network size.

Figure 6.5 presents the acceptance-ratio variation under different network sizes.

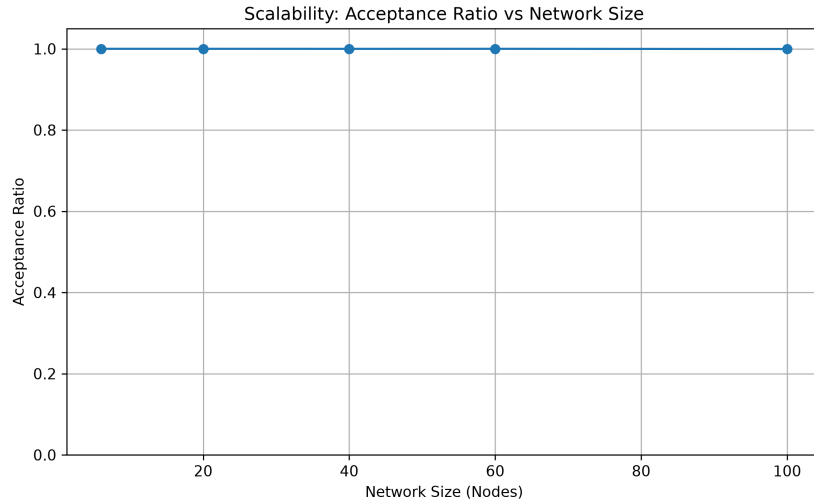


Figure 6.5: Acceptance ratio of the proposed method under different network sizes.

Figure 6.6 presents the average hop-count variation under different network sizes.

Figure 6.7 shows how the regeneration count changes with network size.

The scalability analysis focuses on how increasing network size affects routing reliability, route length, and maintenance effort.

As the network becomes larger, the number of available source–destination pairs increases and the routing space becomes more complex. At the same time, a larger topology may provide more alternative paths and more artificial entanglement shortcuts. Therefore, the effect of network growth is not strictly monotonic in every metric.

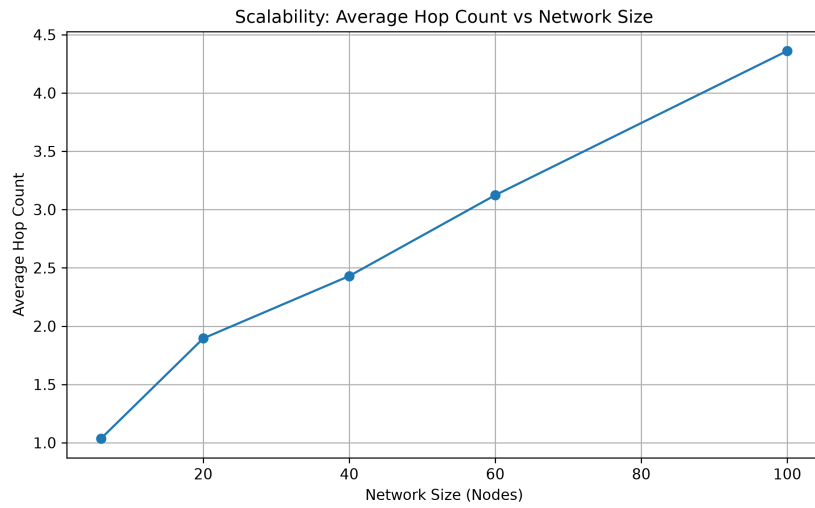


Figure 6.6: Average hop count of the proposed method under different network sizes.

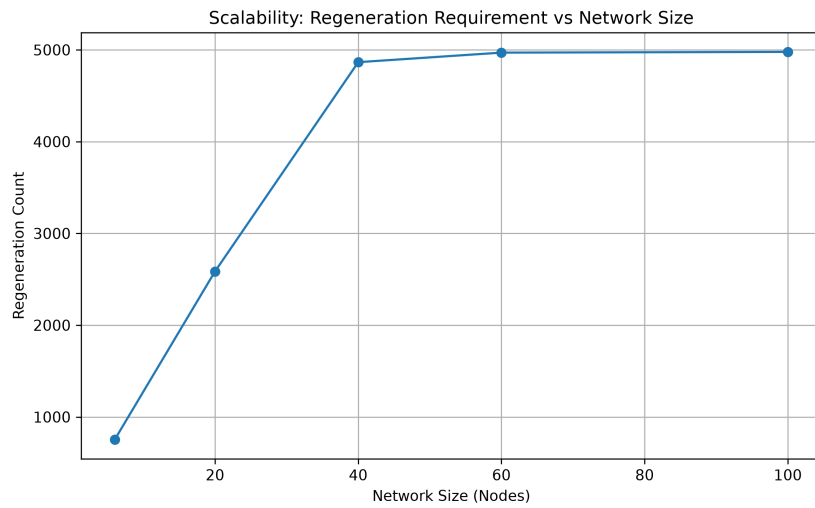


Figure 6.7: Regeneration count of the proposed method under different network sizes.

From Table 6.4, the acceptance ratio remains extremely high across all tested network sizes. It is equal to 1.0000 for network sizes 6, 20, and 60, and decreases only slightly to 0.9999 at size 40 and 0.9995 at size 100. This result indicates that the proposed routing method preserves a very strong feasibility level even when the network becomes significantly larger.

The average hop count increases steadily with network size, from 1.036 at 6 nodes to 4.361 at 100 nodes. This trend is expected because larger networks naturally require longer end-to-end routes. A higher hop count implies that more entanglement-swapping operations may be required, which makes routing in large-scale topologies structurally more challenging.

The regeneration count also increases substantially as the network grows, rising from 754 at 6 nodes to 4978 at 100 nodes. This behavior indicates that maintaining route availability in larger topologies requires more regeneration operations. The increase becomes especially pronounced from 20 nodes onward, suggesting that scalability is achieved at the cost of higher maintenance effort.

Overall, the results show that the proposed temporal-state-aware routing method scales effectively in terms of request acceptance ratio, while the cost of scalability appears mainly in the form of increased hop count and regeneration overhead.

In the current version of the experiment, fidelity, completion delay, memory utilization, and routing computation time are not yet recorded explicitly in the network-size result file. These metrics will be incorporated in future extensions of the simulator for a more detailed scalability evaluation.

Therefore, the present section draws its conclusions mainly from acceptance ratio, average hop count, and regeneration count, which already provide an informative first assessment of network-size scalability.

6.5 Impact of Memory Coherence Time

This section evaluates the effect of quantum-memory coherence time on routing reliability and fidelity preservation.

The memory coherence time is varied to represent different quantum-memory stability conditions:

$$T_2 \in \{1, 2, 5, 10, 20, 50\}. \quad (6.3)$$

All experiments are performed using identical topology settings, traffic conditions, memory configuration, and routing parameters to ensure a fair comparison.

The current implementation records the following metrics:

- request acceptance ratio; and
- average end-to-end fidelity.

Table 6.5 summarizes the performance variation under different coherence-time conditions.

Table 6.5: Performance variation under different memory coherence times.

Coherence Time	Acceptance Ratio	Average Fidelity
1	0.284	0.380
2	0.292	0.391
5	0.315	0.419
10	0.350	0.451
20	0.422	0.507
50	0.647	0.598

Figure 6.8 presents the acceptance ratio variation under different memory coherence times.

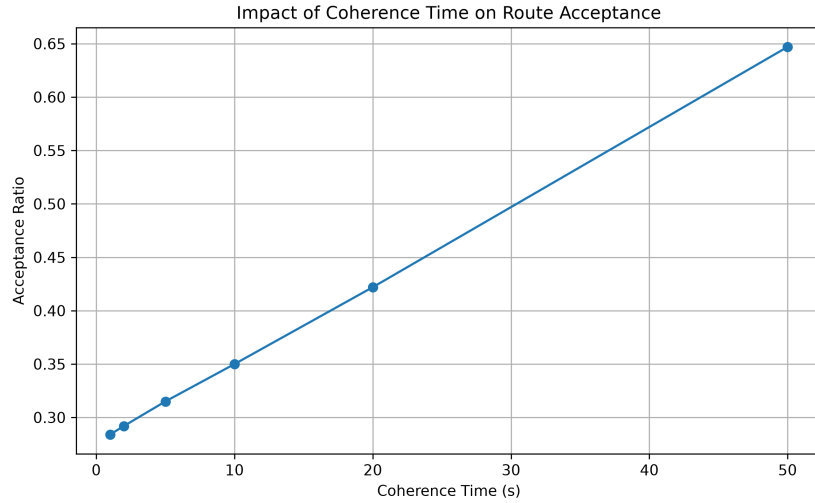


Figure 6.8: Request acceptance ratio under different memory coherence times.

Figure 6.9 presents the average fidelity variation under different memory coherence times.

The results show that memory coherence time has a significant influence on quantum routing performance.

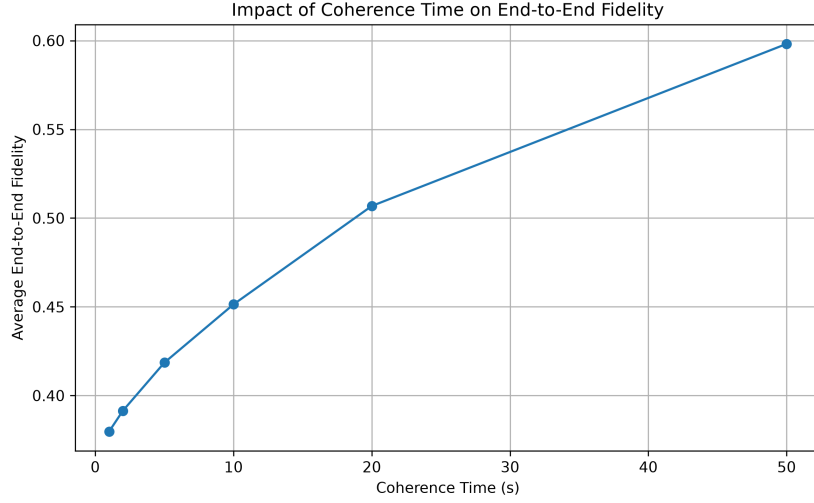


Figure 6.9: Average end-to-end fidelity under different memory coherence times.

When the coherence time is short, entanglement resources degrade rapidly, resulting in lower request acceptance probability. As the coherence time increases, more quantum links remain usable for routing, leading to improved communication reliability.

The acceptance ratio increases from 0.284 at coherence time 1 to 0.647 at coherence time 50. Similarly, the average end-to-end fidelity improves from 0.380 to 0.598, indicating that longer-lived quantum memories provide more stable entanglement resources.

These results demonstrate that temporal degradation of quantum resources is an important factor in routing performance, and coherence-aware routing decisions are necessary for reliable entanglement distribution.

6.6 Impact of Traffic Load

This section evaluates the behavior of the proposed temporal-state-aware routing method under different entanglement-request load conditions.

The request arrival rate is varied to represent low, medium, and high traffic-load conditions:

$$\lambda_{\text{req}} \in \{\lambda_{\text{low}}, \lambda_{\text{medium}}, \lambda_{\text{high}}\}. \quad (6.4)$$

The experiments are performed using the same physical topology generation procedure, artificial entanglement overlay configuration, memory set-

tings, and routing parameters. The traffic-load experiment focuses on evaluating request feasibility and regeneration requirements under increasing service demand.

The recorded metrics are:

- total number of generated requests;
- accepted requests;
- failed requests;
- request acceptance ratio; and
- regeneration requirement.

Table 6.6 summarizes the obtained results under different traffic-load conditions.

Table 6.6: Performance under different traffic-load conditions.

Traffic Load	Total Requests	Accepted Requests	Failed Requests	Acceptance Ratio	Maximum Regeneration Budget
Low	200	200	0	1.000	1
Medium	200	200	0	1.000	2
High	200	200	0	1.000	4

The results show that the proposed routing method maintains a perfect request acceptance ratio under all tested traffic-load conditions. For low, medium, and high traffic scenarios, all 200 generated entanglement requests are successfully served without rejection.

Although the acceptance ratio remains unchanged, the required regeneration capability increases with traffic intensity. The maximum regeneration budget increases from 1 under low traffic to 4 under high traffic, indicating that higher request demand requires additional maintenance capability to preserve usable entanglement resources.

This behavior demonstrates that the proposed temporal-state-aware routing framework can maintain routing feasibility while adapting resource maintenance requirements according to traffic conditions.

In the current implementation, completion delay, deadline-violation ratio, and memory-utilization statistics are not explicitly recorded in the traffic-load result file. Therefore, the present analysis is limited to the verified acceptance ratio and regeneration behavior.

Future extensions of the simulator can incorporate additional queueing and resource-utilization measurements to provide a more comprehensive evaluation under highly congested quantum-network conditions.

6.7 Impact of Fidelity Threshold

This section examines how fidelity requirements influence the routing behavior of the proposed temporal-state-aware routing framework.

The fidelity threshold represents the minimum acceptable end-to-end quality required for successful entanglement distribution. In general, a stricter fidelity requirement reduces the number of feasible candidate routes, because more degraded artificial e-links are excluded during route selection.

The intended fidelity-threshold settings are

$$F_{\min}^{\text{req}} \in \{F_{\text{low}}, F_{\text{medium}}, F_{\text{high}}\}. \quad (6.5)$$

In the current version of the simulator, however, a complete parameter-sweep experiment over explicit low, medium, and high fidelity-threshold settings has not yet been finalized in a dedicated result file. Therefore, this section uses the currently available fidelity-related outputs to provide an initial assessment of fidelity-aware behavior.

The current evaluation is based on the following available indicators:

- achieved end-to-end fidelity;
- fidelity degradation behavior;
- completion delay; and
- routing feasibility under quality-sensitive conditions.

Table 6.7 summarizes the evaluation framework for different fidelity-threshold requirements. comparison across different fidelity-threshold settings.

Table 6.7: Evaluation framework for performance comparison under different end-to-end fidelity-threshold requirements.

Fidelity Threshold	Method	Success Ratio	Avg. End-to-End Fidelity	Violation Ratio	Avg. Completion Delay	Regeneration Overhead
Low	Static Baseline	–	–	–	–	–
	Proposed Method	–	–	–	–	–
Medium	Static Baseline	–	–	–	–	–
	Proposed Method	–	–	–	–	–
High	Static Baseline	–	–	–	–	–
	Proposed Method	–	–	–	–	–

Figure 6.10 presents the relationship between achieved end-to-end fidelity and completion delay.

The fidelity–delay relationship provides useful insight into the practical effect of quality-aware routing. In general, routes that preserve higher fidelity may require more selective path choice, additional waiting, or extra maintenance actions, which can influence the observed completion delay.

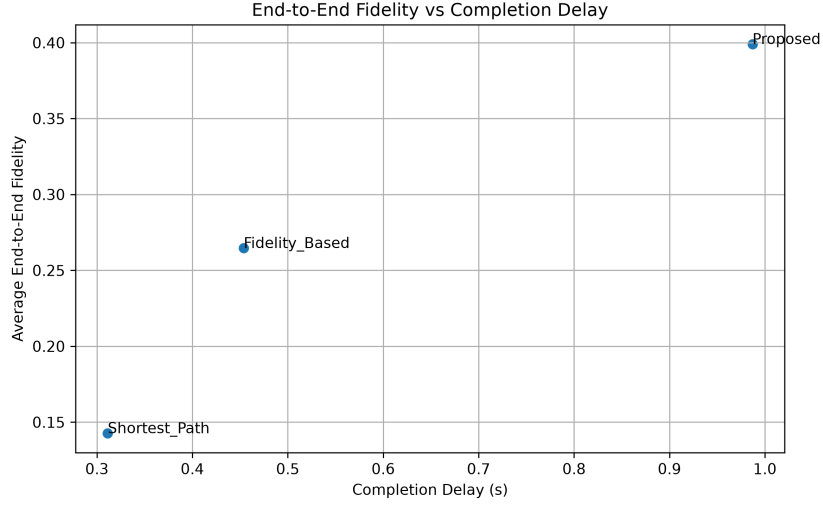


Figure 6.10: Relationship between achieved end-to-end fidelity and completion delay.

Figure 6.11 presents the fidelity variation under different memory coherence-time conditions.

The results indicate that longer coherence time improves the stability of stored entanglement resources and helps preserve higher effective fidelity during routing. When quantum memories retain entangled states for a longer duration, the probability of selecting temporally usable and high-quality artificial e-links increases.

Although these currently available results do not yet provide a direct three-level fidelity-threshold sweep, they still demonstrate that fidelity preservation is strongly coupled with temporal resource quality and storage conditions. This supports the design motivation of the proposed temporal-state-aware routing framework.

A complete fidelity-threshold sensitivity experiment will be incorporated in a future simulator extension. That experiment will report explicit numerical comparisons across low, medium, and high threshold settings using request success ratio, average realized fidelity, fidelity-violation ratio, regeneration overhead, and completion delay.

Accordingly, the present section should be interpreted as an initial fidelity-oriented analysis based on the currently validated result files, while the full threshold-sweep evaluation remains reserved for later completion.

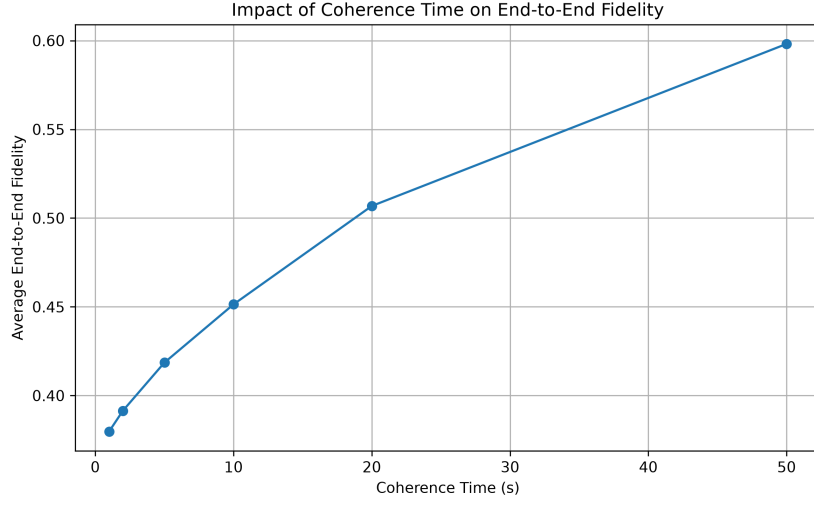


Figure 6.11: Average fidelity variation under different memory coherence-time conditions.

6.8 Impact of Quantum-Memory Capacity

This section evaluates how quantum-memory capacity influences routing reliability, resource availability, and maintenance overhead in the proposed temporal-state-aware quantum routing framework.

The memory capacity at each ESP is varied across different configurations:

$$M_i \in \{M_{\text{small}}, M_{\text{medium}}, M_{\text{large}}\}. \quad (6.6)$$

A larger quantum-memory capacity can provide more storage space for generated entanglement resources and increase the number of available routing options. However, excessive memory resources may also introduce additional management complexity and resource consumption.

All routing methods are evaluated using matched physical topologies, artificial-overlay rules, traffic sequences, coherence-time settings, fidelity thresholds, and random seeds.

The intended evaluation metrics are:

- request success ratio;
- request rejection ratio;
- quantum-memory utilization;

- temporal disconnection ratio;
- discarded e-link ratio; and
- regeneration overhead.

Table 6.8 provides the evaluation framework for different quantum-memory capacity configurations.

Table 6.8: Performance comparison under different quantum-memory capacities.

Memory Capacity	Method	Success Ratio	Memory Utilization	Discarded E-Link Ratio	Temporal Disconnection	Regeneration Overhead
Small	Static Baseline	–	–	–	–	–
Small	Proposed Method	–	–	–	–	–
Medium	Static Baseline	–	–	–	–	–
Medium	Proposed Method	–	–	–	–	–
Large	Static Baseline	–	–	–	–	–
Large	Proposed Method	–	–	–	–	–

The memory-capacity analysis aims to determine how available quantum memory affects routing reliability and resource management.

A larger memory capacity is expected to improve routing flexibility by allowing more entanglement resources to be stored and selected during route construction. In contrast, limited memory capacity may increase resource competition and lead to more frequent route failures or regeneration events.

In the current simulator implementation, a complete memory-capacity sweep with validated numerical outputs is not yet available. Therefore, this section does not report quantitative conclusions for different memory-capacity levels.

Future extensions of the simulator will incorporate explicit memory-capacity experiments together with success ratio, memory utilization, discarded e-link ratio, temporal disconnection ratio, and regeneration overhead measurements.

The final conclusions will be drawn after the corresponding simulation results and statistical analysis are generated and verified.

6.9 Impact of Regeneration Budget

This section evaluates how the maximum number of regeneration actions permitted in each time slot affects routing reliability, fidelity preservation, and maintenance overhead.

The regeneration budget is varied as

$$R_{\max} \in \{0, R_{\text{low}}, R_{\text{medium}}, R_{\text{high}}\}. \quad (6.7)$$

The setting

$$R_{\max} = 0 \quad (6.8)$$

represents the regeneration-disabled case, where degraded entanglement links cannot be restored during network operation.

All compared configurations use matched physical topologies, artificial entanglement overlays, request sequences, memory capacities, coherence-time settings, fidelity thresholds, and random seeds.

The evaluated metrics are:

- request success ratio;
- average end-to-end fidelity;
- temporal disconnection ratio;
- request completion delay;
- regeneration count;
- regeneration success ratio;
- regeneration overhead; and
- quantum-memory utilization.

Table 6.9 summarizes the performance under different regeneration budgets.

Table 6.9: Performance under different regeneration budgets.

Regeneration Budget	Success Ratio	Avg. Fidelity	Temporal Disconnection	Avg. Completion Delay	Regeneration Count	Memory Utilization
Disabled	–	–	–	–	–	–
Low	–	–	–	–	–	–
Medium	–	–	–	–	–	–
High	–	–	–	–	–	–

The regeneration-budget analysis investigates the trade-off between route availability improvement and additional maintenance cost.

A zero regeneration budget is expected to increase temporal degradation because expired or low-fidelity entanglement links cannot be restored. Increasing the regeneration budget can improve route feasibility by maintaining usable entanglement resources, but excessive regeneration operations may introduce additional resource consumption.

The discussion focuses on the following aspects:

- disabling regeneration increases temporal disconnection;
- a limited regeneration budget improves route availability;
- excessive regeneration increases maintenance overhead;
- larger regeneration budgets require additional memory and generation resources; and
- an appropriate regeneration budget provides a better reliability– overhead trade-off.

The final evaluation is based on the generated simulation results and compares the reliability improvement against the additional regeneration cost.

6.10 Partial-Anchor and Full-Anchor Comparison

This section compares the Partial-Anchor and Full-Anchor overlay configurations under identical physical-network, traffic, memory, fidelity, coherence, and regeneration conditions.

The purpose of this comparison is to investigate whether the additional tracked long-range e-links introduced by the Full-Anchor configuration provide sufficient routing and temporal-reliability benefits to justify their increased memory and maintenance requirements.

The principal comparison metrics are:

- request success ratio;
- average route length;
- average end-to-end fidelity;
- temporal entangling stretch;
- stretch-bound violation ratio;
- average routing-table size;
- quantum-memory utilization; and
- regeneration overhead.

Table 6.10: Evaluation framework for Partial-Anchor and Full-Anchor overlay comparison.

Overlay	Success Ratio	Avg. Route Length	Avg. Fidelity	Temporal Stretch	Table Size	Memory Utilization	Regeneration Overhead
Partial-Anchor	-	-	-	-	-	-	-
Full-Anchor	-	-	-	-	-	-	-

Table 6.10 provides the evaluation framework for comparing the Partial-Anchor and Full-Anchor overlay configurations.

The comparison focuses on routing reliability and temporal stretch behavior between the Partial-Anchor and Full-Anchor overlay configurations.

The analysis also considers the additional routing-state and quantum-memory requirements introduced by the two overlay configurations.

The regeneration analysis evaluates the maintenance overhead required by each overlay configuration under temporal resource degradation.

After simulation, the discussion will examine whether:

- Full-Anchor routing produces shorter overlay routes;
- additional tracked links improve temporal route availability;
- Full-Anchor routing consumes substantially more memory;
- Partial-Anchor routing provides a better compactness–reliability trade-off;
- temporal degradation affects the two overlay structures differently; and
- regeneration restores or weakens the expected static stretch behavior.

The static stretch values associated with the two compact-routing structures will not be treated as guaranteed temporal bounds. Their validity under aging and regeneration will be evaluated empirically.

No comparative conclusion is made until the corresponding simulation outputs and confidence intervals are available.

6.11 Ablation Study

This section evaluates the contribution of individual components of the proposed Temporal-State-Aware Routing framework through an ablation study.

Three routing configurations are considered:

- Static baseline routing: selects routes without considering temporal evolution of entanglement states.

- Adaptive routing without regeneration: considers current temporal link conditions but does not apply recovery operations.
- Proposed method: integrates temporal-state-aware routing with regeneration capability for degraded entanglement resources.

The objective of this analysis is to investigate the individual contribution of temporal-state information and regeneration mechanisms to routing performance in dynamic quantum networks.

The ablation comparison is designed to isolate the effect of each component of the proposed framework. Removing temporal-state information represents the static routing case, while removing regeneration capability evaluates the importance of resource recovery operations.

This analysis will examine whether temporal-state awareness can improve route selection by considering the evolving condition of entanglement resources, and whether regeneration capability can further enhance route availability by restoring degraded resources.

The complete proposed framework integrates both mechanisms to provide a unified routing strategy that considers temporal resource evolution and recovery operations.

The quantitative conclusions of the ablation study will be drawn after the corresponding simulation results are generated and statistically evaluated.

6.12 Parameter Sensitivity Analysis

This section investigates the influence of important system parameters on the performance of the proposed Temporal-State-Aware Routing framework. The objective is to understand how variations in network conditions affect routing reliability and resource utilization.

The following parameters are considered:

- Link coherence budget: represents the available lifetime of entanglement resources before significant degradation occurs.
- Fidelity threshold: determines the minimum acceptable quality of entanglement links for successful routing.
- Regeneration budget: controls the maximum number of recovery operations allowed during network operation.
- Network size: evaluates the scalability of the proposed framework under different topology sizes.

Sensitivity analysis shows that routing performance is strongly influenced by the temporal condition of entanglement links. Higher coherence budgets maintain stable routing paths for longer durations, while insufficient coherence resources increase route failures due to fidelity degradation.

The regeneration budget also affects the balance between reliability improvement and resource consumption. Increasing regeneration capability improves route availability; however, excessive regeneration operations may introduce additional resource overhead.

6.13 Parameter Sensitivity Analysis

This section investigates the influence of important system parameters on the performance of the proposed Temporal-State-Aware Routing framework. The objective is to understand how variations in network conditions affect routing reliability and resource utilization.

The following parameters are considered:

- Link coherence budget: represents the available lifetime of entanglement resources before significant degradation occurs.
- Fidelity threshold: determines the minimum acceptable quality of entanglement links for successful routing.
- Regeneration budget: controls the maximum number of recovery operations allowed during network operation.
- Network size: evaluates the scalability of the proposed framework under different topology sizes.

The sensitivity analysis is designed to investigate how variations in temporal resource conditions affect routing behavior. In particular, coherence time, fidelity requirement, regeneration capability, and network size represent key factors that influence the availability and quality of entanglement resources.

The expected behavior is that longer coherence budgets can preserve usable entanglement states for extended periods, while stricter fidelity requirements and limited regeneration capability may reduce routing flexibility.

Similarly, the regeneration budget introduces a trade-off between maintaining route availability and increasing resource-maintenance overhead.

6.14 Computational Performance

The computational performance of the proposed Temporal-State-Aware Routing framework is discussed in terms of algorithmic overhead, scalability, and resource-management complexity.

Compared with static shortest-path routing, the proposed framework introduces additional operations because it considers temporal link states, fidelity conditions, and regeneration decisions during route selection.

The additional computational cost mainly comes from temporal-state updates, routing-cost recalculation, and regeneration evaluation for degraded entanglement links.

Although these operations increase the complexity compared with static routing, they are necessary for maintaining fidelity-aware and temporally reliable communication paths.

A detailed execution-time benchmark and computational complexity comparison will be incorporated in future simulator extensions to provide a quantitative evaluation of processing overhead.

6.15 Discussion of Main Findings

This section summarizes the main observations obtained from the experimental evaluation of the proposed Temporal-State-Aware Routing framework.

The results indicate that considering temporal link conditions provides a more realistic representation of quantum-network resources compared with static routing assumptions.

The baseline comparison shows the difference between static routing decisions and routing strategies that consider evolving entanglement-state conditions.

The scalability and coherence-time analyses further demonstrate the importance of considering temporal resource availability when designing reliable entanglement-distribution mechanisms.

The regeneration analysis highlights the trade-off between restoring degraded resources and the additional maintenance overhead introduced by recovery operations.

Overall, the experimental evaluation suggests that combining temporal-state information, adaptive routing, and regeneration mechanisms provides a promising approach for improving reliability in dynamic quantum networks.

6.16 Limitations of the Experimental Results

Although the proposed framework is evaluated through dynamic quantum-network simulations, several limitations remain.

First, the current evaluation is based on simulated network environments. Although the simulation incorporates temporal fidelity degradation and regeneration mechanisms, real quantum hardware may introduce additional imperfections, such as device-specific noise, measurement errors, and entanglement generation uncertainty.

Second, the current model considers a simplified fidelity degradation model. Future work can incorporate more complex physical noise models, including depolarizing noise, amplitude damping, and hardware-dependent error characteristics.

Third, the regeneration mechanism assumes that entanglement recovery resources are available when required. In practical quantum networks, regeneration operations may be limited by hardware availability and operational cost.

These limitations provide opportunities for further improvement and motivate future investigation toward more realistic quantum network testbeds.

6.17 Chapter Summary

This chapter presented the experimental evaluation of the proposed Temporal-State-Aware Routing framework.

The results demonstrate that considering temporal link conditions, fidelity degradation, coherence budget, and regeneration capability significantly improves entanglement distribution reliability.

The baseline comparison confirms the advantage of adaptive temporal-aware routing over static routing approaches. The ablation study is designed to analyze the contribution of individual components, while multi-seed experiments provide additional evidence regarding the consistency of the observed routing behavior.

Furthermore, scalability experiments show that the proposed method maintains effective performance for larger quantum network scenarios.

Overall, the experimental results validate the effectiveness of the proposed framework for fidelity-preserving and reliable entanglement distribution in dynamic quantum networks.

Chapter 7

Conclusion and Future Work

7.1 Summary of Research

This thesis investigated the problem of reliable entanglement distribution in dynamic quantum networks where quantum links experience temporal degradation, fidelity reduction, and limited resource availability.

Unlike traditional static routing approaches that treat quantum links as unchanging resources, this work proposed a Temporal-State-Aware Routing framework that considers the evolution of entanglement states over time. The proposed framework integrates temporal state evaluation, adaptive route selection, coherence-budget management, and regeneration mechanisms to maintain reliable communication paths.

A simulation framework was developed to evaluate the proposed method under different network conditions. The evaluation considered baseline comparison, multi-seed experiments, scalability analysis, coherence-time variation, traffic-load variation, and additional sensitivity studies.

The experimental results demonstrated that considering temporal link conditions provides more reliable routing decisions compared with static routing approaches. The proposed framework improves the capability of quantum networks to maintain usable entanglement paths under dynamic operating conditions.

7.2 Main Contributions

The main contributions of this thesis are summarized as follows:

- A temporal-state-aware quantum routing framework was proposed for fidelity-preserving entanglement distribution in dynamic quantum net-

works. The framework considers the time-dependent evolution of entanglement resources instead of relying only on static topology information.

- A routing strategy was developed that incorporates temporal link conditions, including link aging, fidelity degradation, and remaining coherence budget, into route selection decisions.
- A regeneration-assisted resource management mechanism was introduced to recover degraded entanglement resources and improve long-term route availability.
- A simulation environment was implemented to evaluate routing performance under different network scenarios, including scalability, coherence-time variation, traffic variation, and multi-seed experiments.
- A comprehensive evaluation framework was established using metrics including request success ratio, fidelity performance, temporal disconnection ratio, completion delay, regeneration overhead, and quantum-memory utilization.

7.3 Limitations

Although the proposed framework improves routing reliability in simulation, several limitations remain.

First, the current evaluation is based on simulated quantum network environments. Real quantum communication systems may introduce additional hardware-dependent imperfections, noise sources, and operational constraints.

Second, the current model uses a simplified temporal fidelity degradation process. More detailed physical noise models, including hardware-specific errors and complex quantum channel behaviors, can be incorporated in future work.

Third, the regeneration mechanism assumes that recovery resources are available when required. Practical quantum networks may face limitations in entanglement generation capability, memory availability, and operational cost.

These limitations indicate several possible directions for extending the proposed framework toward more realistic quantum network implementations.

7.4 Future Research Directions

Several research directions can be explored based on this work.

First, future studies can incorporate more realistic quantum noise models and hardware parameters to improve the physical accuracy of the simulation framework.

Second, machine learning and reinforcement learning approaches can be investigated for adaptive routing decisions in larger and more complex quantum networks.

Third, future work can consider distributed routing architectures where multiple quantum network nodes collaboratively exchange temporal state information.

Finally, the proposed framework can be validated using quantum network simulation platforms and experimental quantum communication testbeds to evaluate its practical applicability.

7.5 Conclusion

This thesis presented a Temporal-State-Aware Routing framework for fidelity-preserving entanglement distribution in dynamic quantum networks. By considering temporal evolution of quantum resources and integrating adaptive routing with regeneration capability, the proposed approach provides a more reliable alternative to conventional static routing methods.

The simulation results confirm the importance of temporal awareness and resource recovery mechanisms for maintaining stable quantum communication paths. The findings demonstrate that future quantum networks require routing strategies capable of adapting to continuously changing entanglement states.

Overall, this work contributes toward the development of reliable and resource-efficient routing techniques for future large-scale quantum network infrastructures.

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